

Designing a Stablecoin Settlement Platform

for Existing Real-Time Payment Networks

A Reference Architecture Using Zelle as a Public Case Study

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Evidence disclaimer. Zelle is used only as a public case study. This paper does not describe or infer confidential Early Warning Services architecture, vendors, data stores, message contracts, controls, service levels, or launch mechanics. Public facts, documented technology behavior, inferred logical capabilities, proposed architecture, operational assumptions, and unknowns are explicitly distinguished.

Existing real-time payment networks create a customer expectation of immediate, familiar money movement, while stablecoin settlement introduces asynchronous issuers, custodians, signing systems, blockchains, foreign-exchange providers, and payout rails. This paper asks how a bank-distributed network can add that settlement capability without turning its customer experience into a cryptocurrency wallet or treating a blockchain receipt as complete financial truth.

The proposed answer is a regulated control plane centered on an internal double-entry ledger and a durable, evidence-gated payment state machine. Participant institutions retain authentication, account posting, and customer support. The control plane accepts idempotent payment intents; coordinates identity, policy, fraud, sanctions, quotes, routing, and liquidity; and delegates chain-native work through explicit settlement, custody, node, issuer, foreign-exchange, and payout adapters. Transactional outbox and inbox patterns, independent observation, reconciliation, and compensating entries construct exactly-once business effects across systems that cannot share one transaction. Blockchain finality, legal settlement finality, customer-visible completion, and accounting finality remain separate policy decisions.

Zelle provides a public case study because its bank distribution, scale, and announced international stablecoin direction make these tensions concrete. The paper neither knows nor claims its private implementation. Public facts, documented technology behavior, inferred logical roles, proposed design, operational assumptions, and unresolved questions are kept distinct.

The recommended pilot uses one approved regulated asset, one approved public network, one corridor, bounded participants, prefunded institutional liquidity, no consumer wallet, and no bridge. Java and Spring fit the enterprise control plane; smart contracts, node clients, and qualified signing remain in their strongest supported ecosystems. Stablecoin settlement is therefore an additional governed rail, not the operational ledger of record.

Keywords. stablecoin settlement; real-time payments; double-entry ledger; payment state machine; finality; Java; reconciliation; Zelle.

How to read this paper. Readers seeking the recommendation and boundary should read sections [1](#), [4](#) and [19](#). Payment and reliability specialists should continue through sections [6](#), [7](#), [11](#) and [12](#). Technology decision makers should use sections [15](#) to [17](#) together with the evidence matrix in section [D](#).

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1. Introduction: a settlement problem, not a wallet feature

Decision resolved. Preserve the participant institution as the customer channel and place stablecoin settlement behind a new institutional control plane. **PROPOSED**

Real-time payment products compress the customer-visible portion of money movement into seconds. Stablecoin networks can move tokenized value continuously. Neither fact makes the combined system atomic. A participant bank, network operator, issuer, custodian, signing service, blockchain, foreign-exchange (FX) provider, and destination payout rail each owns a different state, clock, failure domain, and legal obligation. The engineering problem is therefore to coordinate accountable financial effects under uncertainty—not simply to call a token contract.

Zelle makes the problem concrete without supplying a private implementation blueprint. **PUBLIC** EWS reports that Zelle is owned and operated by EWS, reached more than 2,300 financial institutions, and processed more than \$1.2 trillion across 4.2 billion transactions during 2025 (Early Warning Services 2026b; Early Warning Services 2026c). **PUBLIC** Consumers primarily use Zelle inside a participating bank or credit union experience and address recipients through an enrolled email address or U.S. mobile number (Zelle 2026b; Early Warning Services 2026d). Those facts establish a participant-distributed channel; they do not establish service topology, databases, contracts, queues, or recovery objectives. CLM-ZEL-001,CLM-ZEL-002,CLM-ZEL-003,CLM-ZEL-004,CLM-ZEL-005

Two announcements establish direction, with an important boundary. On 2025-10-24 EWS announced an initiative to use stablecoins for international Zelle money movement (Early Warning Services 2025b). On 2026-06-11 EWS named India as the first announced international destination and separately announced ZelleUSD for possible future capabilities in other markets (Early Warning Services 2026a). **PUBLIC** The latter announcement does not state that the India route uses ZelleUSD. It also does not disclose the issuer, authorization, reserves, redemption, network, contracts, custody, keys, controls, partners, or operating date of that asset. CLM-ZEL-011,CLM-ZEL-012,CLM-ZEL-013,CLM-ZEL-014,CLM-UNK-002

1.1. Method and evidence boundary

Material statements use one evidence class: **PUBLIC** validated public fact, **CODE** official source-code behavior, **DOC** official documentation, **INFERRED** logical decomposition, **PROPOSED** target architecture, **ASSUMPTION** operating posture to approve, or **UNKNOWN** fact requiring validation. The canonical claim ledger is generated from evidence/evidence-matrix.csv; section D gives the exact mapping. Technology documentation proves what a technology says or does, not that EWS deploys it. Proposed components are recommendations, not inferred private boxes.

The method is obligation-first. The paper identifies customer, participant, accounting, legal, security, and operating obligations; assigns authority for each; defines evidence-gated transitions; then evaluates implementation choices. This ordering prevents a familiar technology from becoming the architecture by default.

1.2. Consumer wallet versus invisible settlement

The invisible rail is not “crypto hidden from the customer.” It is a deliberate allocation of duties: the participant remains accountable for its channel and account; the platform remains accountable for payment state, internal obligations, policy evidence, recovery, and participant truth; external providers remain accountable within explicit contracts and trust boundaries.

Table 1: The paper selects invisible institutional settlement for the pilot; a visible wallet is a different product and risk decision.

Dimension	Consumer-wallet framing	Invisible institutional-settlement framing
Customer instrument	Customer holds or controls tokens and addresses	Customer sees the institution's existing account and payment contract
Key and gas burden	Product must explain custody, recovery, fees, and transaction authorization	Qualified institutional custody and treasury absorb chain mechanics
Completion promise	Often coupled to chain status	Defined independently through product, payout, legal, and accounting gates
Support and disputes	New wallet and irreversible-transfer operating model	Existing participant support retained, with route-specific case evidence
Pilot value	Tests a new consumer proposition	Tests settlement economics, controls, recovery, and corridor operations

2. Problem context and obligation map

Decision resolved. Treat international stablecoin settlement as an additional set of systems of record and legal actors, not as a replacement for current payment obligations. **PROPOSED**

Public Zelle material shows participant institutions, enrolled aliases, a network role, support responsibilities, risk controls, and domestic settlement options. An EWS product brief depicts sending and receiving institutions settling through ACH or RTP, but does not prove that every payment follows one route (Early Warning Services 2025c). The RTP network describes final interbank settlement, while FedNow describes continuous processing and final settlement for eligible institutions (The Clearing House 2026; Federal Reserve Financial Services 2026a; Federal Reserve Financial Services 2026b). Nacha permits ACH reversals only for defined conditions and timeframes (Nacha 2026). These distinct behaviors already show why “complete” cannot be one universal status.

INFERRED A logical domestic path includes a customer channel, participant authentication and account posting, alias resolution, network coordination, fraud/risk decisions, a domestic settlement rail, notification, and support. This decomposition explains obligations; it is not a claim that these are independent deployed services. A stablecoin path adds issuer and redemption rights, custody and signing authority, network and node evidence, transaction fees, contract controls, chain analytics, FX, treasury inventory, liquidity, a destination payout provider, and jurisdiction-specific compliance.

Table 2. Role and obligation map. External evidence must be translated into platform state, never accepted as unqualified business truth.

System or role	Owner class	Authoritative evidence	Principal exposure	Evidence
Participant channel/account	Financial institution	Authenticated request, account posting, participant case record	Unauthorized payment, insufficient funds, inconsistent customer status	PUBLIC
Alias/network coordination	Network role	Enrollment/recipient resolution and network operation record	Misrouting, stale enrollment, duplicate intent	PUBLIC / INFERRED
Domestic settlement rail	Rail and member banks	Rail acknowledgement and settlement record	Cutoff, return/reversal constraints, participant position	PUBLIC

System or role	Owner class	Authoritative evidence	Principal exposure	Evidence
Control plane	Proposed operator	Intent, policy decision, workflow transition, ledger journal	Premature completion, duplicate effect, control bypass	PROPOSED
Stablecoin issuer	External regulated entity	Terms, reserve/redemption records, contract events	Depeg, freeze, redemption denial, insolvency, upgrade	UN- KNOWN until selected
Custody and signer	Qualified provider/approved owner	Policy decision, signed bytes, key audit, transaction identifier	Key compromise, unauthorized or unavailable signing	PROPOSED boundary
Blockchain and nodes	Network/provider	Independently observed transaction, receipt, block and finality evidence	Reorganization, censorship, congestion, inconsistent nodes	DOC / selection unknown
FX and treasury	Approved providers/operator	Quote, fill, inventory, settlement and exposure records	Expiry, slippage, shortfall, counterparty concentration	PROPOSED
Destination payout	Payout provider/institution	Idempotent payout operation and recipient-credit evidence	Final chain success but failed or delayed local credit	PROPOSED
Compliance/case operations	Accountable legal and operations roles	Versioned policy decision, screening inputs, case disposition	False positive/negative, post-payment designation, hold handling	PUBLIC obligation / PROPOSED control

The domestic route and the stablecoin route can share intent, policy, ledger, state, notification, audit, and case semantics. They cannot share an assumption that all settlement providers behave alike. The stablecoin route adds non-atomic value movement and an observation problem: the submission call, provider callback, node view, issuer record, custody balance, and internal journal can disagree temporarily or permanently. The architecture must preserve that disagreement long enough to resolve it safely.

Consequence. Domestic payment availability must not depend on digital-asset services. Stablecoin-provider failure disables or degrades only routes that need that provider. Existing participant, fraud, support, and settlement duties remain in force.

3. Goals, non-goals, and design principles

3.1. Testable goals

Decision resolved. Define success as safe, recoverable, and reconcilable payment obligations with a familiar participant experience—not merely successful token transfer. **PROPOSED**

Context: public record versus proposed settlement extension

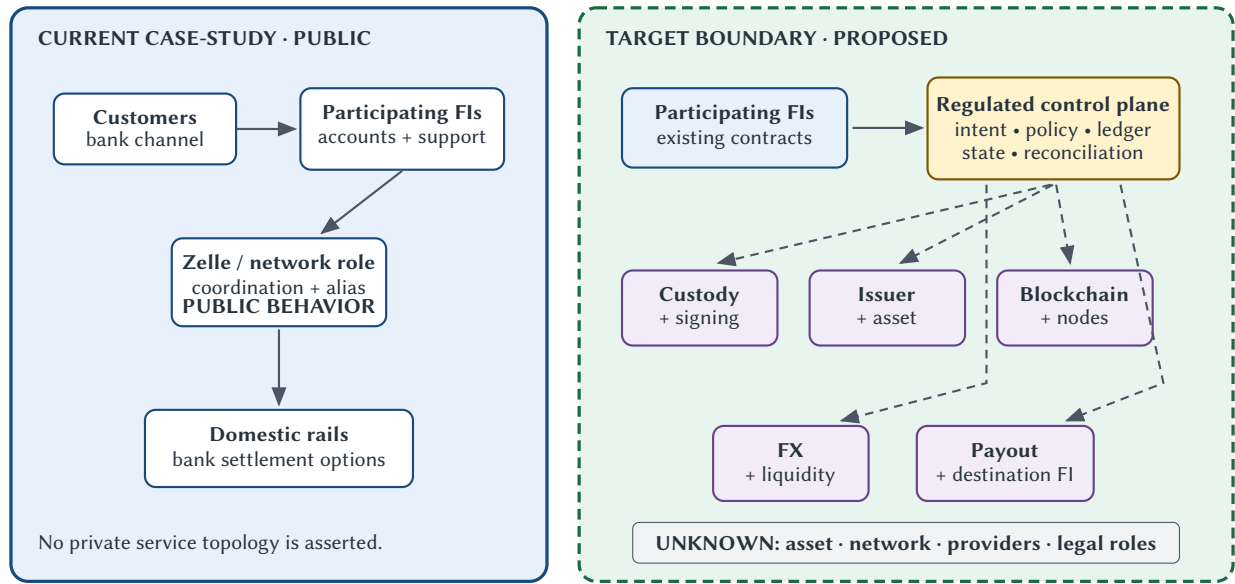


Figure 1: Public case-study context and the proposed target boundary. The PUBLIC region records only documented roles; all internal service boxes are explicitly PROPOSED. *Evidence treatment*: in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

Table 3. Goals are measurable at a decision gate even when an approved numeric target remains a Phase 0 unknown.

ID	Goal	Verification signal or gate	Owner
G1	Preserve bank-owned authentication, account, support, and customer status contracts.	Contract tests show no wallet, key, gas, or chain vocabulary in the participant API; route failure does not disable domestic payment.	Product / participant API
G2	Make every accepted request idempotent.	Same key plus same canonical payload returns the same intent; same key plus conflicting payload is rejected; concurrency tests show one effect.	Payment intent
G3	Preserve double-entry integrity.	Every journal balances per asset/currency; no direct balance mutation; operation IDs are unique; corrections are compensating entries.	Ledger / accounting
G4	Gate external effects through durable state and policy evidence.	Each effect has a recorded operation, expected version, authorization, timeout, and inquiry path.	Workflow / risk
G5	Isolate asset, issuer, chain, custody, FX, and payout providers.	Provider-specific adapters expose capabilities and risk; kill switches contain a provider without corrupting shared state.	Integration / SRE

ID	Goal	Verification signal or gate	Owner
G6	Construct exactly-once business effects across at-least-once delivery.	Duplicate, late, reordered, and replayed events cannot duplicate a posting, signature, submission, payout, or notification.	Platform / service owners
G7	Reconcile internal obligations to independent external evidence.	Every external balance/mutation has owner, cadence, tolerance, break state, case workflow, and aging alert.	Reconciliation / treasury
G8	Separate four finalities.	Product/corridor policy independently records chain, legal, customer, and accounting decisions and their evidence.	Product / legal / accounting / chain risk
G9	Bound operational and financial exposure.	Limits, liquidity buffers, pause scopes, degraded modes, and drills pass approved pilot gates.	Treasury / operations / security
G10	Retain asset/network exit.	Contract versions and data model can stop a route, reconcile it, and introduce a replacement without rewriting participant contracts.	Architecture / product
G11	Produce auditable evidence and telemetry.	Every transition and operator action carries causation, policy/configuration versions, business operation ID, and tamper-evident audit linkage.	Security / observability
G12	Learn measurably in the pilot.	Approved success, stop, exposure, reconciliation, recovery, customer, cost, and reliability measures determine phase promotion.	Program governance

Speed and certainty are intentionally separate. The system may expose an earlier customer status when an approved party absorbs the interval risk, but that policy does not alter chain, legal, or accounting evidence. Likewise, adapters improve portability only while preserving provider-specific powers such as freeze, pause, upgrade, fee, and finality behavior.

3.2. Non-goals

The initial design does not:

- expose a consumer wallet, seed phrase, address book, gas token, or self-custody recovery process;
- require a customer to receive or hold a stablecoin;
- replace the existing domestic Zelle experience or make it depend on a blockchain;
- preselect ZelleUSD, another stablecoin, a chain, proprietary infrastructure, or Hyperledger Besu;
- build a bridge, new blockchain, or permissioned network as the default pilot;
- claim universal asset, chain, bank, or corridor interoperability;
- solve every jurisdiction, data-residency rule, disclosure, refund, tax, or legal-finality question;
- provide legal advice or describe the current private EWS implementation; or

- implement production software or smart contracts.

These are scope boundaries, not permanent prohibitions. They isolate the value of the first experiment: can a regulated operator move institutional settlement value through one selected route while keeping accounting, policy, recovery, and customer obligations coherent? A later phase may change a boundary only through evidence and a separately owned product/architecture decision.

3.3. Principles that constrain the design

Table 4. Design principles have concrete consequences and prevent named failure modes.

Principle	Consequence	Prevented failure	Applied in
Preserve participant contracts.	Add route/version fields behind compatible payment-intent semantics; keep customer status provider-neutral.	Chain details leak into every institution and make exit impractical.	§3, §6
Operational ledger is off-chain and double-entry.	Every reservation, fee, FX, pay-out, custody, and suspense obligation is journaled internally.	Token ownership is mistaken for customer or participant entitlement.	§5
Separate intent, authorization, execution, observation, reconciliation.	Each stage has distinct evidence, owner, operation ID, and retry policy.	A timeout or callback silently repeats value movement.	§4–8
External mutation is uncertain.	Timeouts enter inquiry/manual-review states; blind resubmission is prohibited.	Both original and retry succeed.	§6, §8
Construct business-effect exactly once.	Combine idempotency, ACID, versions, outbox/inbox, inquiry, compensation, and reconciliation.	Delivery guarantees are overstated across systems.	§7
Four finalities never collapse.	Product policy records chain, legal, customer, and accounting gates separately.	A confirmation count becomes an unsupported legal or customer promise.	§8
Adapters expose material differences.	A stable interface includes capabilities, evidence semantics, error classification, and control powers.	Least-common-denominator abstraction hides freeze, reorg, or redemption risk.	§3–4
Minimize on-chain data.	Use opaque intent references; keep personal and policy data off-chain.	Irreversible disclosure, correlation, and deletion conflicts.	§9
Bound exposure by default.	Prefund, limit, allowlist, rate-limit, and pause by participant/corridor/asset/provider.	A novel failure scales before recovery is proven.	§8, §12
Manual action is governed software behavior.	Cases require roles, reason codes, four-eyes approval where needed, and immutable evidence.	Ad hoc repair bypasses segregation and audit.	§8–9

Principle	Consequence	Prevented failure	Applied in
Design for issuer, asset, and chain exit.	Version contracts and persist neutral operations plus provider-specific evidence.	Concentration becomes technical lock-in.	§11–12
Use the strongest supported ecosystem.	Java owns enterprise control; Solidity/Rust/Go/native services own chain-specific work where appropriate.	Language preference chooses a weaker node, signer, or contract toolchain.	§10
Promote only evidence-backed claims.	Unknowns remain gates and operational assumptions remain labeled.	A plausible design is misreported as deployed fact.	all chapters

Consequence. The architecture accepts coordination and reconciliation work because that work makes obligation, uncertainty, and authority visible. Safety does not emerge from a fictional end-to-end transaction.

4. Reference architecture

Decision resolved. Place ledger and workflow authority inside a regulated control plane; treat chains and providers as external execution and evidence domains. **PROPOSED**

PROPOSED CLM-PROP-001, CLM-PROP-002, CLM-PROP-003, CLM-PROP-004

The architecture has two interacting planes. The *control plane* accepts intent, makes policy and routing decisions, reserves funds, records obligations, authorizes effects, advances state, communicates participant truth, and reconciles. The *settlement plane* signs and submits transactions, moves token or fiat value, observes networks, obtains issuer/FX/payout evidence, and returns that evidence. A settlement provider never promotes business state directly; it produces an observation that the authoritative workflow evaluates against route and policy versions.

4.1. Layers and authority

Table 5. Architecture layers separate authority from execution.

Layer	Responsibility / owned state	Inputs and outputs	Prohibited responsibility	Evidence
Participant channel	Authentication, customer account, consent, presentation, support	Payment intent/status; account reservation/posting	Holding platform keys or deriving chain finality	PUBLIC context
Edge and intent	Contract version, participant identity, idempotency record, canonical request	Authenticated request; stable intent ID and receipt	Making settlement finality or accounting decisions	PROPOSED

Layer	Responsibility / owned state	Inputs and outputs	Prohibited responsibility	Evidence
Policy and case	Fraud, sanctions, product, limit, corridor, and manual disposition evidence	Versioned facts; decision and obligations	Editing ledger history or silently overriding signer policy	PRO-POSED
Orchestration	Payment aggregate, transaction events, operation plan, timeouts	Commands and observations; durable states	Owning account balances or treating provider status as canonical	PRO-POSED
Ledger and treasury	Journals, accounts, reservations, positions, fees, exposure	Authorized operation; balanced postings and balance version	Calling chains/payout providers inside a ledger transaction	PRO-POSED
Adapters	Provider-specific command, inquiry, evidence normalization, capability version	Operation ID; provider request/observation	Collapsing freeze, re-org, issuer, or payout semantics into generic success	PRO-POSED
Custody/signing	Key policy, approval, canonical-byte verification, signature audit	Authorized unsigned transaction; signed result/refusal	Accepting raw destination/amount outside approved intent	PRO-POSED / vendor DOC
External rails	Token/fiat movement and provider records	Submission; receipts, balances, fills, credits	Declaring participant payment complete	EXTERNAL / selection unknown
Evidence/operations	Independent observation, reconciliation, telemetry, cases, controls	Internal/external records; breaks, alerts, evidence packages	Mutating source records to force agreement	PRO-POSED

The ledger is single-writer by account partition, not necessarily one process or database instance. A partition owner serializes conflicting reservations and postings for its accounts; a journal spanning partitions requires a deterministic coordinator or a posting topology that keeps an atomic journal within one ownership boundary. The exact partition strategy is an implementation ADR after workload and accounting requirements are known. Read models may scale independently but cannot authorize money movement from stale balances.

Domestic and stablecoin routes share edge, intent, policy, ledger, workflow, messaging, audit, and participant status. The domestic adapter does not call the digital-asset plane. Therefore a signer, node, chain, issuer, or custody outage can disable affected routes while domestic service remains available, subject to its own dependencies.

4.2. Trust and deployment boundaries

1. **Participant boundary.** Mutual authentication, participant authorization, request signing, quotas, contract versions, and replay protection apply before intent acceptance.
2. **Regulated operator boundary.** Policy, ledger, workflow, configuration, audit, and case services run under controlled identities. Database ownership is service-scoped; shared-table integration is prohibited.

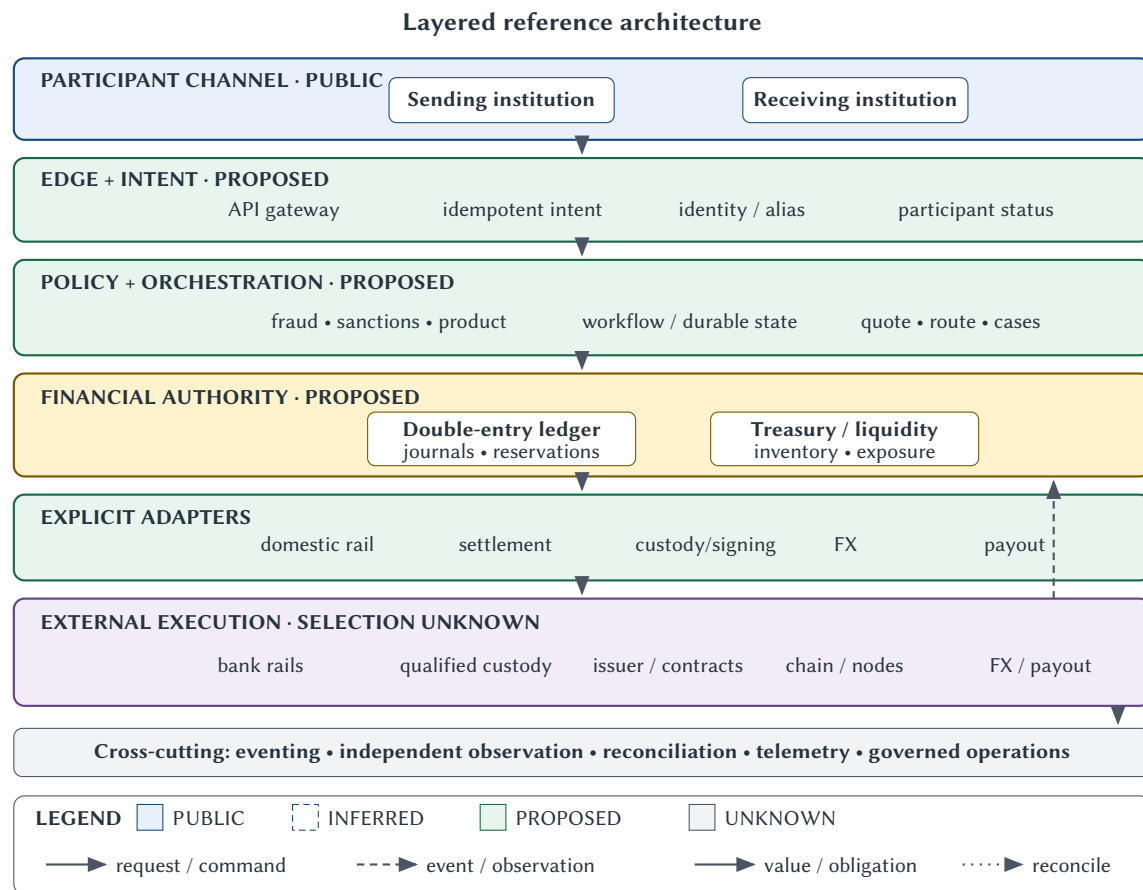


Figure 2: The layered reference architecture. The ledger and workflow are the operational center; external rails execute bounded effects and return evidence. All green components are proposed, not deployed EWS topology. *Evidence treatment:* in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

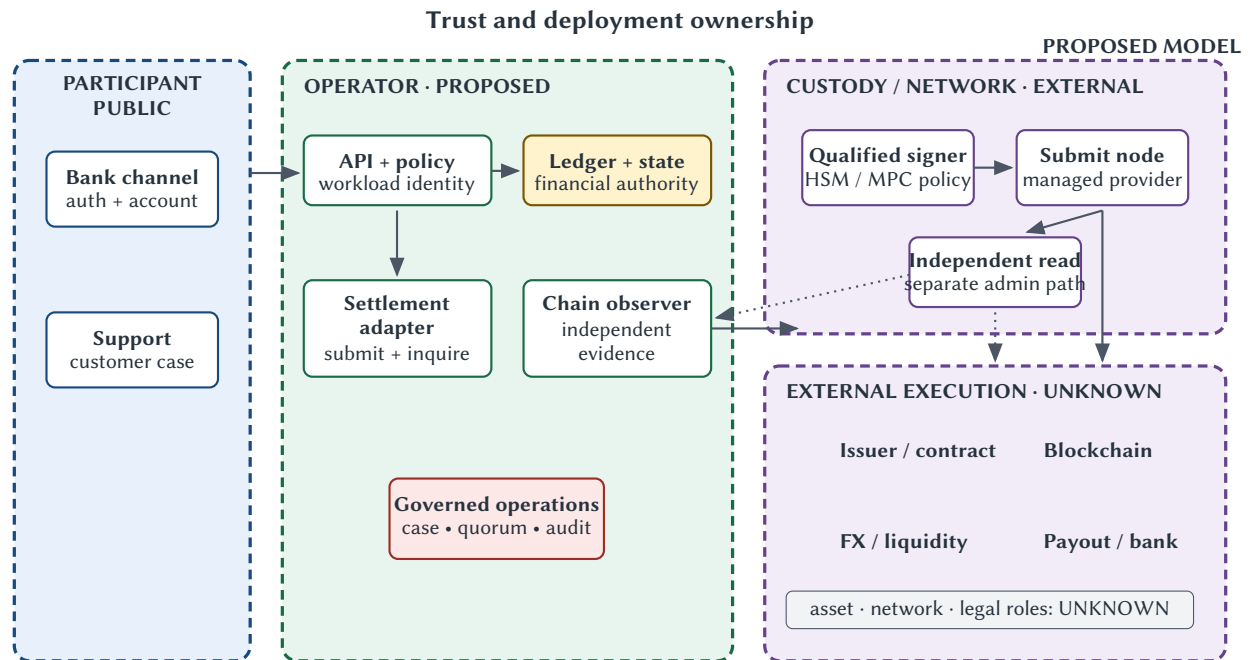


Figure 3: Deployment and trust ownership. No provider callback crosses directly into ledger mutation; custody and independent observation form separate authorization and evidence paths. *Evidence treatment*: in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

3. **Signing boundary.** Private keys remain within an approved HSM, multiparty-computation (MPC), or qualified custody service. The signer independently verifies canonical transaction fields against an authorized intent.
4. **Observation boundary.** Submission and observation are separable. A managed RPC endpoint may submit, but an independent node/provider and custody/issuer records are used to detect inconsistent evidence.
5. **Provider boundary.** Each issuer, FX, payout, chain, analytics, and custody provider has distinct identity, availability, data, legal, and exit obligations.
6. **Human authority boundary.** Operators act through case workflows with least privilege, reason codes, approval policy, immutable audit, and time-bounded elevation.

4.3. Control plane versus chain-centric design

A chain-centric design makes token transfer the primary state and attempts to reconstruct product obligations around it. That model cannot naturally represent participant reservations, fiat accounts, quote expiry, fees, pending payout, post-payment cases, customer status, or accounting close. The proposed model records those obligations before and after the external transaction. Chain evidence remains indispensable, but only for the portion of truth that the chain owns.

Provider adapters are likewise not generic “send money” interfaces. A common operation envelope supplies business operation ID, intent version, amount/asset, route, deadline, and authorization. Provider-specific capabilities report whether inquiry, replacement, nonce control, freeze, burn/mint, memo, finality proof, or payout cancellation exists. Portability comes from stable platform semantics plus explicit capability branches, not from hiding differences.

Consequence. Every later flow maps to one control-plane authority, one external mutation operation, and one evidence path. No callback, node, contract, or provider duplicates the business source of truth.

5. Component contracts and ownership

Decision resolved. Give each component one state authority, each financial effect one business operation ID, and each boundary an explicit ambiguity contract. **PROPOSED**

An accepted participant request receives an immutable payment-intent ID. Every effect below it receives a business operation ID scoped by intent, operation type, and attempt policy—for example PAY-123/CHAIN-SUBMIT/1. The operation ID is stable across transport retries. A new attempt number is permitted only after inquiry proves the prior operation absent or an operator-authorized strategy deliberately creates a distinct replacement.

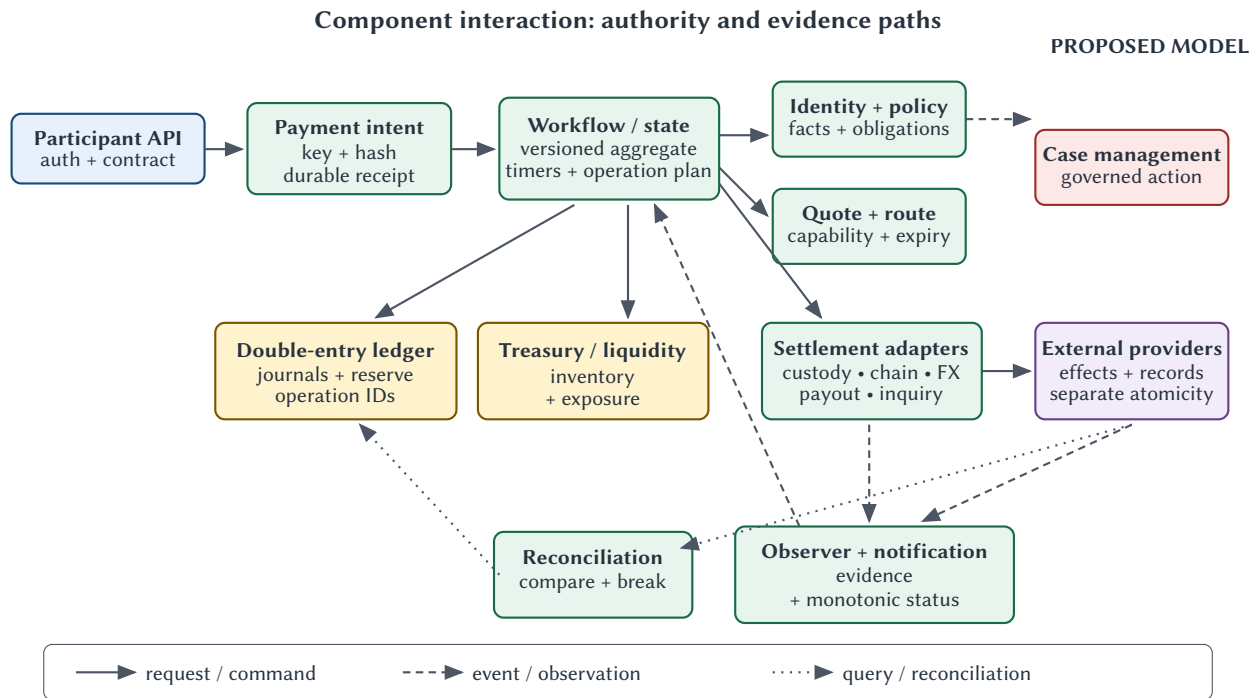


Figure 4: Bounded-context interaction. Synchronous validation ends at durable acceptance; financial execution continues through commands, observations, and reconciliation. *Evidence treatment*: in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

Table 6. Responsibility matrix. Each row defines authority, duplicate control, and failure posture.

Component	Owned data / responsibility	Idempotency or order key	Failure behavior and observability	Prohibited short-cut
Participant API gateway	Participant identity, authorization context, contract/rate policy	Participant + request key	Reject before acceptance; trace authentication and quota decision	Reusing a key across participants; trusting body identity
Payment-intent service	Canonical payload hash, intent receipt, participant idempotency	Participant + idempotency key	Same payload returns same intent; conflicting hash is a deterministic conflict	Starting side effects before durable acceptance
Identity/alias adapter	Versioned resolution request/evidence cache	Intent + resolution version	Unknown/stale result blocks route or revalidates; logs source/time	Claiming directory data ownership
Policy/risk service	Versioned facts, decisions, obligations, expiry	Intent + policy stage + version	Hold/reject/manual review; reason-coded metrics	Mutable opaque “risk passed” flag
Quote/router	Quote, route candidates, capability and expiry snapshot	Intent + quote request	Expired quote requires reauthorization and exposure recheck	Silent substitution after acceptance
Treasury/liquidity	Inventory view, reservations, limits, rebalance requests	Asset/account + operation	Shortfall queues or rejects; exposure and aging alerts	Spending from an eventually consistent balance
Ledger service	Accounts, journals, postings, reservations, account versions	Ledger operation ID; account partition	Atomic balanced post or no post; invariant metrics and audit	Direct balance updates or external call in transaction
Workflow/state service	Aggregate version, transitions, timers, operation plan	Intent ID + expected version	Conflict reloads deterministically; ambiguity enters explicit state	Provider callback writing status directly
Settlement adapter	Provider request map, capability/version, observations	Chain operation ID	Classifies absent/ambiguous/confirmed; exposes inquiry	Blind timeout retry or generic success
Signer/custody adapter	Authorization reference, canonical bytes hash, signature audit	Signing operation ID + bytes hash	Refusal/outage is explicit; quorum and latency telemetry	Private key in application or signer trusting caller fields
Chain observer	Node observations, block/receipt lineage, confidence/finality facts	Network + transaction/block	Conflicts create evidence incident; lag and divergence alarms	Advancing business state itself

Component	Owned data / responsibility	Idempotency or order key	Failure behavior and observability	Prohibited short-cut
Payout adapter	Idempotent payout request, provider mapping, credit evidence	Payout operation ID	Inquiry before retry; terminal/ambiguous codes preserved	Treating API 200 as recipient credit
Reconciliation	Comparison runs, breaks, evidence snapshots, disposition	Source + account + period	Break remains open/aged until evidence-backed disposition	Editing source data to clear a break
Notification	Participant/customer status delivery and receipt	Intent + status version + audience	At-least-once with monotonic versions; query remains authoritative	Notification success defining payment completion
Case management	Case, assigned role, action, approval, evidence links	Case/action ID	SLA, aging and segregation alerts	Unlogged repair or direct data mutation
Configuration/policy registry	Approved versioned route, limits, thresholds, kill switches	Configuration version	Safe default disables undefined route; change audit/rollout metrics	Mutable environment flags without approval history

The responsibility table is complemented by the following contract views. Together, the three rows for each component define all fields required for implementation review; the matrices are separated to preserve a readable print size.

Table 7. Component contract inputs and outputs.

Component	API or event inputs	Outputs
Participant API gateway	Authenticated participant request, contract version, idempotency key	Authorized canonical request or reason-safe rejection; trace context
Payment-intent service	Authorized request, canonical payload, participant key	Durable intent receipt, payload hash, validation command, status URI
Identity/alias adapter	Versioned resolution command, alias token, corridor context	Source-stamped resolution observation or explicit unknown/stale result
Policy/risk service	Versioned payment facts, party/address signals, route stage	Allow, reject, hold, obligation, expiry, and reason-coded decision evidence
Quote/router	Eligible routes, asset/network capability, amount, expiry policy	Versioned quote, selected route snapshot, alternatives, or no-route result
Treasury/liquidity	Reservation/release/rebalance command, position and limit version	Reservation evidence, exposure update, shortfall, or rebalance operation
Ledger service	Authorized balanced journal command, account versions, operation ID	Committed journal/postings and out-box fact, or invariant/version conflict

Component	API or event inputs	Outputs
Workflow/state service	Intent commands; ledger, policy, provider, timer, and case evidence	Versioned transition event, next operation plan, timer, participant status fact
Settlement adapter	Authorized operation, canonical provider payload, inquiry command	Provider request identity and normalized absent/ambiguous/observed evidence
Signer/custody adapter	Authorized canonical bytes/hash, key-policy reference, operation ID	Signature/refusal, quorum evidence, custody operation and audit references
Chain observer	Network/contract allowlist, transaction/block identity, finality policy	Source-stamped receipt/log/canonicity observations and confidence conflicts
Payout adapter	Authorized payout operation, beneficiary token, amount/currency	Provider request identity, inquiry result, recipient-credit or ambiguity evidence
Reconciliation	Versioned statements/observations, ledger snapshot, rule set, period	Match, break, aged exposure, evidence bundle, and governed disposition request
Notification	Monotonic participant status fact, audience and template version	Delivery attempt/receipt evidence; query link remains authoritative
Case management	Evidence snapshot, case type, assigned role, proposed action	Approved/rejected governed action, reason, approvers, expiry, audit event
Configuration/policy registry	Signed change proposal, approvals, effective window	Immutable approved version, roll-out/kill-switch event, read-only snapshot

Table 8. Component authorization, consistency, and scaling contracts.

Component	Authorization boundary	Consistency expectation	Scaling dimension
Participant API gateway	Mutual participant identity plus contract/role/rate policy	Authentication and request admission before durable acceptance	Participant, endpoint, request rate
Payment-intent service	Only gateway principal may create; owner participant scopes reads	Key/hash uniqueness and intent commit are one local transaction	Participant/key partition and creation rate
Identity/alias adapter	Workflow-approved purpose and minimized alias token	Versioned observation; staleness blocks or forces revalidation	Directory/provider quota and alias partition
Policy/risk service	Policy role controls facts; only governed service signs decisions	Decision binds exact fact and policy versions; no mutable pass flag	Stage, participant, model/provider latency
Quote/router	Workflow requests; treasury/risk gate eligible candidates	Quote and route capability snapshot are immutable through expiry	Corridor, asset/network pair, quote rate

Component	Authorization boundary	Consistency expectation	Scaling dimension
Treasury/liquidity	Treasury policy authorizes limits and rebalance; workflow requests reserve	Single writer/version per inventory partition; no overspend from replica	Asset, account, custody location, corridor
Ledger service	Narrow journal-command roles; segregation for correction/close	Serializable or locked account partition; balanced atomic commit	Account partition, contention, journal volume
Workflow/state service	Domain handler plus expected aggregate version; manual actions are role-approved	One versioned aggregate transition per commit; evidence may arrive at least once	Intent partition, timers, history, active cases
Settlement adapter	Route-specific service identity; command must reference approved operation	Local request-map commit precedes effect; observations are deduplicated	Provider/network quota, operation/inquiry rate
Signer/custody adapter	Independent key policy/quorum validates canonical bytes and intent binding	Signature identity immutable; no assumption of atomicity with workflow	Key class, approval queue, signer latency
Chain observer	Read-only network credentials and approved contract/network registry	Multiple provenance-preserving observations; canonicity can change before threshold	Network, block/log rate, reorg window
Payout adapter	Corridor/provider role and approved beneficiary/operation binding	Request identity durable before send; credit requires contractual evidence	Corridor/provider quota and payout/inquiry rate
Reconciliation	Read-only source access; disposition requires accounting/case role	Rerunnable versioned comparison over frozen source snapshots	Source, account, period, history volume
Notification	Audience-scoped delivery credential; cannot authorize payment state	At-least-once delivery with monotonic status version	Audience, channel, delivery rate
Case management	Assigned operator role, dual approval for material action	Append-only actions; case projection may lag but approval evidence may not	Case type, queue, SLA, analyst capacity
Configuration/policy registry	Segregated proposer/approver/deployer roles	Immutable signed version and effective-time ordering	Route/version count, rollout cohort, read fan-out

5.1. System-of-record matrix

5.2. Contract and failure semantics

Participant APIs use synchronous validation only for checks that can complete within a bounded request budget. Durable acceptance returns an intent ID and status link; policy, funding, signing, submission, finality, and payout continue asynchronously. This avoids keeping a client connection open across systems whose completion may take minutes or require manual review.

Table 9: Each question has one authoritative source and named corroborating evidence.

Question	Authority	Corroboration / recovery source
Was the participant request accepted?	Payment-intent record and canonical hash	Signed ingress/audit record
May the effect execute?	Versioned workflow transition plus policy obligations	Configuration snapshot and ledger reservation
What does the platform owe?	Balanced ledger journals and reservations	Participant account posting, treasury subledger
Was a transaction authorized?	Signer/custody authorization record and bytes hash	Workflow operation and policy version
Was it submitted?	Independent chain inquiry by transaction hash/nonce and custody record	Submission provider response
What happened on-chain?	Finality policy applied to independent block/receipt observations	Managed provider, explorer only as tertiary evidence
Was the recipient credited?	Destination institution/payout evidence under contract	Provider inquiry and participant confirmation
What status may the customer see?	Product/corridor completion policy applied by workflow	Underlying ledger and external evidence
Is the period/account closed?	Reconciliation disposition and accounting close	Chain, custody, issuer, payout, bank, FX statements

Commands express requested effects: `ReserveFunds`, `AuthorizeSigning`, `SubmitTransfer`, or `InitiatePayout`. Events express durable facts: `FundsReserved`, `SignatureRefused`, `TransactionObserved`, or `RecipientCreditConfirmed`. Observations from providers are not events until validated, deduplicated, and durably recorded by the owning adapter or observer.

Versioned contracts use additive evolution by default. Envelopes include schema version, aggregate version, business operation ID, causation/correlation IDs, producer, occurred time, and policy/configuration version. Unknown future versions quarantine rather than guess; exact duplicates acknowledge without reapplying; a different payload at the same aggregate version becomes a conflict incident.

Combining ledger and workflow in one mutable record looks simpler but creates ambiguous authority: accounting corrections and payment-state recovery have different invariants, retention, and ownership. The design separates them and relates them through immutable operation IDs and atomic local commands where co-location is deliberately chosen. A workflow state never substitutes for a journal; a journal never declares a customer-visible state.

Consequence. Financial mutation authority is narrow. Provider uncertainty is retained, observable, and recoverable rather than normalized into a misleading Boolean success.

6. Internal ledger

Decision resolved. The internal double-entry ledger is authoritative for business obligations; blockchain balances are external assets to reconcile. **PROPOSED** CLM-PROP-002

A chain can establish control of tokens at addresses. It cannot, by itself, represent a participant's reserved fiat, an unexpired quote, accrued fee, FX payable, recipient-credit obligation, sanctions hold, suspense investigation,

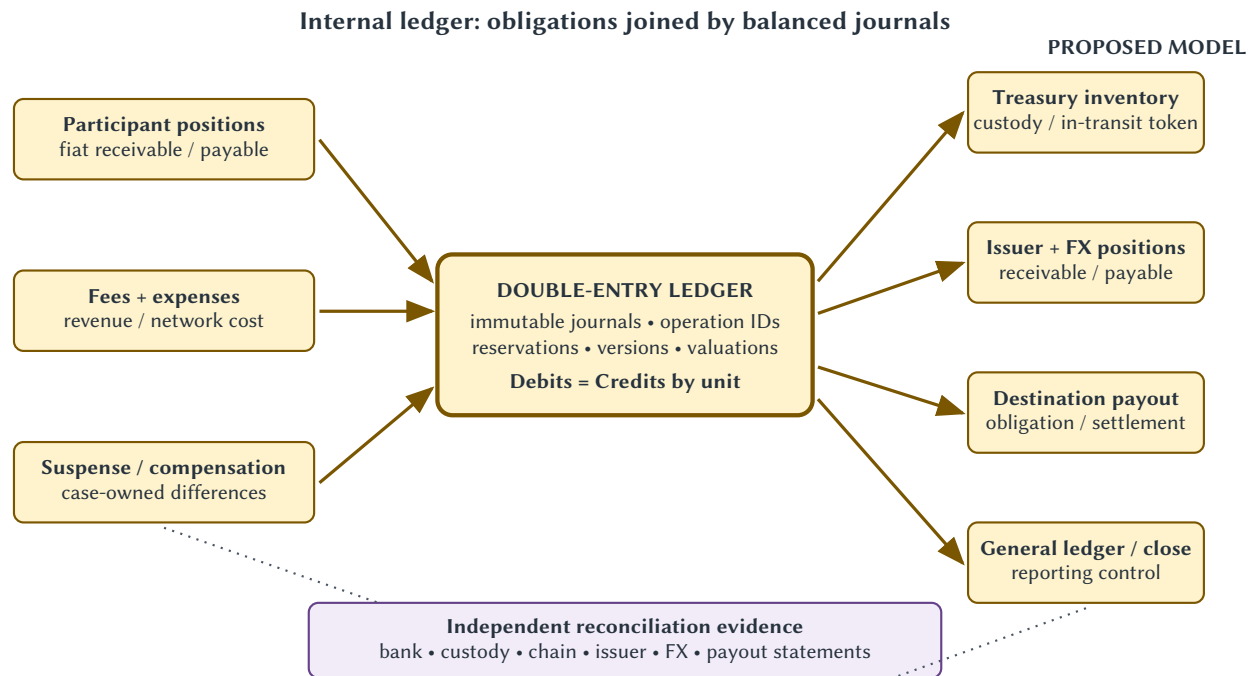


Figure 5: Ledger relationships. Participant positions, custody inventory, payout obligations, fees, FX, and suspense are separate accounts joined by balanced journals and reconciliation. *Evidence treatment*: in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

or period close. Those obligations span multiple asset and legal domains. The platform therefore records immutable journals composed of balanced postings. A balance is a derived projection, not an independently editable field.

An account is keyed by legal entity, owner/participant, asset or currency, account purpose, and where needed provider/custody location. Quantity and reporting-currency valuation are separate. Reservations reduce available capacity without rewriting settled quantity. A journal correction references the original and posts a compensating journal; “rollback” means a local database transaction did not commit, never deletion of a committed financial fact.

6.1. Illustrative chart of accounts

Table 10. Illustrative accounts. Final legal ownership and accounting treatment require accounting and counsel approval.

Account	Normal side	Dimension	Reconciles to	Purpose
Participant fiat receivable	Debit	Legal entity, participant, USD	Bank/participant statement	Funds due from sending participant
Participant settlement obligation	Credit	Legal entity, participant, USD	Participant settlement record	Amount due from or allocated to the participant settlement path

Account	Normal side	Dimension	Reconciles to	Purpose
Participant funds available	Credit	Legal entity, participant, USD	Participant pre-funding/settlement account	Prefunded participant liability still eligible for reservation
Customer payment funds reserved	Credit	Participant, intent, USD	Participant authorization	Memo/control account for unavailable committed capacity
Treasury stablecoin inventory	Debit	Custodian, wallet class, asset, chain	Custody + independent chain + issuer	Controlled token asset
Stablecoin in transit	Debit	Operation, asset, chain	Chain transaction/contract events	Submitted but not yet economically final quantity
Stablecoin settlement consumption	Debit	Payment, route, asset, chain	Final chain effect plus approved valuation	Economically final token quantity delivered beyond platform control
Settlement cost / delivered value	Debit	Payment, corridor, reporting currency	Valuation record and general ledger	Reporting-currency value consumed to create the destination obligation
Issuer redemption receivable	Debit	Issuer, asset, jurisdiction	Issuer statement/redemption	Claim awaiting fiat redemption
FX receivable/payable	Debit/Credit	Provider, quote, currency pair	FX fill and settlement	FX principal and unsettled position
Destination payout payable	Credit	Provider, corridor, currency	Payout/institution statement	Fiat owed to recipient delivery path
Payout provider cash/settlement asset	Debit	Provider, corridor, currency	Provider and bank statement	Controlled cash or settlement asset consumed when payout is proven
Fee receivable	Debit	Participant, product, currency	Participant invoice/settlement	Contractual fee due
Fee revenue	Credit	Product, legal entity, currency	General ledger	Recognized platform fee
Network/custody fee expense	Debit	Provider, operation, asset/currency	Chain/custody invoice	External execution cost
Suspense—external break	Debit/Credit	Source, case, asset/currency	Reconciliation evidence	Controlled temporary difference
Compensation expense/recovery	Debit/Credit	Incident, responsible party, currency	Case and recovery records	Loss absorption and recovery

6.2. Balanced posting examples

The examples use a \$100 payment, a \$1 participant fee, and separate journals per currency/asset. They illustrate mechanics rather than a legal/accounting conclusion. Debit and credit totals balance within each journal; the asset quantity and reporting valuation are not mixed.

Table 11: J1A—accept the participant settlement obligation (USD 100.00).

Entry	Account	Dimension	Debit USD	Credit USD
J1A.1	Participant fiat receivable	FI-A / PAY-123	100.00	—
J1A.2	Participant settlement obligation	FI-A / PAY-123	—	100.00
Total			100.00	100.00

Table 12: J1B—reserve prefunded participant capacity for the accepted intent (USD 100.00).

Entry	Account	Dimension	Debit USD	Credit USD
J1B.1	Participant funds available	FI-A / USD	100.00	—
J1B.2	Customer payment funds reserved	FI-A / PAY-123	—	100.00
Total			100.00	100.00

J1B is the committed ledger evidence for FUNDS_RESERVED: it reduces available prefunded capacity in the same balanced transaction that records the reservation operation and outbox. A credit-capacity product may instead use a ledger-native encumbrance against an approved limit, but it must enforce the same atomic available-capacity reduction. The accounting model and exact account names are Phase 0 decisions; workflow never creates the reservation by itself.

Table 13: J2—designate 100.00 units of stablecoin inventory for an authorized chain operation.

Entry	Account	Dimension	Debit token	Credit token
J2.1	Stablecoin in transit	CHAIN-OP-1	100.00	—
J2.2	Treasury stablecoin inventory	CUST-A / ASSET-X	—	100.00
Total			100.00	100.00

J2 is posted when the approved accounting policy changes inventory classification, not merely because an RPC returned HTTP 200. If submission is ambiguous, the in-transit classification and workflow ambiguity remain until inquiry resolves them.

Table 14: J3A—economic chain finality clears stablecoin inventory in transit (100.00 token units).

Entry	Account	Dimension	Debit token	Credit token
J3A.1	Stablecoin settlement consumption	PAY-123 / ASSET-X	100.00	—
J3A.2	Stablecoin in transit	CHAIN-OP-1	—	100.00
Total			100.00	100.00

J3A is the required disposition of J2: once the approved chain-finality policy proves delivery, the token quantity can no longer remain in transit. The valuation record links this token-unit journal to the separately balanced reporting and payout-currency journal below.

Table 15: J3B—apply reserved participant funding to the destination payout obligation (USD 100.00).

Entry	Account	Dimension	Debit USD	Credit USD
J3B.1	Customer payment funds reserved	FI-A / PAY-123	100.00	—
J3B.2	Destination payout payable	PAYOUT-OP-1	—	100.00
Total			100.00	100.00

J3B consumes the exact reservation created by J1B; it does not create a second source of value. The actual design may recognize an FX asset, issuer receivable, intercompany position, or reporting-currency valuation between token and destination currency. Those route-specific journals remain separate from the participant-funding journal. Each leg must balance in its own unit, carry the approved valuation record, and leave no operation-scoped transit balance without a named exception or reconciliation break.

Table 16: J3C—close the accepted participant settlement control pair after funding is applied (USD 100.00).

Entry	Account	Dimension	Debit USD	Credit USD
J3C.1	Participant settlement obligation	FI-A / PAY-123	100.00	—
J3C.2	Participant fiat receivable	FI-A / PAY-123	—	100.00
Total			100.00	100.00

J1A and J3C are an illustrative acceptance-control pair, not a claim about required external-accounting recognition. A production chart may omit that pair when the reservation and funding-application journals alone represent the approved legal and accounting event. If retained, both accounts must close on success or through the matching failure disposition; neither may remain as an unexplained successful-payment balance.

Table 17: J4—destination provider proves recipient credit and settles the obligation (USD 100.00).

Entry	Account	Dimension	Debit USD	Credit USD
J4.1	Destination payout payable	PAYOUT-OP-1	100.00	—
J4.2	Payout provider cash/settlement asset	PROVIDER-B	—	100.00
Total			100.00	100.00

Table 18: J5—recognize the participant fee (USD 1.00).

Entry	Account	Dimension	Debit USD	Credit USD
J5.1	Fee receivable	FI-A / PAY-123	1.00	—
J5.2	Fee revenue	PRODUCT-X	—	1.00
Total			1.00	1.00

Table 19: J6A—compensate a failed route by releasing the participant obligation (USD 100.00).

Entry	Account	Dimension	Debit USD	Credit USD
J6A.1	Participant settlement obligation	FI-A / PAY-123	100.00	—
J6A.2	Participant fiat receivable	FI-A / PAY-123	—	100.00
Total			100.00	100.00

Table 20: J6B—after proof of no chain effect, restore the in-transit token classification (100.00 token units).

Entry	Account	Dimension	Debit token	Credit token
J6B.1	Treasury stablecoin inventory	CUST-A / ASSET-X	100.00	—
J6B.2	Stablecoin in transit	CHAIN-OP-1	—	100.00
Total			100.00	100.00

J6B is permitted only when independent inquiry proves that no chain transfer occurred, or that a reverted transaction left the platform-controlled inventory intact. If the chain transfer actually succeeded, neither J6A nor J6B is safe: the platform would release participant funding or recreate inventory while retaining an external loss. That case requires loss allocation and perhaps a compensation-expense/recovery journal, not a mechanical reversal.

Table 21: J6C—release a reservation only after the route is terminated without consuming participant funding (USD 100.00).

Entry	Account	Dimension	Debit USD	Credit USD
J6C.1	Customer payment funds reserved	FI-A / PAY-123	100.00	—
J6C.2	Participant funds available	FI-A / USD	—	100.00
Total			100.00	100.00

J6C is the exact failure disposition of J1B. It is forbidden after J3B has consumed the reservation unless a separately authorized compensation journal funds the loss. This prevents a successful external effect from both consuming treasury value and restoring the participant's spendable prefunding.

Table 22: Worked happy-path closeout. Operation-scoped control balances return to zero; economic assets decrease by the delivered amount.

Account	Opening / created	Closing entry	End-state meaning
Participant fiat receivable	J1A debit 100	J3C credit 100	Zero for PAY-123
Participant settlement obligation	J1A credit 100	J3C debit 100	Zero for PAY-123
Customer payment funds reserved	J1B credit 100	J3B debit 100	Zero; not still reserved
Participant funds available	J1B debit 100	None	Available prefunding reduced by 100
Stablecoin in transit	J2 debit 100 token	J3A credit 100 token	Zero for CHAIN-OP-1
Destination payout payable	J3B credit 100	J4 debit 100	Zero after proven recipient credit
Treasury inventory / settlement consumption	J2 credit / J3A debit	None	Controlled token decreases; delivered quantity is retained for valuation and audit
Payout provider cash/settlement asset	Existing debit balance	J4 credit 100	Controlled payout asset decreases by 100

Table 23: J7—record an approved unresolved custody reconciliation shortage in suspense (1.00 token unit).

Entry	Account	Dimension	Debit token	Credit token
J7.1	Suspense—external break	CASE-77 / ASSET-X	1.00	—
J7.2	Treasury stablecoin inventory	CUST-A / ASSET-X	—	1.00
		Total	1.00	1.00

J7 is not posted merely because two sources differ. It requires accounting authorization and evidence that the external controlled quantity is lower than the ledger quantity at the comparison cutoff. The open case retains both source snapshots. Recovery reverses J7; a proven loss clears suspense to the approved compensation/loss account through a separate journal.

6.3. Invariants and atomicity

Table 24. Testable ledger invariants.

ID	Invariant	Enforcement/evidence
INV-001	For every journal and unit, total debits equal total credits before commit.	Database constraint/service invariant plus property test
INV-002	A business operation ID posts at most one effective journal of a given type.	Unique constraint including legal entity and operation type
INV-003	Committed postings are immutable; corrections reference original postings.	Append-only permissions and audit reconciliation
INV-004	Available balance equals policy-eligible settled balance minus active reservations and restrictions; it is never read from an unauthoritative replica to approve spend.	Partition-owner check in the posting transaction

Table 25: External-to-ledger reconciliation map.

External source	Internal accounts	Comparison	Break owner/posture
Participant/bank statement	Receivables, payables, prefunding	Amount, currency, value date, operation/reference	Participant settlement; suspense while investigated
Custody statement	Treasury inventory, fees	Asset, address class, available/restricted quantity	Treasury/custody; route limit or pause on unexplained deficit
Independent chain data	Inventory in transit, network fees	Transaction, canonical block, logs, nonce, amount, contract	Chain operations; ambiguity/manual review
Issuer record	Inventory/redemption receivable	Supply/holdings, mint/burn, freezes, redemption status	Treasury/legal; asset kill switch
FX confirmation	FX receivable/payable	Quote/fill IDs, amounts, rate, fee, settlement	Treasury; revalue or suspense
Payout report	Payout payable/provider cash	Operation, recipient-credit status, amount, currency	Payout operations; retry/inquiry/case
General ledger/close	All control/subledger totals	Trial balance, valuation, cut-off, aged breaks	Accounting; no close while material break unresolved

ID	Invariant	Enforcement/evidence
INV-005	An account has one writer/partition version at a time; expected-version conflict cannot silently overwrite.	Pessimistic lock or serializable/optimistic version strategy proven by contention tests
INV-006	Asset quantity and reporting-currency valuation are distinct dimensions with a timestamped valuation source.	Schema constraint and valuation journal
INV-007	Each external controlled address/account maps to ledger accounts and a reconciliation owner.	Configuration registry plus coverage check
INV-008	Workflow completion cannot bypass required journal and reconciliation gates.	Transition guard queries authoritative operation evidence

Postings for one journal, their outbox record, account versions, and the ledger-operation result commit in one local ACID transaction. PostgreSQL documents transaction isolation within such a database boundary, not across external providers (PostgreSQL Global Development Group 2026). **DOC**^{CLM-INT-004} No database transaction spans the signer, chain, issuer, FX provider, payout provider, or participant. That limitation is a design input, not an implementation defect.

7. Durable payment state machine

Decision resolved. A versioned, evidence-gated state machine governs payment truth; provider statuses are observations mapped into it. **PROPOSED**^{CLM-PROP-003}

Each payment aggregate stores state name, schema/version, route version, policy/configuration versions, last transition evidence, next permitted action, timers, and optimistic aggregate version. A transition appends an immutable event and updates the current projection in one local transaction. A handler supplies the expected version; a conflict reloads and reevaluates instead of overwriting.

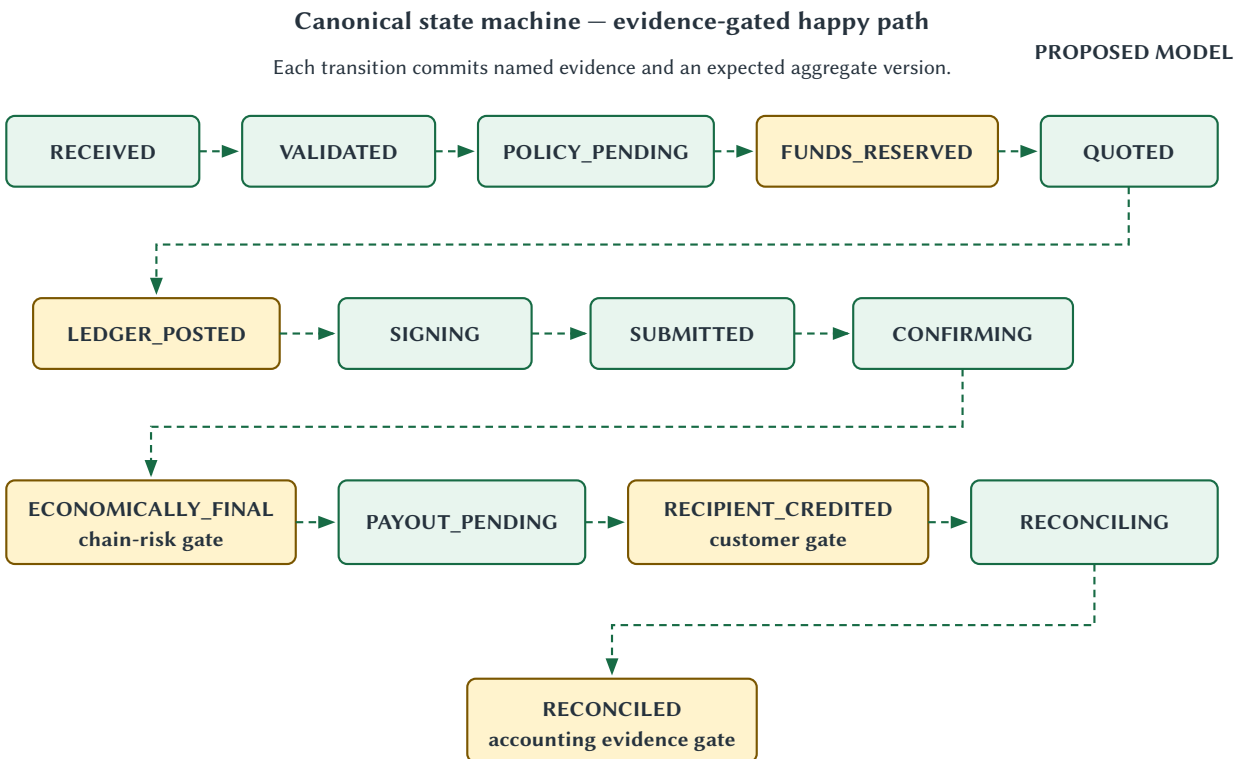
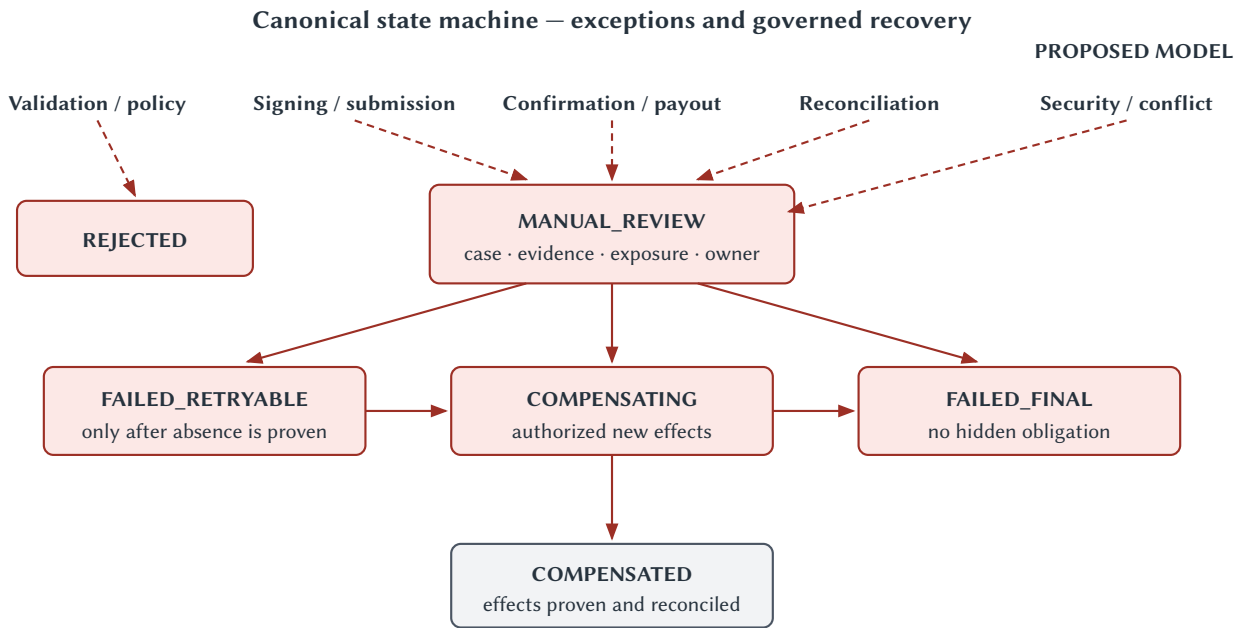


Figure 6: Happy-path state model. Routes may skip inapplicable states, but each transition requires named evidence and an expected aggregate version. *Evidence treatment*: in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.



No generic FAILED state: uncertainty remains visible until evidence supports disposition.

Figure 7: Exception state model. Ambiguity and governed manual review are stable states; retry and compensation are explicit transitions, not hidden loops. *Evidence treatment*: in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

Table 26. Canonical state dictionary. Customer wording is illustrative and route policy controls visibility.

State	Entry evidence / guard	Owner	Customer view	Timeout / termination
RECEIVED	Durable intent, canonical hash, participant authorization	Intent	Received	Short validation timer
VALIDATED	Contract, identity, route prerequisites valid	Workflow	Processing	Policy timer
POLICY_PENDING	Versioned facts submitted for decisions	Policy	Processing/under review	SLA to hold/reject/review
FUNDS_RESERVED	Ledger reservation operation committed	Ledger/workflow	Processing	Reservation expiry
QUOTED	Accepted unexpired quote and route snapshot	Quote/router	Amount/terms confirmed	Quote expiry
LEDGER_POSTED	Required authorization journal committed	Ledger/workflow	Processing	Ready for execution
SIGNING	Authorized canonical transaction and signer operation recorded	Custody/signing	Processing	Outage/refusal to retry/review

State	Entry evidence / guard	Owner	Customer view	Timeout / terminality
SUBMITTED	Submission evidence or transaction identity sufficient for inquiry	Settlement adapter	Sent/processing	Never blind retry after timeout
CONFIRMING	Independent canonical-chain observation below finality threshold	Chain observer/workflow	Processing	Route-specific delay escalation
ECONOMICALLY_FINAL	Chain-risk policy satisfied against independent evidence	Workflow/chain risk	Processing or sent	Not legal/accounting final by implication
PAYOUT_PENDING	Authorized payout operation recorded	Payout adapter	Delivering	Inquiry/repair SLA
RECIPIENT_CREDITED	Contractually sufficient recipient-credit evidence	Workflow/product	Completed (policy-dependent)	Reconciliation remains
RECONCILING	External effects complete enough for comparison	Reconciliation	Completed/processing	Break aging thresholds
RECONCILED	All required comparisons within tolerance or approved disposition	Accounting/reconciliation	Completed	Terminal for route period
MANUAL_REVIEW	Ambiguous, policy, security, or material break case opened	Case owner	Under review	Governed SLA; nonterminal
COMPENSATING	Recovery plan authorized and operations recorded	Workflow/ledger	Correcting	Each effect separately idempotent
COMPENSATED	Compensating journals/effects proven and reconciled as required	Workflow/accounting	Canceled/refunded	Terminal for failed route
FAILED_RETRYABLE	Proven transient failure with safe retry conditions	Owning adapter/workflow	Delayed	Bounded attempts/backoff
FAILED_FINAL	Permanent failure with no authorized compensation pending	Workflow/case	Failed	Terminal
REJECTED	Pre-effect policy/validation decision	Policy/intent	Not sent	Terminal

Business state and provider state are intentionally different. “Provider completed” may map to SUBMITTED, CONFIRMING, or RECIPIENT_CREDITED depending on the provider contract and independent evidence. FAILED_RETRYABLE means the platform has evidence that retrying a particular operation is safe; it does not mean a generic transport error occurred. RECIPIENT_CREDITED may permit a completed customer message, but RECONCILED remains a separate accounting/evidence gate.

Table 27: Representative transition guards. The complete transition matrix appears in table 58.

Transition	Required guard	Side effect after commit
POLICY_PENDING → FUNDS_RESERVED	All decisions allow; unique reservation operation committed; expected aggregate version	Request quote against the committed reservation
QUOTED → LEDGER_POSTED	Quote unexpired; capacity and policy rechecked; balanced authorization journal committed	Publish signing or route-execution command through the journal outbox
SIGNING → SUBMITTED	Signer verified canonical bytes; signed transaction identity recorded; submission policy valid	Idempotent provider submit command
SUBMITTED → CONFIRMING	Independent node observes expected transaction/contract/amount	Schedule finality observations
CONFIRMING → ECONOMICALLY_FINAL	Chain-specific threshold and conflict checks pass	Ledger classification and payout authorization evaluation
PAYOUT_PENDING → RECIPIENT_CREDITED	Destination evidence satisfies corridor product policy	Versioned participant notification
Any external ambiguity → MANUAL_REVIEW	Inquiry cannot prove absent/success, or evidence conflicts	Case with exposure, evidence snapshot, and action limits
Recoverable failure → COMPENSATING	Original effects known; plan authorized; loss and legal constraints evaluated	Individually idempotent compensation operations

Consequence. A late callback can add evidence but cannot move a stale aggregate past its guards. A timeout preserves uncertainty rather than manufacturing a duplicate financial effect.

8. Payment lifecycle and flow catalog

Decision resolved. All products use one intent, ledger, state, evidence, and recovery model while selecting route-specific postings and finality gates. **PROPOSED**

The common lifecycle is: authenticate participant; resolve the idempotency key; validate the canonical intent; resolve recipient/identity; obtain versioned fraud, sanctions, product, and corridor decisions; reserve funding; quote and select a route; post authorization journals; construct and simulate any chain transaction; authorize signing; submit once; observe independently; apply finality policy; initiate destination payout where required; publish monotonic participant status; reconcile every external position; and route exceptions to bounded retry, compensation, or governed case work.

Each flow below names the distinguishing funds path and completion evidence. All inherit the same participant authorization, policy, operation-ID, audit, notification, and reconciliation requirements. Detailed concern coverage appears in table 30.

8.1. Six business flows

Table 28. Business-flow catalog. “Complete” is always a product-policy outcome, not a provider status.

Flow	Initiator and funds path	Ledger and external effects	Finality / notification	Recovery focus
FLOW-01 Existing domestic	Sending participant; source account to recipient institution through approved domestic route	Reserve/post participant position; domestic rail operation; no chain service	Rail/product evidence governs customer status; accounting close after domestic reconciliation	Duplicate rail request, return/reversal constraints, participant mismatch
FLOW-02 Stablecoin funded, fiat delivered	Institution supplies approved stablecoin; recipient receives destination fiat	Custody receipt/inventory, FX/redemption or sale, payout payable/settlement	Source token economic finality plus destination credit; legal/accounting gates separate	Wrong asset/network, incoming reorg, issuer/freeze, FX and payout failure
FLOW-03 Fiat invisible stablecoin settlement	Sending participant supplies fiat; treasury token settles behind the product	Participant receivable/reservation, treasury inventory/in-transit, destination payout payable	Customer sees conventional payment; chain threshold and payout evidence are internal gates	Ambiguous submit, inventory shortfall, finalized transfer but payout failure
FLOW-04 Cross-border with FX and payout	U.S. participant funds USD; recipient receives destination currency	USD obligation, quote/fill, token transfer if selected, destination-currency payout	Quote expiry, chain/FX/payout evidence and corridor completion policy	Slippage, FX settlement mismatch, payout delay, sanctions hold
FLOW-05 Stablecoin to stablecoin	Approved institutional sender and recipient use supported assets/networks	Source custody/token obligation, transfer or issuer-mediated conversion, destination token obligation	Both source and destination evidence; no assumption that bridge message is final	Asset mismatch, contract/admin powers, cross-network ambiguity
FLOW-06 Tokenized deposit	Bank-issued deposit token interfaces with platform/settlement asset	Bank deposit liability/claim mapped separately from stablecoin inventory and participant obligation	Issuing-bank rules, network evidence, redemption/convertibility and accounting gates	Singleness/conversion, bank/network availability, legal claim mismatch

For FLOW-04 the participant request identifies source amount/currency, recipient, destination currency, corridor, and terms; it never supplies an arbitrary token contract or destination wallet. Policy produces an allow/hold/reject decision with obligations. Treasury reserves both participant capacity and route inventory. The quote record fixes rate, fees, expiry, provider, tolerance, and re-quote policy. A transaction builder derives the approved contract, destination, amount, chain ID, nonce and fee cap from the route, then simulates. The signer independently checks those fields and binds the signature to the intent/policy versions. Chain observation and finality authorize the next economic step but do not prove foreign exchange or destination

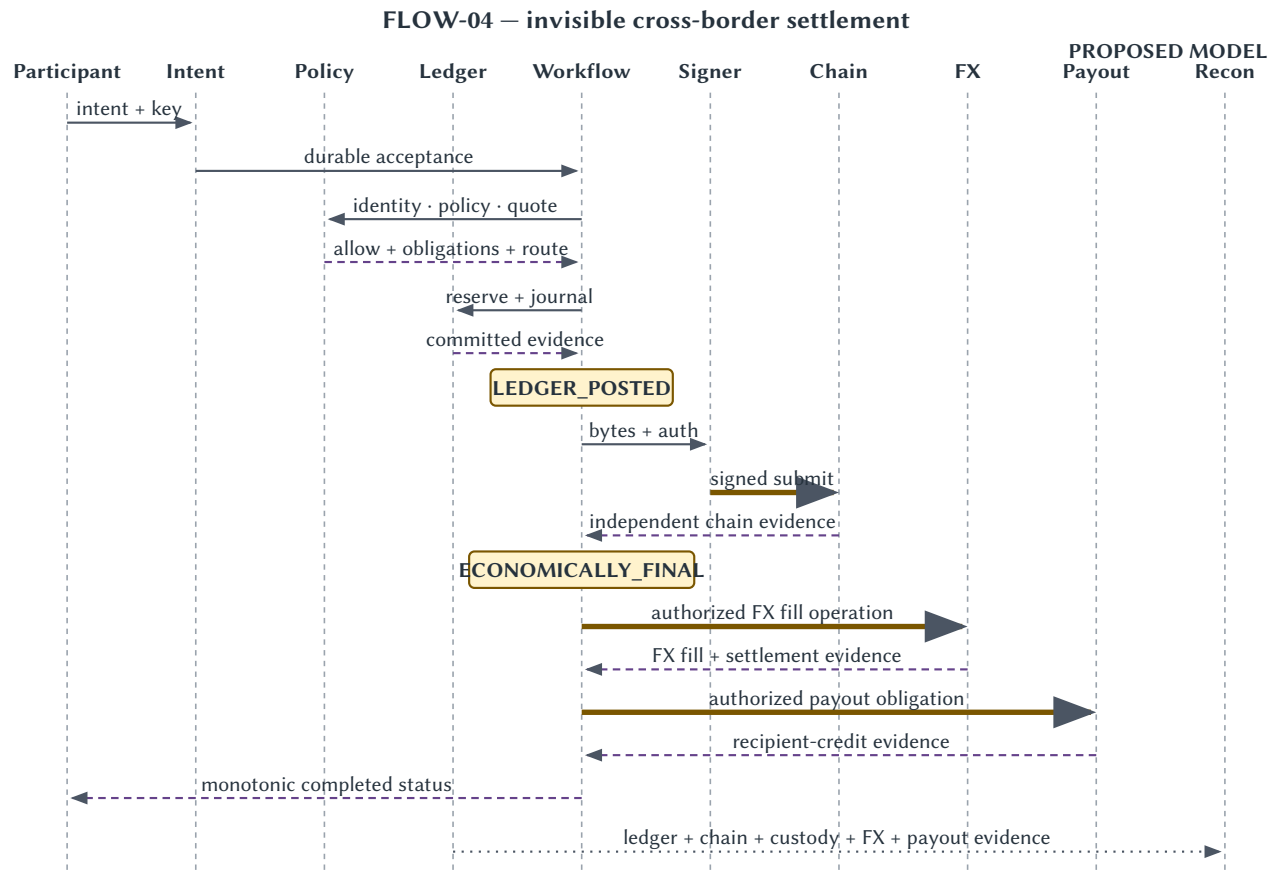


Figure 8: FLOW-04 invisible cross-border settlement. Ledger authorization precedes chain execution; payout and reconciliation remain separate after economic chain finality. *Evidence treatment*: in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

payout. A separately authorized FX operation records quote/fill identity, principal, rate, fee, expiry and settlement evidence before the payout obligation is released. The payout adapter then uses its own operation ID and source-of-truth inquiry. Recipient-credit evidence advances customer status; custody, chain, issuer, FX, payout, bank, and ledger records later reconcile.

FLOW-01 is deliberately important: it proves the common control plane is not synonymous with blockchain. Its route omits SIGNING, SUBMITTED, and CONFIRMING, maps domestic-rail evidence into the canonical model, and remains available while digital-asset routes are paused.

FLOW-02 validates incoming-asset controls. A custody notification is insufficient: the observer verifies supported network and contract, amount, destination ownership, transaction provenance required by policy, and finality threshold before inventory is recognized as available. FLOW-05 adds a strict rule against generic bridge use. An issuer-mediated burn/mint protocol such as CCTP illustrates how source finality, attestation, and destination mint are separate evidence steps; it is a USDC-specific documented mechanism, not a neutral architecture fact (Circle 2026a). [DOC](#) ^{CLM-DA-004}

FLOW-06 keeps a tokenized commercial-bank deposit distinct from a stablecoin. BIS analysis highlights different issuer claims, singleness, liquidity, and settlement implications (Bank for International Settlements 2023; Bank for International Settlements 2026). [DOC](#) ^{CLM-DA-003} Interoperability therefore needs a conversion and legal/accounting contract, not just token-interface compatibility.

8.2. Seven failure flows

Table 29. Failure-flow catalog. Inquiry precedes retry whenever an external effect may exist.

Flow	Trigger	State / ledger posture	Recovery, evidence, and audit
FLOW-07 Chain fail- ure/com- pensation	Simulation rejects; submit proven absent; transaction reverts	FAILED_RETRYABLE or COMPENSATING; reserva- tion/in-transit handled according to proven effect	Correct route defect or release/com- pensate with new operation; retain revert/absence evidence
FLOW-08 Reorg or delayed finality	Observed transaction loses canonicity or threshold stalls	Stay CONFIRMING; if previ- ously risk-accepted, case and exposure record	Observe independent nodes; do not payout unless approved risk policy allows; reorg drill and loss path
FLOW-09 Compli- ance hold	New/changed fact be- fore payout or post- payment concern	POLICY_PENDING or MANUAL_ REVIEW; funds restricted, not erased	Versioned screening evidence, blocked/rejected property procedure, legal disposition, disclosure controls
FLOW-10 Liquidity shortfall	Required token, gas, fiat, FX, or payout ca- pacity unavailable	Before external effect: FAILED_ RETRYABLE/queued/re- jected; retain reservation policy	Rebalance separately; never borrow implicitly or reroute after quote with- out reauthorization

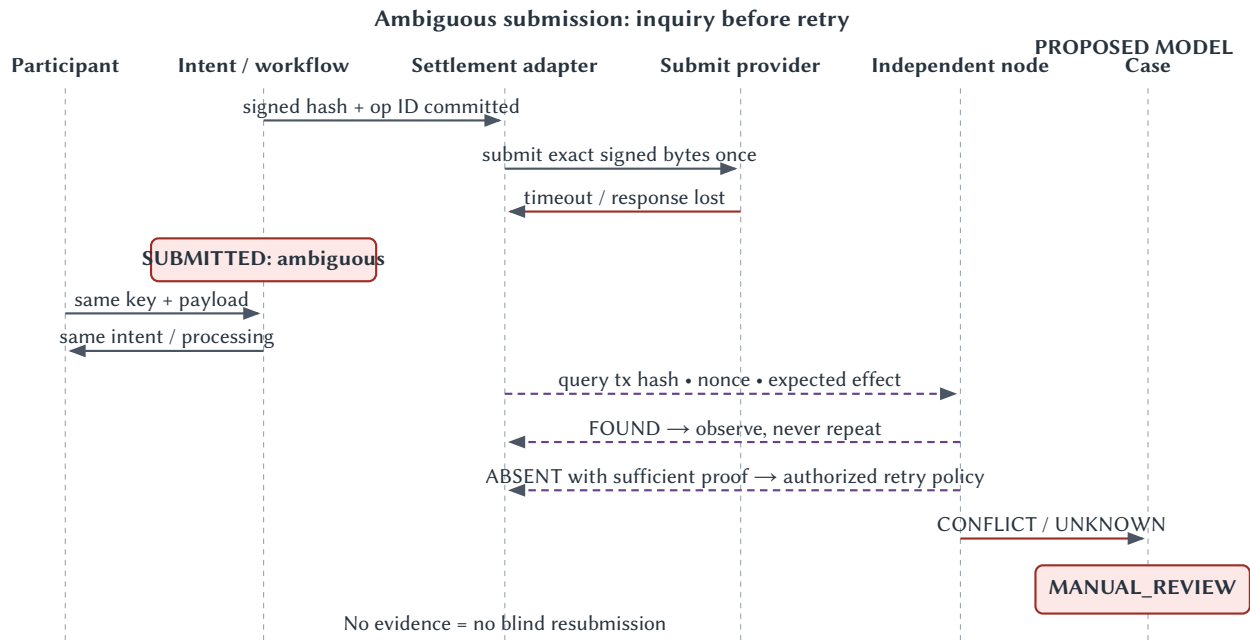


Figure 9: FLOW-07/11 ambiguous submission and duplicate retry. A timeout leads to inquiry by stable transaction identity; only proved absence permits a new attempt. *Evidence treatment*: in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

Flow	Trigger	State / ledger posture	Recovery, evidence, and audit
FLOW-11 Dupli- cate re- quest/de- livery	Same participant key, event, callback, or pay-out retry arrives	Same intent/operation result; no duplicate journal/effect	Compare canonical hash; in-box/unique constraint; return monotonic state; conflicting payload is incident
FLOW-12 Recon- ciliation mismatch	Internal and chain/customer/issuer/FX/pay-out/bank evidence differ	RECONCILING or MANUAL_REVIEW; suspense only by authorized journal	Snapshot both sources, quantify exposure, pause scope, case disposition and compensating entry if approved
FLOW-13 Sign- ing/key incident	Signer unavailable, refuses policy, or key compromise suspected	SIGNING, FAILED_RETRYABLE, or security MANUAL_REVIEW; no submission	Disable affected key/route, invoke quorum/recovery, identify signed set, rotate/migrate, independently reconcile

Ambiguous submission is the canonical distributed-systems failure. If the provider accepted a signed transaction but the response was lost, the database may show “command outstanding” while the chain later includes it. The workflow must persist the signed transaction hash and relevant nonce before submission. After a timeout it queries independent nodes, custody, and where applicable the provider by operation ID. Observation resumes if found. A new submission is allowed only if absence is sufficiently proven and route policy defines

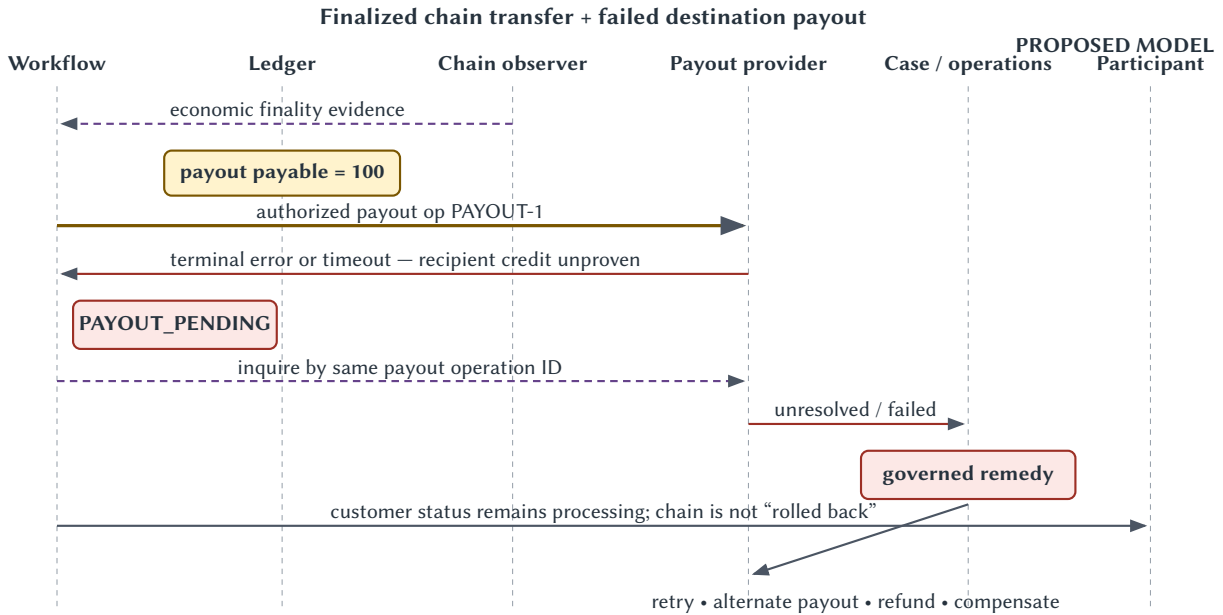


Figure 10: FLOW-04 failure branch: finalized token movement with failed destination payout. The chain effect is not rolled back; the payout obligation remains, with controlled retry, reroute, refund, or compensation. *Evidence treatment:* in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

nonce/replacement behavior; otherwise the payment remains in manual review. A participant retry returns the same intent because its idempotency record is independent of the submit response.

Finalized-chain/payout failure proves why a chain receipt cannot define payment completion. The platform still owes destination value or a contractually approved remedy. It preserves the payout payable, suspends any route that would increase unbounded exposure, inquires by payout operation ID, and follows corridor policy for safe retry, alternative payout, refund/redemption, or compensation. Customer messaging reflects what is known. An immutable case connects the chain transaction, payout evidence, ledger posture, disclosure, loss owner, and final disposition.

8.3. Flow coverage

Table 30. Flow coverage matrix. Every flow inherits intent authentication, versioned policy, audit, participant query, and reconciliation; cells name the distinguishing concern.

Flow	Funding	Ledger	External mutation	Chain gate	Payout	Duplicate control	Recovery	Audit evidence
01 Domestic	Bank or participant	Position + fee	Domestic rail	Not applicable	Recipient FI	Intent + rail op	Return or case	Rail + bank
02 Token to fiat	Incoming token	Custody + FX + payable	Custody and FX	Incoming finality	Fiat provider	Deposit + payout ops	Hold or redeem	Chain, issuer, payout
03 Fiat, token rail	Fiat + treasury	Receivable + inventory	Chain transfer	Economic	Fiat provider	Intent, chain, payout	Inquiry or compensate	Full evidence set
04 Cross-border	Fiat + quote	FX + inventory + payable	FX, chain, payout	Route-specific	Destination	Per-leg ops	Requote or case	Quote, fill, chain, credit
05 Token to token	Source token	Per-asset positions	Transfer or conversion	Source + destination	Token delivery	Per-network ops	Contain or migrate	Both chains + issuer
06 Tokenized deposit	Bank claim	Deposit + conversion	Bank network	Network + legal	Bank or recipient	Issuer ops	Convert or suspend	Bank + network
07 Chain failure	Reserved	In-transit or suspense	Submit or revert	Absence or revert	Not started	Stable tx or op ID	Retry or compensate	Inquiry + revert
08 Reorg or delay	Reserved, in transit	No premature release	Observation	Canonicity	Held or risk-accepted	Block or tx identity	Wait, case, loss	Independent nodes
09 Compliance hold	Restricted	Reservation or restriction	None or contained	Evidence retained	Held	Decision version	Legal case	Facts, decision, actions
10 Liquidity shortfall	Insufficient inventory	Reservation retained or released	Rebalance separate	Not applicable	Not started	Rebalance op	Queue or reject	Inventory + limits
11 Duplicate	Original funding	One journal	One external op	Original evidence	One payout	Key, hash, inbox	Return original	Duplicate + conflict log
12 Reconcile break	Existing	Suspense if approved	Inquiry only	Re-observe	Reconcile	Run or break ID	Case or entry	Source snapshots
13 Key incident	Frozen capacity	No new external post	Signer disabled	Observe signed set	Route paused	Sign operation ID	Rotate or migrate	Signer, key, chain audit

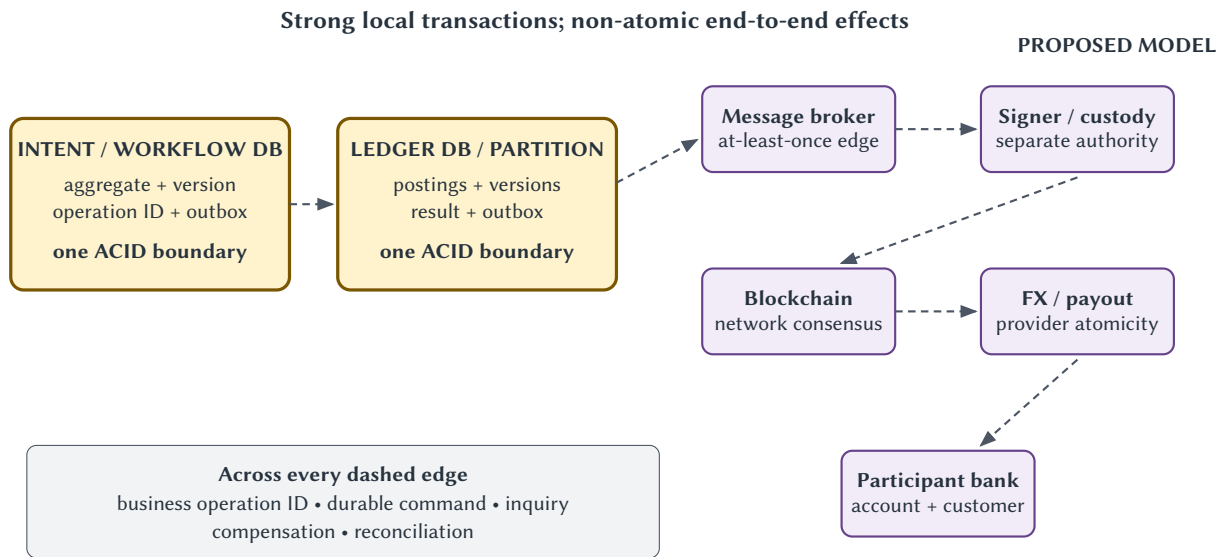


Figure 11: Transaction boundaries. The ledger/workflow database can commit business state and outbox locally; every provider edge is asynchronous and potentially ambiguous. *Evidence treatment:* in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

Consequence. Product variants change funding, settlement, payout, and policy gates. They do not create alternative sources of truth or bypass the shared recovery model.

9. Consistency: narrow atomicity, end-to-end recoverability

Decision resolved. Promise exactly-once business effects through composable controls; do not claim distributed exactly-once delivery. **PROPOSED** ^{CLM-PROP-007}

No transaction manager can atomically commit the ledger database, message broker, signer, chain, issuer, FX provider, payout provider, and participant account. The architecture makes each local boundary strong and every cross-boundary gap durable, observable, and reconcilable.

Exactly-once business effect is constructed as follows:

1. participant-scoped idempotency key and canonical payload hash produce one intent;
2. a deterministic business operation ID names each financial effect;
3. local ACID transactions commit an aggregate version, balanced journal or provider-operation record, and an outbox row together;
4. a publisher may send more than once; consumers use an inbox/unique effect constraint and expected aggregate version;
5. external requests reuse the operation ID or stable transaction identity, and adapters inquire before uncertain retry;
6. reconciliation compares independent evidence and opens a controlled break when records disagree; and

7. compensation uses new authorized, idempotent operations and journals rather than deleting history.

AWS's idempotent-API guidance emphasizes token recording and semantic equivalence; an idempotency key alone does not make arbitrary retries safe (Featonby 2026). Debezium documents the transactional-outbox pattern, whose publication remains asynchronous and whose consumers still require idempotent handling (Debezium 2026). Spring Kafka's exactly-once semantics apply to Kafka read-process-write transactions; they do not include a ledger database or external rail (Spring 2026). [DOC](#) CLM-INT-001,CLM-INT-005,CLM-JAVA-011

Table 31. Guarantee matrix. Strong guarantees are deliberately local.

Operation	Consistency scope	Duplicate control	Ambiguity resolution	Reconciliation
Intent acceptance	One intent store transaction	Participant/key unique + payload hash	Query by key/intent	Signed ingress audit
Ledger posting	One journal/account partition transaction	Ledger operation unique + balanced journal	Transaction result by operation ID	Trial balance/external sources
State transition	One aggregate transaction	Expected version + transition/effect key	Reload events/current projection	Operation/state invariant scan
Outbox publication	State + outbox local commit; broker separate	Event ID; publisher retries	Broker acknowledgment plus outbox state	Outbox age and broker offsets
Consumer effect	Consumer-owned local transaction	Inbox event ID + effect constraint	Query consumer result	Event/effect coverage
Signing	Custody boundary	Sign operation + canonical bytes hash	Provider inquiry/audit; never reconstruct a different payload	Signer versus authorized set
Chain submission	Chain/provider boundary	Stable signed tx hash, nonce, op ID	Independent node/custody inquiry	Chain/custody/ledger
Payout	Provider boundary	Payout operation ID	Provider/destination inquiry before retry	Payout report/bank/ledger
Notification	Notification store/provider	Intent + audience + status version	Participant status query authoritative	Delivery status versus state

Account mutations require a partition owner, serializable behavior, or equivalent explicit locking proven by contention tests. A quote is accepted only while its expiry, route version, available capacity, and policy remain valid in the authorization transaction. Stale read models can display progress but cannot release a reservation, choose a nonce, or authorize spend. Participant status is monotonic in product semantics even when underlying evidence regresses: a reorg after a risk-accepted customer credit becomes an incident/compensation state, not a deceptive backward UI transition.

The word *saga* is useful only if it does not imply that financial effects can always be undone. A finalized token transfer may be irreversible at the protocol layer. Recovery might create a new payout, redemption, refund, or compensation expense, each with its own legal and liquidity consequences. The design therefore names compensating operations explicitly.

10. Messaging and event contracts

Decision resolved. Messaging transports durable facts and commands; it never replaces ledger authority, state guards, or source-of-truth inquiry. [PROPOSED](#)

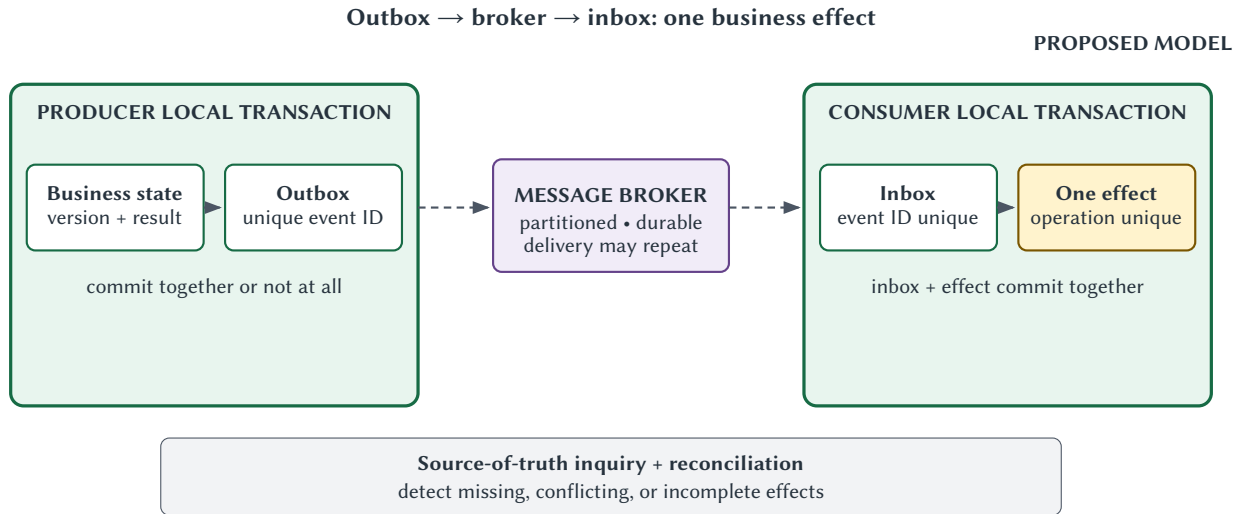


Figure 12: Outbox/broker/inbox path. At-least-once transport is converted into one local business effect through unique event and operation identities. *Evidence treatment:* in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

10.1. Envelope and ordering

CloudEvents supplies a useful interoperable envelope vocabulary but does not define business authority, ordering, or delivery guarantees (Cloud Native Computing Foundation 2026). [DOC](#) ^{CLM-INT-002} The platform envelope contains:

Payment events partition by payment-intent ID because aggregate transitions require order. Ledger commands partition by account owner or shard because balance conflicts, not payment affinity, determine safe serialization. A journal involving multiple account partitions follows the ledger’s approved coordination design. Consumers never mutate another service’s table.

10.2. Retry, replay, and poison handling

Table 33. Retry and replay policy separates transport recovery from business authority.

Condition	Automated action	Business posture	Operator evidence
Transient broker/consumer outage	Bounded exponential back-off with jitter; preserve partition order	Effect not assumed; source state queryable	Attempt/lag/age metrics
Exact duplicate	Inbox acknowledges existing effect/result	No new transition, journal, or provider call	Duplicate count and original link
Late lower aggregate version	Retain/audit, no regression; query owner if needed	Monotonic business state preserved	Late-event diagnostic
Same version, different payload	Quarantine and security/data-integrity incident	No guessed winner	Both hashes/producers
Unknown future schema	Quarantine; do not dead-letter into oblivion	Owner upgrades or explicitly rejects	Schema, producer, payload retention policy
Poison/corrupt message	Stop affected partition or isolate per policy	No unbounded retry storm	Sanitized payload, parser error, owner

Condition	Automated action	Business posture	Operator evidence
Technical dead letter	Controlled queue with owner/SLA, then source query	Not a business rejection	Retry history and disposition
Business exception	Case/event in business exception workflow	Funds/state remain under explicit posture	Decision, limits, exposure
Authorized replay	New replay job ID; original event IDs preserved	Inbox prevents duplicates; missing effects may execute	Approver, scope, before/after counts

Retention must exceed the longest approved recovery, dispute, replay, and audit window, subject to privacy and legal policy. The exact duration is **UNKNOWN** until product/corridor requirements are approved. Replay requires scoped authorization and dry-run reporting; it is not an unrestricted broker console action.

Participant webhooks carry intent ID, monotonic status version, occurred time, event ID, contract version, and a query link. They are signed with rotated participant-specific credentials, deduplicated by event ID, retried on an approved schedule, and do not expose policy reasons or sensitive chain data beyond contract need. A webhook is notification; the authenticated status API is the participant source of truth.

Consequence. Duplicate, late, reordered, incompatible, and poison messages each have deterministic handling. At-least-once transport remains visible rather than being renamed exactly once.

11. Failure model and containment

Decision resolved. Uncertainty triggers inquiry and containment; it never triggers blind repetition of a financial effect. **PROPOSED**

Failures are classified by what is known, not by transport exception class. A *safe transient* is proven to have no external effect and may retry within bounds. An *ambiguous* result may have an effect and requires inquiry. A *business retry* requires new authorization, quote, capacity, or customer facts. A *permanent* failure cannot execute under the current intent. A *poison/corrupt* input threatens deterministic processing. A *security incident* invokes containment, evidence preservation, and a separate response authority.

Table 34. Failure-domain taxonomy maps every lifecycle boundary to a class, posture, containment, and owner.

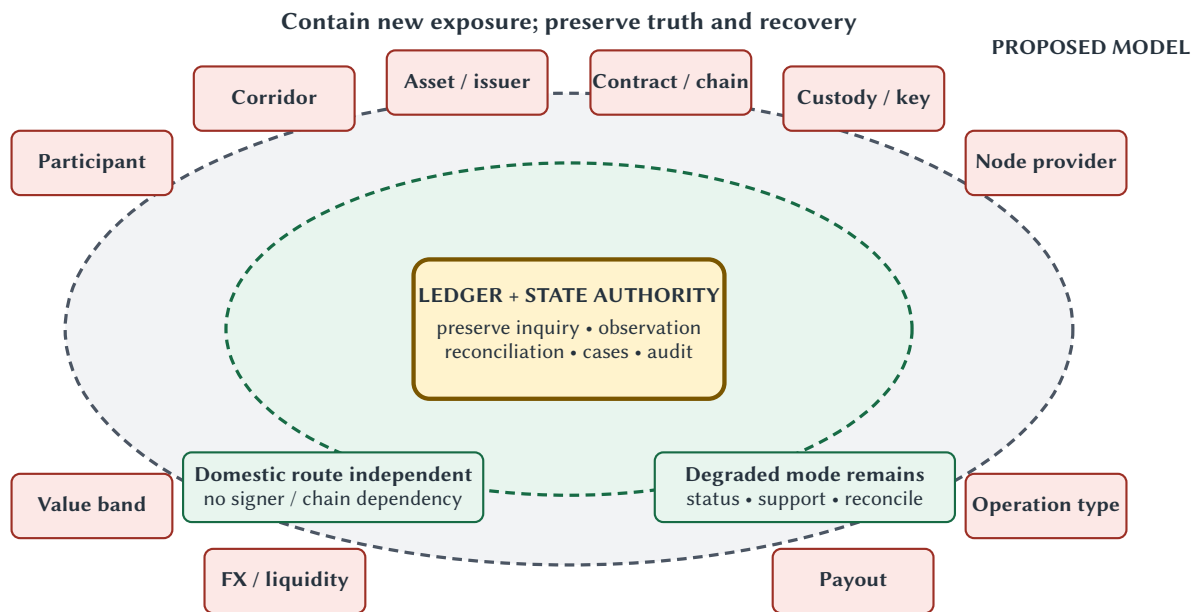
Domain	Likely class/state	Evidence and containment	Primary owner
Validation/contract	Permanent or poison; REJECTED	Deterministic schema/auth result; isolate corrupt parser input and version conflict	Participant API/intent
Policy/eligibility	Business retry, hold, or permanent; MANUAL_REVIEW/REJECTED	Preserve facts/rule versions and obligations; no downstream effect while unresolved	Risk/compliance
Funding/reservation	Safe transient or business retry; FAILED_RETRYABLE	Ledger operation result and account version; bound reservation age and exposure	Ledger/participant
Quote/route	Safe transient or business retry	Quote identity/expiry and capability snapshot; never substitute silently after acceptance	Router/treasury
Liquidity/treasury	Business retry or permanent for route	Position/reservation/limit versions; queue, rebalance, cap, or disable route	Treasury/risk

Domain	Likely class/state	Evidence and containment	Primary owner
Signing/custody	Safe transient, ambiguous, permanent, or security incident	Signature identity, policy/quorum record, custody inquiry; disable key/route on uncertainty	Security/custody
Submission	Ambiguous unless absence is proven; SUBMITTED/MANUAL_REVIEW	Stable transaction/operation identity and independent inquiry; never blind replay	Settlement adapter
Confirmation/finality delay	Safe waiting or incident; CONFIRMING	Independent block/receipt lineage and network health; stop promotion, not observation	Chain risk/observer
Reorganization/conflict	Ambiguous or security/network incident	Preserve competing lineage; contain network/value band and quantify downstream exposure	Chain risk/incident command
Issuer/contract control	Permanent or security/legal incident	Admin/code/freeze/issuer evidence; pause asset and retain redemption/migration options	Treasury/security/legal
FX and payout	Safe transient, ambiguous, business retry, or permanent; PAYOUT_PENDING/MANUAL_REVIEW	Quote/fill/payout operation and provider/destination inquiry; retain payable	Treasury/payout operations
Notification	Safe transient; business state unchanged	Monotonic status version and delivery receipt; query remains authoritative	Participant operations
Reconciliation	Ambiguous/conflict; RECONCILING/MANUAL_REVIEW	Freeze source snapshots, age/quantify break, suspense only by approved journal	Accounting/reconciliation
Region/platform	Safe transient or disaster; writes paused until fenced ownership	Fencing token, recovery point, invariants, backlog and provider exposure	SRE/ledger owner
Operator action	Security incident or governed business repair	Immutable proposal/approval/action evidence; revoke scope and reconstruct effects	Case/security/accounting

Transport handling for these classes is defined in table 33; that table is subordinate to this business classification and never authorizes an external effect by itself.

Table 32: Required message envelope.

Field	Constraint	Purpose
event ID	Globally unique, immutable	Inbox deduplication and audit
aggregate ID/version	Payment/account identity; monotonic version	Ordering and conflict detection
business operation ID	Stable across delivery retry	Exactly-once effect identity
causation/correlation IDs	Required ancestry and end-to-end trace	Reconstruct decision/effect chain
schema version/type	Registered, compatibility-governed	Safe evolution and quarantine
occurred time/recorded time	Source time plus durable-ingest time	Ordering diagnosis and latency
producer	Authenticated workload identity	Ownership and authorization
policy/configuration version	Immutable approved snapshot IDs	Decision reproducibility
evidence references	Opaque internal links, no secrets/PII payload	Trace external observation safely



Every switch is versioned, scoped, approved, auditable, time-reviewed, and exercised.

Figure 13: Failure containment and degraded modes. Kill switches are scoped by participant, corridor, asset, issuer, contract, chain, signer, provider, and operation; domestic service remains independent. *Evidence treatment*: in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

Table 35. Failure truth table. The database and blockchain success/failure quadrants are represented explicitly.

Observed condition	State and ledger posture	Inquiry/evidence	Retry rule	Customer status	Operator action/exposure
Local commit failed; external absent	Prior durable state; no journal/effect operation committed	Local transaction result, outbox absent, provider/chain absence if a call could have escaped	Retry whole deterministic command with same operation ID after absence proven	Processing/delayed	Alert if commit outcome itself uncertain; exposure zero if absence proven
Local commit failed; external succeeded (blockchain success/database failure)	MANUAL_REVIEW; create recovery journal/state only after external identity linked	Independent chain/customer/provider evidence, logs, signed hash/nonce	Never repeat external effect; repair internal record through governed recovery	Processing; do not claim failure	Contain route, quantify unrecorded asset movement, four-eyes repair
Local commit succeeded; external failed (database success/blockchain failure)	Reservation/journal and operation remain; FAILED_RETRYABLE or COMPENSATING	Provider error plus independent proof of absence/revert	Same signed transaction/operation only if safe; otherwise new authorized attempt or compensate	Delayed/failed after policy	Age reservation; release only with proven posture
Submitted; provider timeout	SUBMITTED or MANUAL_REVIEW; in-transit posture retained	Transaction hash/nonce, independent nodes, custody/provider inquiry	No blind resubmit; absence threshold and replacement policy required	Processing	Monitor ambiguity age and max exposure
Confirmed then reorganized	CONFIRMING; or incident if risk-accepted payout occurred	Multiple node/block lineage observations and custody/contract state	Observe canonical outcome; do not replay blindly	Processing; if already credited, preserve customer status and handle incident	Freeze affected route/value, quantify at-risk payout
Economically final; payout failed	PAYOUT_PENDING/MANUAL_REVIEW; payout payable remains	Payout provider plus destination institution inquiry	Retry with same payout ID only if provider contract guarantees safety; else authorized alternative	Processing/delayed; completed only if policy already absorbed risk	Bound outstanding obligations; reroute/refund/compensate per corridor
Payout complete; later compliance concern	Recipient credit/journals remain; restricted case/recovery accounts as legally directed	Original facts, designation times, screening version, legal instructions	No automatic reversal; legal/case decision	Do not misstate completed history; required communication only	Sanctions/AML incident process, evidence preservation, reporting
Conflicting external evidence / reconciliation mismatch	RECONCILING or MANUAL_REVIEW; suspense only via balanced authorized entry	Independent node/provider/issuer/custody/bank sources with timestamps	No mutation retry until authority/conflict resolved	Completed or processing per known customer gate	Pause narrow scope, assign owner, age and quantify difference

Table 36: Degraded-mode capability. A safe mode narrows new exposure while preserving inquiry, evidence, and customer support.

Trigger	Disabled	Remains available	Exit evidence
Signer/custody outage	New affected signatures	Domestic route, status, observation, reconciliation, support	Provider recovery plus signing/inquiry drill
Node disagreement/chain incident	New submissions and finality promotion on network	Independent observation, cases, unaffected routes	Canonical convergence and chain-risk approval
Issuer/contract control incident	New asset use, mint/redeem/transfer as scoped	Reconciliation, existing-customer support, alternative approved routes	Issuer evidence, exposure reconciliation, legal/security approval
FX/payout outage	New affected corridor after capacity limit	Existing payout inquiry, status, alternative corridors if independently approved	Backlog reconciled and recovery test
Reconciliation integrity loss	Exposure-increasing operations for affected source/account	Read/query, domestic/unaffected routes, incident evidence	Source integrity restored and break set resolved
Policy service outage	New intents requiring unavailable decision	Query/support; routes with explicitly approved fail-safe policy only	Current policy service and decision audit healthy
Regional control-plane outage	New writes according to ledger ownership/failover policy	Read/status where consistent; alternate region only after fenced takeover	Fencing, recovery point, invariants and owner approval

The most dangerous quadrant is external success with missing local commit. It can occur when an application reaches a provider before durably recording the operation, or when a provider-side action succeeds while the acknowledgement and local recovery state are lost. The architecture reduces this risk by persisting authorization and stable transaction identity before submission, but cannot erase it. Recovery imports external truth through a case-controlled transition and balanced journal. Operators do not insert a success flag by hand.

Kill switches are versioned configuration with named approvers, expiry/review time, reason, scope, and audit. A global “off” is necessary but insufficient. Operators need narrower controls for participant, corridor, asset, issuer, contract, network, custodian, signer key, node provider, FX provider, payout provider, value band, and operation type. Every switch is exercised in simulation and pilot drills; an untested runbook is not a control.

12. Four independent finalities

Decision resolved. Blockchain, legal settlement, customer-visible completion, and accounting finality are four independently owned policy gates. **PROPOSED** CLM-PROP-004

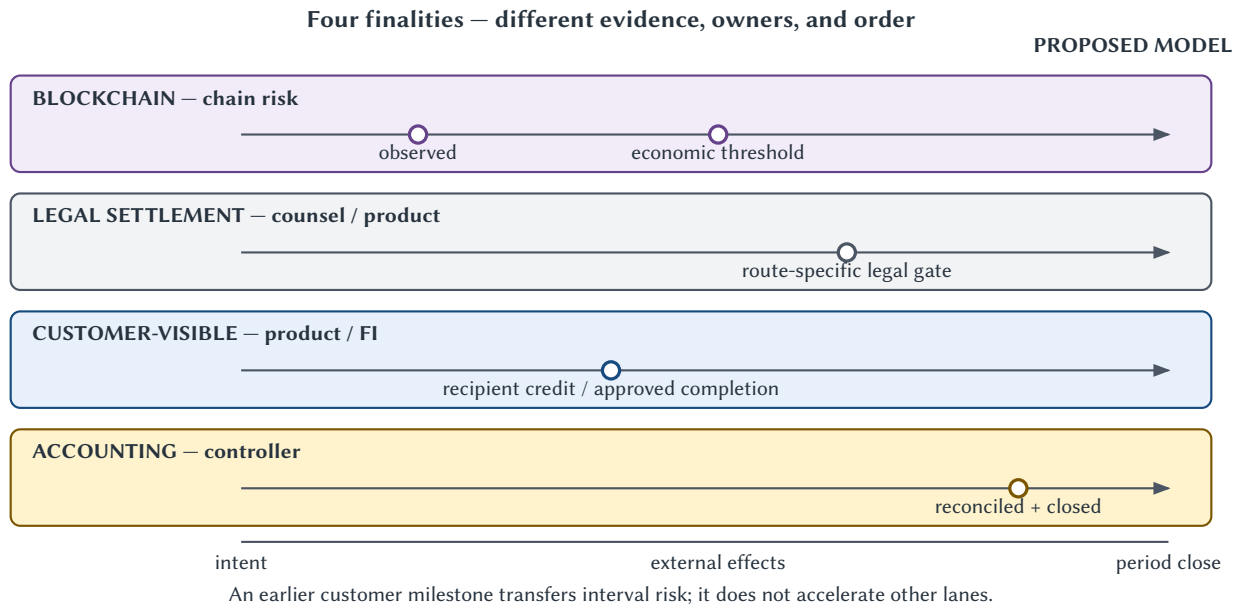


Figure 14: Four-finality timeline for a cross-border flow. The lanes can complete in different orders; a policy-approved early customer credit transfers interval risk to an accountable party. *Evidence treatment:* in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

Table 37. Finality decision matrix. Threshold values are corridor and route decisions, not universal constants.

Finality	Required evidence	Policy owner	Reversal/remediation	Customer consequence
Blockchain finality	Expected transaction/contract/log, canonical block lineage, independent node agreement, chain-specific threshold, no disqualifying contract/network incident	Chain risk / engineering	Protocol may reorganize before/under threshold; after threshold loss handled by route incident policy	Normally internal; may gate payout or status
Legal settlement finality	Product/corridor legal analysis of parties, instrument, rail, terms, insolvency, revocation and applicable law	Legal/product governance	Rights/remedies follow governing agreements and law; paper gives no opinion	Defines disclosures, rights, loss allocation and when obligation is legally discharged
Customer-visible completion	Product policy plus required source debit/reservation, recipient credit or approved risk-absorption evidence, disclosure state	Product/participant operations	Customer remedy/compensation may be owed even if chain is irreversible	Controls “completed,” “processing,” “failed,” and support response
Accounting finality	Balanced journals, external reconciliations, valuation, cutoff, approved break disposition and close	Controller/accounting	Later corrections are new dated journals; closed history not edited	Usually invisible; supports statements, financial reporting and audit

Table 38: Product/corridor finality policy template. Every selected route must complete this record before production.

Field	Required decision
Identity	Product, corridor, source/destination institutions, legal entities, asset, issuer, network, contracts, custody, FX and payout route versions
Blockchain gate	Observation sources, threshold algorithm, value bands, network-health constraints, reorg handling and escalation
Legal gate	Counsel-approved rule, governing agreements, settlement moment, revocation/insolvency treatment, unknowns and review date
Customer gate	Status terms, recipient-credit evidence, disclosure, error/dispute path, early-credit risk owner and maximum exposure
Accounting gate	Journals, statements, valuation, tolerance, cutoff, break disposition, close owner and evidence retention
Pause/recovery	Kill-switch scope, safe modes, compensation authority, notification, drills and restart approvers

Ethereum documents proof-of-stake finality as depending on validator votes and notes that finality can be delayed when a blocking share does not vote (Ethereum Foundation 2026). **DOC**^{CLM-DA-005} This is protocol evidence, not a legal conclusion. “Twelve confirmations” or any other number is only a route policy parameter supported by chain-specific analysis, transaction value, reorg history, network health, contract behavior, and loss tolerance.

Two orderings illustrate the model:

1. **Conservative payout.** Chain threshold is met; destination payout succeeds; the recipient is credited and customer status becomes completed; reconciliation then establishes accounting finality. Legal finality conditions may be evaluated across more than one of these events.
2. **Risk-accepted early credit.** A destination institution credits the recipient before the source chain reaches the platform’s economic threshold. Product policy identifies the party absorbing reorg/default risk and caps exposure. Blockchain finality follows later; accounting close follows reconciliation. The earlier customer status does not accelerate the chain.

Conversely, a chain transfer can be economically final while payout is still pending, as figure 10 shows. Customer-visible completion must then remain “processing” unless an approved product contract defines another outcome. A payout can complete before all accounting sources reconcile; the customer should not be regressed because a back-office break opens, but the break remains an operational and accounting obligation.

Consequence. `ECONOMICALLY_FINAL`, `RECIPIENT_CREDITED`, and `RECONCILED` are distinct because they answer different questions. Legal finality remains an explicit corridor-specific unknown until qualified counsel and governing contracts decide it. **UNKNOWN**^{CLM-UNK-003}

13. Security architecture

Decision resolved. Make authorization of canonical transaction bytes independent of application intent, and make key/custody authority a segregated trust boundary. **PROPOSED**

Threat actors include an external attacker, malicious or compromised participant, compromised application workload, provider insider, operator with excessive privilege, software-supply-chain attacker, chain/contract attacker, and colluding approvers. Protected assets include participant credentials, personal and screening data, ledger authority, route/configuration policy, treasury inventory, signing keys, contract administration, audit evidence, and recovery mechanisms.

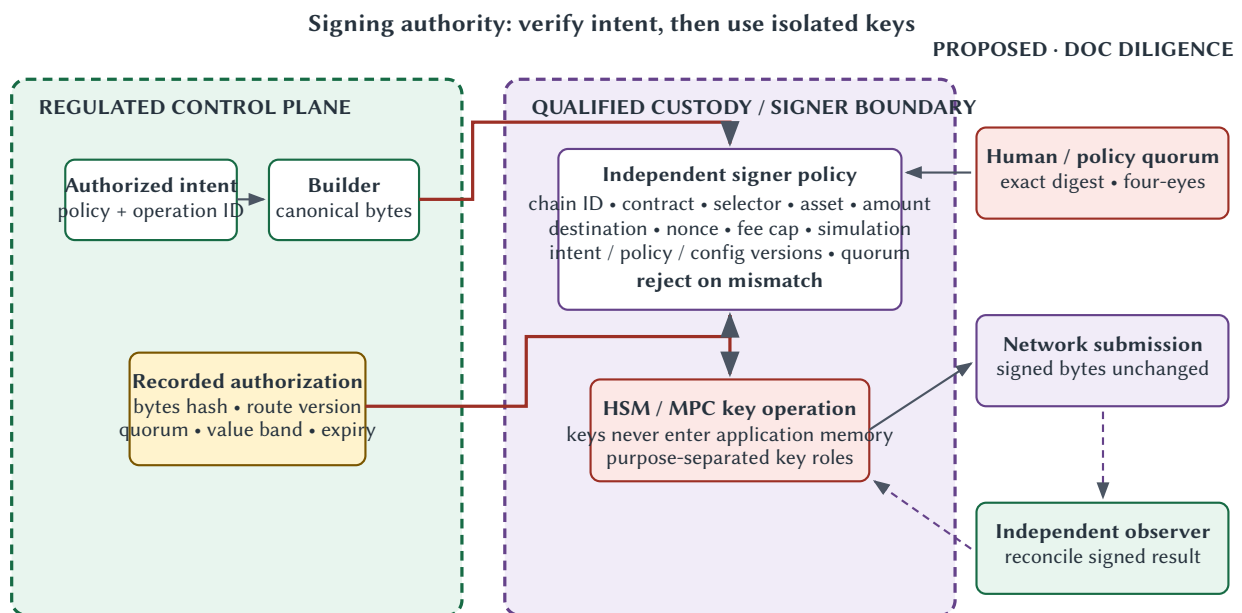


Figure 15: Key and custody authority. The application proposes canonical bytes; an independent policy boundary verifies intent, chain, contract, selector, amount, destination, nonce, fee, and approvals before any key operation. *Evidence treatment:* in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

13.1. Key roles and signer policy

Keys are separated by production/test environment, legal entity, network, asset, treasury/settlement purpose, fee payment, contract administration, and recovery. Treasury inventory does not share a key with contract administration; recovery authority does not perform daily settlement. Generation and use occur in an approved hardware security module (HSM), MPC system, or qualified custody boundary. Private key material never enters application memory, logs, CI/CD, tickets, backups, or operator workstations.

NIST FIPS 140-3 defines cryptographic-module security requirements, and NIST SP 800-57 provides lifecycle guidance (National Institute of Standards and Technology 2019; National Institute of Standards and Technology 2020). **DOC** ^{CLM-SEC-001,CLM-SEC-002} Module validation is necessary evidence for a selected control posture but does not define custody governance, quorum, recovery, or transaction authorization.

As one provider-diligence example, Fireblocks documents role-based access, transaction policies, administrative quorum, MPC key shares, co-signers, and a recovery process (Fireblocks 2026b; Fireblocks 2026a). **DOC** ^{CLM-CUST-001,CLM-CUST-002} These are documented product behaviors, not a provider recommendation or proof of independent assurance. A selected custodian must still prove contractual asset segregation, exact authorization semantics, key/share ownership, recovery and export, incident evidence, financial condition, jurisdiction, and tested exit.

For every signing operation the signer independently verifies:

- authorized business operation and immutable intent/policy/configuration versions;
- exact canonical unsigned bytes and their cryptographic hash;
- network/chain ID, transaction type, nonce, validity window, and replacement policy;
- allowlisted contract and code/version identity where available;
- function selector/instruction, asset, amount, decimals, destination, and memo/opaque intent reference;
- fee/gas ceiling and fee-payer authority;
- simulation result and prohibited state changes;
- value-band quorum, segregation of duties, rate/velocity limits, and kill-switch state; and
- an append-only authorization result usable for independent reconciliation.

The signer rejects caller-supplied fields that do not reproduce the canonical bytes. For higher value bands a second human or system authority approves the exact transaction digest, not a screenshot or mutable form. Recovery keys require a distinct quorum and time delay where supported; their use automatically pauses affected routes and triggers a full signed-transaction and balance reconciliation.

13.2. Contract, node, application, and data controls

ERC-20 standardizes token interface behavior but says nothing about reserves, redemption, custody, or legal rights (Vogelsteller and Buterin 2015). Upgrade patterns such as EIP-1967 make implementation/admin slots discoverable, while libraries such as OpenZeppelin document role and delayed-administration patterns (Worthington et al. 2019; OpenZeppelin 2026). **DOC** ^{CLM-DA-006,CLM-DA-007,CLM-SEC-004} The platform inventories contract code hashes, proxy implementations, administrators, pause/freeze/denylist roles, upgrade events, decimals, and approved function selectors. A contract or administrator change pauses promotion until reviewed; a library name never substitutes for governance analysis.

Managed nodes use mutually authenticated or strongly authenticated channels, per-workload credentials, egress restrictions, rate limits, and response integrity monitoring. Independent observation uses a different failure and administrative domain. Nodes and custody callbacks enter a validation queue; they do not reach ledger credentials. Application services use least-privilege workload identities, service-owned schemas, encrypted transport and storage, secret rotation, dependency pinning, signed artifacts, software bills of materials, provenance, and policy-as-code review. NIST's Secure Software Development Framework supplies a baseline, not a system-specific threat model (National Institute of Standards and Technology 2022). **DOC** ^{CLM-SEC-003}

On-chain data is limited to what the approved transaction and protocol require. Personal identifiers, sanctions results, participant names, case reasons, and reusable customer aliases remain off-chain. An opaque reference is non-guessable, non-personal, scoped, and not a direct database key. Logs redact secrets and sensitive payloads; access to chain analytics and case data is purpose-bound and audited. Retention/deletion policy accounts for the impossibility of deleting public-chain data.

Table 39. Representative threat/control/evidence matrix.

Threat or abuse	Prevent/detect controls	Recovery/containment	Evidence
Participant replay or payload swap	Mutual auth, signed request, key+hash idempotency, quotas	Return original or reject conflict; credential containment	Ingress signature, canonical hash, idempotency record
Application requests wrong transfer	Route-derived bytes, simulation, signer independent field checks, allowlists	Sign refusal; disable work-load/route	Intent, policy, bytes hash, signer decision
Key compromise	HSM/MPC/custody isolation, quorum, purpose/value limits, anomaly monitoring	Pause key/asset/network; rotate/migrate; reconcile signed set	Key audit, all signed hashes, chain/custody state
Contract upgrade/admin abuse	Contract inventory, admin/upgrade event monitoring, code-hash policy, delay where available	Pause asset/contract; value containment; migration plan	Pre/post code/admin state and review
Node lies or is stale	Independent providers/nodes, block/receipt lineage, lag/divergence metrics	Stop finality promotion/submission as scoped	Signed/recorded observations and comparison
Operator bypass	RBAC/ABAC, just-in-time elevation, four-eyes actions, immutable case/audit	Revoke access, contain affected route, reconstruct actions	Identity, approvals, before/after state
Supply-chain compromise	Pinned dependencies, provenance, signed builds/SBOM, isolated CI, secret scanning	Roll back signed artifact, rotate credentials, incident response	Build provenance, manifests, deployment identity
Denial of service/fraud burst	Participant/corridor/value rate limits, queues, capacity reservations, backpressure	Degraded mode, priority inquiry/settlement queues	Rate/queue/exposure telemetry

Table 40. Privileged-operation approval matrix: authority and limits.

Action	Initiator	Authorization and quorum	Limits and automatic alerts
Settlement signing	Workflow service identity	Independent signer policy; value-band system/human quorum over exact digest	Asset, destination, function, value, velocity, fee, validity window; mismatch/refusal alert
Contract administration	Segregated contract-admin role	Named multisignature/quorum plus delayed change where supported; never treasury key	Approved code hash/function and maintenance window; any admin event pauses promotion
Key/share recovery	Offline recovery custodians	Distinct recovery quorum, time delay, dual control, incident authorization	No routine settlement permission; any invocation pauses key scope and pages security
Route or asset kill switch	On-call security/risk role	Pre-authorized narrow stop; dual approval to re-enable	Scope, reason, expiry, affected exposure; alert participants/operations

Action	Initiator	Authorization and quorum	Limits and automatic alerts
Compensating journal or state repair	Case owner proposes	Accounting plus product/legal/risk approval as required; ledger/workflow service executes	Case/value/corridor limits; invariant and customer-remedy alert
Payout reroute or beneficiary change	Payout case owner	Re-screened beneficiary and corridor-specific dual approval	Same economic obligation; no silent destination change; high-risk route alert
Configuration/policy release	Authorized proposer	Segregated reviewer/approver/deployer with signed version	Cohort, effective window, rollback, policy-diff and stale-version alerts
Bulk evidence/data export	Audit/compliance purpose owner	Purpose-bound role plus data-owner approval and logged destination	Class, period, subject count, region, expiry; anomalous export alert

Table 41. Privileged-operation audit and recovery requirements.

Action	Mandatory audit evidence	Recovery or containment
Settlement signing	Intent/policy/config versions, canonical bytes/hash, approvals, signature identity, signed result	Disable workload/key/route; reconcile every signed hash and controlled balance
Contract administration	Proposal, code/admin before/after state, approvers, delay, execution transaction	Pause asset/contract, revoke role where possible, migrate only through approved plan
Key/share recovery	Incident, custodians, recovered scope, ceremony transcript, new key/address mapping	Rotate/migrate, revoke old authority, reconcile historical and new controlled addresses
Route or asset kill switch	Actor, scope, reason, time, exposure snapshot, notifications, re-enable approval	Preserve inquiry/reconciliation; resume only after evidence-backed disposition
Compensating journal or state repair	Original effects, plan, legal/accounting rationale, journal IDs, state versions, customer action	Reverse only by new journals/events; retain loss/recovery and reconciliation case
Payout reroute or beneficiary change	Original/new destination tokens, screening, approvals, provider operation IDs	Stop duplicate path, inquire both providers, retain payable until one outcome is proven
Configuration/policy release	Signed artifact, diff, tests, approvals, deployment cohort, observed metrics	Roll back to signed prior version or disable route; preserve both decision versions
Bulk evidence/data export	Purpose, query, classes, region, recipient, approval, checksum, deletion/expiry proof	Revoke link/credential, contain destination, privacy/security incident process

Security drills cover signer unavailable, key loss, suspected key compromise, malicious transaction construction, node inconsistency, contract upgrade, custody exit, and restoration from independently verified balances. Vendor exit must identify transaction/authorization export, address/key migration, record retention, recovery cooperation, and service continuity before onboarding.

Table 42: Signing technology and key-temperature comparisons.

Option	Strength for this design	Limitation and operating burden	Pilot posture
Hardware security module (HSM)	Mature hardware boundary, standardized module assurance, direct key control	High-availability clusters, backup/ceremony, firmware, integration and specialist staffing remain operator duties	Accept only inside approved custody/provider or proven institutional service
Multi-party computation (MPC)	Distributes signing shares and can express organizational quorum without reconstructing one key	Protocol/provider implementation, share governance, recovery/export, latency and assurance require independent diligence	Accept conditionally through qualified provider; do not treat “MPC” as assurance by itself
Hot operational key	Lowest signing latency for bounded routine settlement	Highest online exposure; requires narrow purpose, value/velocity limits, rapid disable and continuous monitoring	Only for tightly capped operational inventory
Warm approval key	Additional human/system quorum and delay reduce autonomous compromise impact	Higher latency and on-call/approval dependency can block time-sensitive settlement	Higher value bands, administrative actions and controlled replenishment
Offline/recovery key	Strong separation from routine workloads	Ceremony, availability, geographic custody, obsolescence and untested recovery risk	Recovery only; periodic witnessed exercise and automatic route pause on use

14. Compliance as versioned decision evidence

Decision resolved. Compliance is an overlay across intent, funding, execution, payout, monitoring, and case handling—not a one-time screening API. **PROPOSED**

The architecture records a decision as facts considered, rules/model versions, timestamp, accountable service/role, outcome, reason category, obligations, expiry/re-screen trigger, and evidence references. It avoids putting sensitive source data in broad event payloads. Policy can allow, reject, hold, require enhanced review, cap value, constrain route, require information exchange, or mandate post-payment monitoring. A decision expiring before signing or payout is re-evaluated; the new result never rewrites the prior decision.

The GENIUS Act became U.S. Public Law 119-27 on 2025-07-18 and contains a payment-stablecoin issuer framework including permitted-issuer, reserve, redemption, disclosure, and Bank Secrecy Act provisions (United States Congress 2025). **PUBLIC** ^{CLM-REG-001,CLM-REG-002} FinCEN’s analysis of convertible virtual-currency business models depends on the facts and roles involved (Financial Crimes Enforcement Network 2019). OFAC states that sanctions obligations apply to virtual-currency transactions as they do to fiat transactions (Office of Foreign Assets Control 2021). FATF guidance addresses risk-based controls and information sharing for virtual-asset transfers (Financial Action Task Force 2025). Regulation E establishes U.S. consumer electronic-fund-transfer protections for covered transfers (Consumer Financial Protection Bureau 2026). **PUBLIC** ^{CLM-REG-003,CLM-REG-004,CLM-REG-005,CLM-REG-006} These sources motivate controls; none is a legal conclusion for a particular structure.

Table 43. Compliance responsibilities remain open until legal entities, roles, products, and corridors are selected.

Obligation	Lifecycle gates	Architecture evidence	Required owner decision	Status
Participant/customer identity	Enrollment, intent, recipient resolution	Identity assertions, assurance, participant source, expiry	Which party performs/relies on which checks	UNKNOWN / Phase 0

Obligation	Lifecycle gates	Architecture evidence	Required owner decision	Status
Sanctions screening	Intent, address/contract, pre-sign, pre-payout, designation change	List/version, facts, decision, hold/case, transaction linkage	Screening scope, false-positive, blocked/rejected handling	UN-KNOWN / Phase 0
Fraud and scams	Enrollment, intent, velocity, recipient, post-event	Signals, warnings, participant/customer actions, model/rule version	Allocation, intervention, reimbursement, escalation	Existing context + new route design
AML monitoring/reporting	Funding, transaction, payout, linked behavior	Source/destination, asset/network, amounts, counterparties, alerts/cases	Regulated roles, filing/reporting and retention	UN-KNOWN / Phase 0
Travel Rule/information exchange	Before qualifying transfer/payout	Counterparty classification, required fields, secure exchange/receipt	Jurisdiction, threshold, role and data minimization	UN-KNOWN / Phase 0
Consumer protection	Terms, authorization, status, error/dispute, remedy	Consent, disclosures, status history, investigation and disposition	Product coverage, error rules, liability and timing	UN-KNOWN / Phase 0
Privacy/data localization	Collection, processing, sharing, retention/deletion	Purpose, classification, region, access, provider and deletion record	Applicable law and cross-border transfer mechanism	UN-KNOWN / Phase 0
Blocked/restricted property	Hold, post-designation, case disposition	Restricted ledger posture, legal order, access controls, reports	Custodian/issuer/operator ownership and permissible action	UN-KNOWN / Phase 0

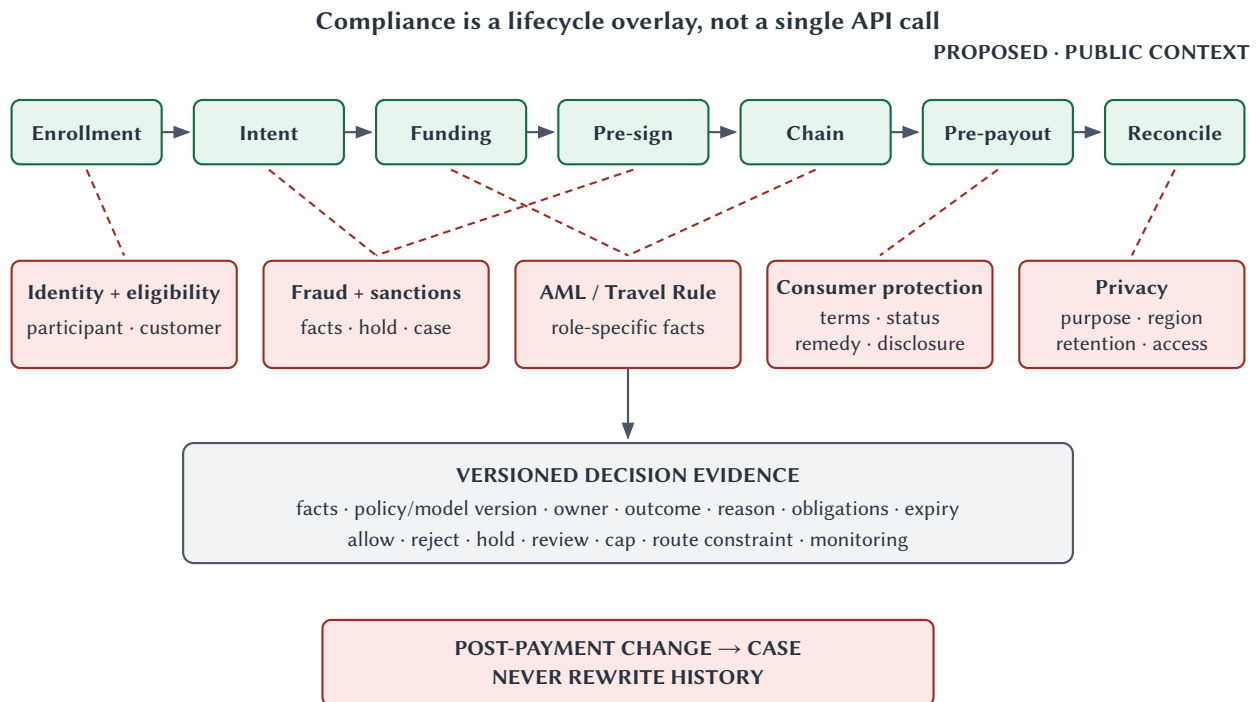


Figure 16: Compliance decision overlay. Identity, sanctions, fraud, transaction monitoring, Travel Rule analysis, consumer protection, privacy, and case obligations are evaluated at different lifecycle gates. *Evidence treatment*: in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

Table 44. Data-classification, residency, and public-chain exposure matrix.

Data class	Region/residency decision	Cross-boundary transfer	Retention and access	On-chain exposure	Owner
Participant/customer identity and aliases	Participant and corridor law; approved processing region	Tokenized/minimized fields only under approved role and mechanism	Product/legal schedule; participant support, identity, compliance roles	Prohibited; no names, email, phone, account, or reusable alias	Product/privacy/participant
Beneficiary and payout instructions	Destination corridor and provider location are Phase 0 gates	Purpose-bound payout payload over approved channel	Through remedy/reconciliation plus legal schedule; tightly scoped payout/case access	Prohibited; destination chain address only when the approved rail requires it	Payout/privacy/legal
Screening, fraud, and case evidence	Region follows regulated-role and provider decisions	Minimum facts under information-sharing and vendor terms	Regulatory/case schedule; need-to-know immutable access audit	Prohibited; no score, match, hold, or case reason	Compliance/fraud
Payment intent and participant status	Operator region plus participant contract	Stable opaque intent/status contract; no unnecessary sensitive facts	Product, dispute, accounting, and audit schedule	Opaque non-personal reference only if required and approved	Product/operations
Ledger journals and valuations	Legal-entity accounting location and approved recovery region	Controlled finance/participant reporting and disaster-recovery copies	Financial-record schedule; append-only finance/audit access	Never as the sole ledger; chain transaction identifier may be referenced	Accounting/treasury
Canonical transaction and signature evidence	Custody/network jurisdictions and operator evidence store	Signer/custody/node interfaces; immutable hash/evidence export	Key/transaction/audit schedule; signer, security, reconciliation access	Protocol-required transaction fields are public and effectively permanent	Security/custody
Chain observations and analytics	Public data plus provider processing locations	Node/analytics providers receive addresses and transaction context	Route/reconciliation schedule; restrict enrichment and model output	Intrinsically public for public-chain routes; never enrich on-chain	Chain risk/privacy
Telemetry and logs	Service region and approved observability platform	Redacted structured signals only; no secrets or raw sensitive payloads	Short operational schedule unless promoted to audit evidence	None	Platform/SRE/privacy
Backups, exports, and recovery artifacts	Same or explicitly approved recovery jurisdiction	Encrypted, inventoried, purpose-bound transfer	Tested expiry/deletion; break-glass access and export audit	None beyond already-public transaction data	Data owner/security

Table 45: Customer-obligation comparison: visible token product versus invisible settlement rail.

Obligation	Customer-visible token product	Invisible settlement rail	Phase 0 owner
Instrument disclosure	Explain issuer, token rights, custody/wallet, network, redemption, fees and admin powers	Explain cross-border payment, exchange/payout terms, status, fees and remedies without implying customer token ownership	Product/legal
Authorization and security	Wallet/key or custodial authorization, address risks, token-transfer consequences	Existing bank authentication plus payment intent; no seed phrase, gas, or direct address handling	Participant/product/security
Finality and status	Distinguish chain confirmation, legal rights, redemption and possible freeze/reorg conditions	Customer completion follows corridor policy and recipient-credit evidence, not chain status alone	Product/legal/risk
Error, dispute, and remedy	Define token/custody error paths and limits on reversing public-chain effects	Preserve bank support path and applicable transfer investigation/remedy while external settlement may be irrevocable	Participant/support/legal
Privacy	Disclose public address/transaction visibility and analytics	Keep personal data off-chain; disclose provider/cross-border processing appropriate to the product	Privacy/product
Exit and recovery	Customer redemption/transfer, wallet or custody migration, lost-access treatment	Operator route disable, payout remedy, reconciliation and provider migration remain behind stable participant contract	Treasury/operations/legal

Address analytics is a risk signal, not identity or legal truth. The system records provider/model/version, input addresses and transaction context, result, timestamp, threshold, and human disposition. A post-payment concern cannot erase a completed journal or promise a reversal of finalized value. It invokes the approved legal/case process, restricts assets or routes where authorized, preserves evidence, and records any compensation or reporting separately.

Consequence. Legal, sanctions, consumer, privacy, and role decisions remain visible Phase 0 gates. Architecture makes those decisions executable and auditable; it does not make them on behalf of counsel or regulators.

15. Java architecture and language boundaries

Decision resolved. Use Java/Spring for the enterprise control plane; choose chain-native, custody, cryptographic, and node implementations by ecosystem and operating fit. **PROPOSED** ^{CLM-PROP-006}

Java is appropriate where the dominant work is authenticated APIs, domain invariants, orchestration, relational transactions, policy integration, event handling, reconciliation, observability, and long-lived enterprise operations. It is not a reason to implement private-key cryptography, an execution or consensus client, or smart contracts in Java when the selected ecosystem has a stronger native implementation and security toolchain.

15.1. Module and dependency model

A practical starting point is a modular control-plane deployment, split only where ownership, availability, security, scaling, or data authority differs. Domain modules define payment intent, money/asset quantities, journal commands, state transitions, route capability, evidence, and operation IDs without depending on Spring or a chain SDK. Application

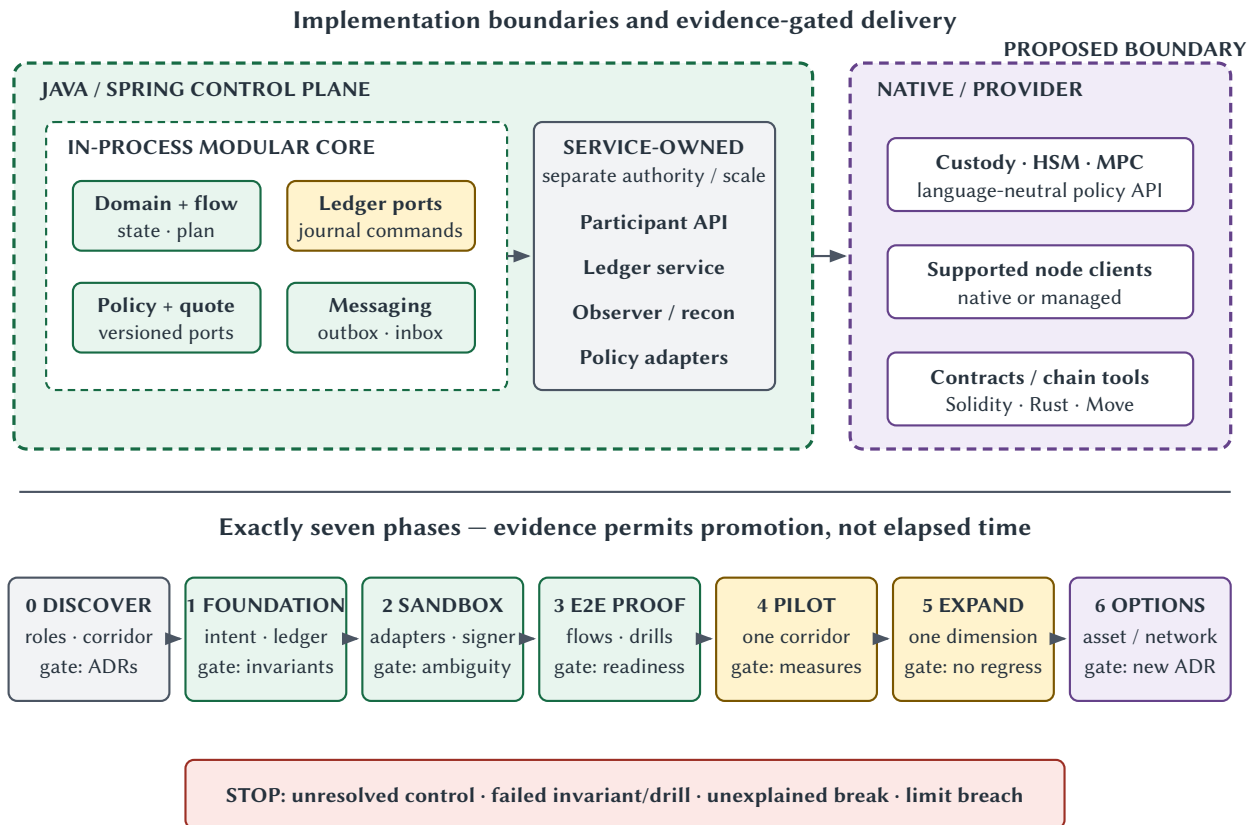


Figure 17: Java/native boundary and phased delivery. The upper panel assigns enterprise control to Java and chain-native execution to supported ecosystems; the lower panel makes promotion evidence-gated across exactly seven phases. *Evidence treatment*: in-figure labels identify public, documented, inferred, proposed, assumed, or unknown elements as applicable; proposed elements are reference design, not deployed Early Warning Services topology.

modules implement command handlers and ports. Adapters implement HTTP/event/persistence/provider contracts. Deployment modules wire configuration, security, telemetry, and runtime concerns.

Ledger service, signer/custody boundary, and chain observer deserve separate security and availability ownership even if other modules initially share a deployable unit. A service split must not create distributed transactions between objects that require one invariant. Conversely, a modular monolith must not share tables or in-process bypasses that obscure future ownership.

Table 46. Language and build/buy responsibility matrix.

Capability	Java role	Non-Java / provider role	Pilot posture	Rationale
Participant API and intent	Spring HTTP/security, validation, idempotency, domain handlers	Participant client in its own stack	Build	Enterprise contract and business authority
Workflow/state	Durable Java handlers and timers through an engine abstraction	Managed/self-hosted workflow engine may use another runtime	Build domain; evaluate engine	State semantics are ours; scheduling can be bought
Ledger-facing services	Java domain/API, SQL transactions, outbox	Relational database engine	Build authority on proven database	Strong local transactions and audit
Policy integration	Java orchestration, versioned decision contract	Fraud/sanctions/identity vendors and participant systems	Integrate	Roles and models are external, evidence contract is ours
Messaging	Java producers/consumers, schema and inbox/outbox	Broker/schema registry	Integrate	Mature ecosystem; delivery is not business authority
Transaction construction	Java adapter/SDK where complete and testable	Chain-native simulation/tooling	Conditional build	Keep business operation binding in control plane
Smart contracts	Interface, allowlist, events, risk metadata	Solidity for EVM; Rust/Move/other selected chain toolchain	Reuse audited standard contracts; minimal custom code	Native audit/testing ecosystem and bytecode fidelity
Node clients	JSON-RPC/gRPC clients and observation normalization	Go, Rust, Java, C++ or vendor-supported client per network	Managed + independent read path	Client language is subordinate to protocol fit and operations
Signing/custody	Language-neutral authorized request, canonical hash and evidence validation	Qualified custody/HSM/MPC implementation	Buy for pilot	Keys outside application memory and specialist controls

Capability	Java role	Non-Java / provider role	Pilot posture	Rationale
Reconciliation/operations	Java comparison/case services, SQL/analytics jobs, telemetry	Provider statements/data exports	Build core; integrate sources	Obligation ownership cannot be outsourced

Web3j can construct EVM transactions, bind contracts, call JSON-RPC, decode events, and integrate with HSM interfaces (LF Decentralized Trust Web3j Project 2026b; LF Decentralized Trust Web3j Project 2026a). **DOC CODE** ^{CLM-JAVA-003,CLM-JAVA-004} It is appropriate inside a bounded EVM adapter when its version supports the selected network and transaction types and conformance tests compare encoded bytes with native tooling. It is not a node, signer policy, operational ledger, or end-to-end reliability mechanism.

Java virtual threads can support high-throughput thread-per-request code with compatible blocking libraries (Pressler and Bateman 2023). **DOC** ^{CLM-JAVA-012} They do not remove provider latency, database contention, connection-pool limits, or backpressure. Select conventional blocking plus virtual threads, reactive APIs, or a mix only after measuring workload and library behavior. Durable workflow, not an open thread, represents a payment waiting on finality or a case.

Testing layers include pure domain/property tests for balance and transitions; database integration and concurrency tests; contract/schema compatibility tests; golden transaction encoding and simulation vectors; provider sandbox/record-replay tests; fault injection for ambiguous effects; reconciliation fixtures; and end-to-end drills with duplicate, late, reorged, and conflicting evidence. OpenTelemetry provides standard traces, metrics, and logs (OpenTelemetry 2026); business audit remains a separate immutable record. **DOC** ^{CLM-INT-003}

16. Technology evaluation

Decision resolved. Select technologies after ownership, invariant, failure, regulatory, and exit obligations; a familiar implementation language is not a selection criterion. **PROPOSED**

All evaluations are as of 2026-07-13. “Performance” below identifies the scaling axis and required measurement; it does not repeat incomparable vendor benchmarks. Fatal constraints—custody/legal approval, data authority, recovery, licensing, or missing required protocol behavior—cannot be averaged away.

16.1. Enterprise control and data plane

Table 47. Evaluation summary for control/data-plane classes. Each status follows the common evaluation framework.

What / obligation	Why use	Why not / operational risk	Alternatives / change trigger	Performance and maintenance	Recommendation
Java 25 + Spring class; API/orchestration/integration	Strong typing, mature security/data/messaging/telemetry ecosystem, supportability	Framework/version complexity; blocking mistakes; not chain-native	Kotlin/Go/.NET; change if platform ownership or measured fit dominates	Measure request, event and DB contention; patch JDK/framework, contracts and dependencies	recommended

What / obligation	Why use	Why not / operational risk	Alternatives / change trigger	Performance and maintenance	Recommendation
Relational database class; ledger/state local authority	ACID, constraints, mature backup/recovery and SQL reconciliation	Horizontal write scaling and cross-region single-writer complexity	Existing approved relational platform; specialized ledger only if invariants/export/operations proven	Benchmark account contention, journals, recovery; own schema, upgrades, PITR	recommended conditionally
Event-streaming broker; durable transport	Partition order, replay, ecosystem and decoupled consumers	Delivery not business exactly once; capacity/security/retention burden	Existing enterprise broker/queue; change on ordering/replay/operating needs	Measure partition skew, lag, recovery; manage schemas, ACLs, upgrades	recommended conditionally
Workflow engine; timers/retries/cases	Durable long waits, visibility and deterministic orchestration support	New critical platform, versioning/operating semantics; engine state not ledger	Explicit DB state/timer scheduler; managed or self-hosted products	Measure workflow history, task lag, failover; pin deterministic code	evaluate in pilot
OpenTelemetry class; cross-service telemetry	Vendor-neutral signals and context propagation	Cardinality/privacy/cost; not audit or reconciliation	Existing standards-compatible platform	Measure ingest/query cost; govern semantics and retention	recommended
Reconciliation SQL/data jobs; external truth	Transparent comparison, aging and evidence export	Bad data can automate false clearing; large history cost	Existing finance data/recon platform	Partition by source/period; rerunnable versioned rules, lineage and access	recommended conditionally

The relational database recommendation is a class, not an unconditional PostgreSQL procurement choice. PostgreSQL documentation establishes relevant transaction behavior (PostgreSQL Global Development Group 2026); the organization should prefer an already approved relational platform if it proves the unique constraints, isolation, partition ownership, recovery, audit, and performance model. Developer experience must include deterministic local tests and inspectable SQL. A workflow product is valuable only if it preserves domain-owned state semantics, deterministic versioning, export, and incident recovery.

16.2. Distributed-ledger and integration options

Table 48. DLT and SDK options illuminate strategic choices; none is preselected for the pilot.

What / obligation	Why use	Why not / operational risk	Alternatives / change trigger	Ecosystem/DX/performance	Recommendation
Web3j; Java EVM adapter	RPC, contract bindings, events and transaction support in Java	Version/network gaps, SDK is neither node nor custody	Native sidecar or another supported SDK; change on encoding/support gaps	Conformance/golden tests; upgrade with chain/EIP support	recommended conditionally
Hyperledger Besu; EVM execution client	Apache-licensed Java client for public/private EVM networks; familiar operations for some teams	Public Ethereum also needs consensus client; self-operation burden; private network creates governance	Managed multi-provider RPC, other execution clients; change if independent node ownership is justified	Measure sync, RPC, reorg recovery, storage; patch protocol/client	evaluate in pilot, not default
Hedera/Hiero; governed network/client option	Java SDK and Java node source; EVM-compatible surface plus distinct mirror/consensus roles	Governance, fees, finality, data and interoperability differ from EVM public-chain assumptions	Other approved public networks; change if asset/liquidity/corridor fit is superior	Benchmark SDK/node APIs and recovery; maintain network-specific adapter	evaluate in pilot
R3 Corda 5; permissioned application network	Identity, flows, UTXO options and notary uniqueness can fit consortium obligations	Not a public stablecoin network; consortium governance and deployment burden	Existing bank/tokenized-deposit network; public rail	Strong JVM DX for Corda teams; performance must be consortium-specific	not recommended for initial pilot
bitcoinj; Bitcoin protocol library	Mature Java library for Bitcoin-oriented integration	No EVM/token/issuer contract model; does not meet recommended stablecoin path by itself	Selected network SDK/node	Good bounded Bitcoin DX; irrelevant capabilities dominate	rejected for initial scope
Custom/private blockchain	Maximum protocol/governance control in theory	Creates consensus, membership, liquidity, interoperability, security, upgrade and exit obligations	Approved public network; existing permissioned consortium	Must benchmark entire network and operate validator governance	rejected for initial pilot

Besu is an Apache-licensed Ethereum execution client written in Java and can operate with public or private networks; public Ethereum operation also requires a consensus client (Besu Project 2026b; Besu Project 2026a). **DOC CODE** ^{CLM-JAVA-001,CLM-JAVA-002} Those facts make Besu an option to evaluate if independent node operation becomes an evidence or concentration requirement. They do not make a permissioned network or self-operated node the default.

Hedera provides Java SDKs and EVM-compatible contracts while its consensus and mirror-node model remains distinct; the Hiero repository contains the Java consensus-node implementation (Hedera 2026; Hiero Project 2026). Corda documents a permissioned application-network model with identities, flows, UTXO options, and notary uniqueness (R3 2026; R3 2024). bitcoinj supplies Bitcoin protocol/wallet functionality, not a general stablecoin platform (bitcoinj Project 2026b; bitcoinj Project 2026a). **DOC CODE** ^{CLM-JAVA-005,CLM-JAVA-006,CLM-JAVA-007,CLM-JAVA-008,CLM-JAVA-009,CLM-JAVA-010} Source language is never evidence of Zelle use or network suitability.

16.3. Comparable technology dossiers

Table 49 makes the twelve evaluation questions visible for every required technology. “Performance” names an axis to measure; no cell is a benchmark claim. Product selection remains conditional where the organization has not supplied its approved platform, support model, workload, data jurisdiction, or recovery objectives. Versions are the reviewed baselines, not an instruction to bypass enterprise patch governance.

Table 49. Common twelve-question technology dossiers, evaluated 2026-07-13.

What / ver- sion / de- ployment / assumption	Obligation / why use	Why not / alterna- tives	Change trigger / operational risk / maintenance	Performance / de- veloper experience / maturity	Recommen- dation / ratio- nale / owner
Java SE 25; Spring Boot 4.0.6 / Frame- work 7.0.x; supported control-plane runtime; Linux/con- tainer assump- tion (Oracle 2025; Spring Project 2026)	APIs, domain model, orchestra- tion, SQL, event and telemetry integration; typed JVM ecosystem and enterprise operability	Not chain-native and framework/de- pendency graph is material; alterna- tives are supported Java LTS baseline, Kotlin, Go, or .NET	Change on platform support or measured workload mismatch. Risks: CVEs, depen- dency drift, block- ing/pool exhaustion. Maintain JDK/frame- work, SBOM, compat- ibility and contract tests	Measure re- quest/event latency, concurrency, alloca- tion, DB/pool con- tention. Strong local debugging/test tool- ing; mature governed releases, multiple vendors, portable bytecode; GPL+Class- path/Apache-2.0 licensing requires normal review	recom- mended ; control-plane fit without language uniformity. Owner: appli- cation/plat- form architec- ture
Relational database class; PostgreSQL 18 documen- tation as ref- erence; single- writer ledger partitions; approved managed or self-operated deployment (PostgreSQL Global Devel- opment Group 2026)	ACID journals, constraints, ver- sion checks, SQL reconciliation, PITR and in- spectable evidence	Cross-region writes and hot-account contention; alter- natives are existing approved RDBMS or a specialized ledger only after invariant/export proof	Change if recovery, contention or juris- diction cannot meet policy. Risks: corrup- tion, operator error, failover split-brain, lock pressure. Main- tain schema, migra- tions, backups, restore drills and statistics	Measure commit/tail latency, lock waits, partition skew, jour- nal growth, replica lag and restore. Ex- cellent local SQL/test tooling; mature ecosystem, permis- sive license and broad support/porta- bility	recom- mended conditionally ; prefer the approved RDBMS that proves invari- ants. Owner: ledger and SRE

What / version / deployment / assumption	Obligation / why use	Why not / alternatives	Change trigger / operational risk / maintenance	Performance / developer experience / maturity	Recommendation / rationale / owner
Apache Kafka 4.1 class; broker/schema service; self-managed or approved managed deployment (Apache Software Foundation 2026; Spring 2026)	Ordered partitions, durable replay and decoupled outbox/inbox consumers	Broker transactions exclude ledger and provider effects; alternatives are existing enterprise broker/queue or log service	Change on ordering, replay, retention, data or operating fit. Risks: partition loss/skew, ACL error, backlog, correlated managed outage. Maintain brokers/clients, schemas, quotas, certificates, replay drills	Measure produce/consume tail latency, lag, skew, recovery and retention cost. Strong JVM clients and test containers; mature Apache governance/licensing and portable protocol, but operating skill remains	recommended conditionally ; transport, never authority. Owner: messaging/platform
Durable workflow class; Temporal documentation reviewed 2026-07-13; managed/self-hosted product unselected (Temporal Technologies 2026)	Long timers, retries, visibility, versioned orchestration and case handoff	New critical history/worker platform; engine state is not ledger. Alternatives: database state machine plus timer scheduler, or an existing approved engine	Change when history scale, determinism, export or operating proof fails. Risks: hidden retry, non-deterministic upgrade, backlog, lock-in. Maintain worker compatibility, history controls, failover and replay tests	Measure task lag, schedule-to-start, history size, worker saturation and recovery. Good SDK testing if deterministic discipline is enforced; maturity/support/licensing are product-specific and must be reviewed	evaluate in pilot ; domain owns semantics. Owner: workflow/platform
OpenTelemetry specification, accessed 2026-07-13; SDK/collector to approved telemetry backend (OpenTelemetry 2026)	Portable traces, metrics, logs and propagation across adapters	Not an audit ledger or reconciliation source; high-cardinality/privacy/cost risk. Alternative: existing standards-compatible telemetry pipeline	Change if backend or data policy lacks required signals/export. Risks: dropped data, secret/PII leakage, sampling blindness. Maintain semantic conventions, collectors, redaction and retention	Measure collector loss, ingest/query latency, cardinality and cost. Broad language tooling; CNCF governance and Apache-2.0 components support portability, subject to backend features	recommended ; operational signal standard only. Owner: observability/SRE

What / version / deployment / assumption	Obligation / why use	Why not / alternatives	Change trigger / operational risk / maintenance	Performance / developer experience / maturity	Recommendation / rationale / owner
Reconciliation/data infrastructure class; relational rules plus approved finance/data platform; versioned jobs and immutable source snapshots	Independent comparison, aging, lineage, reruns, cases and close evidence	Bad normalization can falsely clear; warehouse latency may miss operational breaks. Alternatives: existing finance reconciliation product or SQL jobs	Change when scale, cutoff, lineage or segregation cannot be proven. Risks: source lateness, rule drift, access error, silent tolerance. Maintain mappings, versions, lineage, backfills and control totals	Measure source freshness, unmatched volume/age, rerun time and close latency. SQL is inspectable/testable; platform maturity and license depend on selected internal/product stack; export is mandatory	recommended conditionally ; own semantics even if platform is bought. Owner: accounting/data/reconciliation
Web3j 6.0.0; library inside a bounded EVM adapter, not a service authority (LF Decentralized Trust Web3j Project 2026b ; LF Decentralized Trust Web3j Project 2026a)	Java RPC, ABI/event decoding, contract bindings and transaction construction	Network/EIP/encoding gaps; not a node or custody control. Alternatives: native sidecar, another supported SDK, direct typed RPC	Change on byte mismatch, unsupported transaction/network or maintenance lag. Risks: dependency/CVE, silent encoding mismatch. Maintain pinned release, golden vectors, simulation and native-tool conformance	Measure encoding/decoding and RPC overhead only within end-to-end adapter tests. Good JVM DX; Apache-2.0 open source and portable, but ecosystem breadth is smaller than native EVM tooling	recommended conditionally ; only after conformance. Owner: chain integration

What / version / deployment / assumption	Obligation / why use	Why not / alternatives	Change trigger / operational risk / maintenance	Performance / developer experience / maturity	Recommendation / rationale / owner
Hyperledger Besu 26.7.0; execution client, managed or self-operated; public use also needs consensus client (Besu Project 2026b; Besu Project 2026a)	Independent EVM execution/RPC and possible private-network evaluation	Node sync/storage/patch burden; private deployment creates governance and liquidity obligations. Alternatives: managed multi-provider RPC or other supported execution client	Change if integrity, concentration, archival or economics justify ownership. Risks: consensus/execution compatibility, reorg handling, disk/state growth, incident staffing. Maintain protocol upgrades, client pair, snapshots and recovery drills	Measure sync/recovery, RPC tail, storage growth and reorg observation under selected network. Java aids some teams; Apache-2.0/Linux Foundation project with protocol portability but client-specific operations	evaluate in pilot, not default ; evidence need must justify. Owner: blockchain platform/SRE
Hedera services plus Hiero consensus node 0.74.0; public governed network with Java SDK/node and mirror model (Hedera 2026; Hiero Project 2026)	Candidate network, contract execution and independent observation pattern	Distinct governance, fee, finality, mirror and interoperability assumptions; asset/corridor liquidity unproved here. Alternatives: other approved public or permissioned networks	Change if legal, asset, liquidity and measured operational fit outrank alternatives. Risks: network governance/upgrades, provider/mirror dependence. Maintain network SDK, adapter, fee/finality policy and migration tests	Measure submission/finality observation, mirror lag, fee behavior and recovery. Strong Java path; Apache-2.0 Hiero governance is open, while network support/portability remains protocol-specific	evaluate in pilot ; no preselection. Owner: architecture/treasury/legal
R3 Corda runtime 5.2.2.0; permissioned application network / UTXO and flow stack (R3 2026; R3 2024)	Strategic tokenized-deposit or consortium identity/unique-ness alternative	Not a public stablecoin rail; membership, notary, deployment and consortium governance burden. Alternatives: existing approved bank network or public rail	Change if a live approved consortium supplies issuer, corridor, liquidity, governance and exit. Risks: member coordination, upgrade compatibility, notary/flow incidents. Maintain network, CorDapps, identities and governance	Measure flow/notary latency, member throughput and recovery only in target consortium. Strong JVM DX; Apache-2.0 open-source runtime with commercial ecosystem, but portability is Corda-specific	not recommended for initial pilot ; strategic interoperability only. Owner: institutional architecture/legal

What / version / deployment / assumption	Obligation / why use	Why not / alternatives	Change trigger / operational risk / maintenance	Performance / developer experience / maturity	Recommendation / rationale / owner
bitcoinj 0.17.1; Java Bitcoin protocol/wallet library in an adapter (bitcoinj Project 2026b ; bitcoinj Project 2026a)	Illuminates SDK-versus-node boundary for Bitcoin-oriented products	Does not provide EVM token/issuer contracts or selected stablecoin functionality. Alternatives: selected network SDK/node	Change only if approved route is Bitcoin-native and satisfies asset/legal design. Risks: wallet/key misuse and protocol/library drift. Maintain consensus vectors, peer/RPC behavior and signing isolation	Measure sync/RPC/transaction handling for that route. Familiar Java DX; Apache-2.0 mature project, but capability portability to account/token chains is low	rejected for initial scope; obligation mismatch. Owner: chain architecture
Institutional custody/signing class; FIPS 140-3 HSM or approved MPC/service; language-neutral policy API; no keys in application (National Institute of Standards and Technology 2019 ; National Institute of Standards and Technology 2020 ; Fireblocks 2026b ; Fireblocks 2026a)	Segregated key use, quorum, policy, recovery, statements and specialist operations	Concentration, opaque MPC/control claims, export/recovery and cost; alternatives: qualified custodian, cloud/on-prem HSM, approved in-house custody	Change if due diligence, recovery drill, policy fidelity, data or economics fail. Risks: key loss/compromise, policy bypass, provider outage. Maintain ceremonies, roles, firmware/service changes, allowlists and recovery exercises	Measure authorization/sign tail, availability, capacity, recovery and audit export without exposing secrets. DX must be testable via sandbox; certification is scoped, vendor maturity/support/license and asset portability require contract review	recommended conditionally to buy for pilot; controls stay owned. Owner: security/treasury

What / version / deployment / assumption	Obligation / why use	Why not / alternatives	Change trigger / operational risk / maintenance	Performance / developer experience / maturity	Recommendation / rationale / owner
Managed submit node plus administratively independent read path; provider/network selected in Phase 0; 2026-07-13 service-class evaluation	Submission availability and independent chain evidence without immediate node operations	Correlated upstream/cloud/admin failures and rate/history limits; alternatives: two truly independent providers or self-operated full/archive node	Change on integrity, concentration, archival, latency, cost or jurisdiction. Risks: false/stale response, outage, data loss. Maintain provenance, quorum policy, provider exit and optional node runbook	Measure RPC tail/error, head divergence, reorg visibility, archive coverage and recovery. Easy API DX can hide failure; maturity/support/licensing are provider/client-specific; raw evidence export and endpoint portability required	recommended conditionally ; independence must be demonstrated. Owner: blockchain platform/SRE
Custom/private blockchain; new validator/member network and protocol governance; no implementation selected	Maximum theoretical membership, privacy and upgrade control	Creates consensus security, validator operations, liquidity, interoperability and exit program. Alternatives: approved public rail or existing consortium	Change only if a legal/business need cannot be met elsewhere and funded governance exists. Risks: capture, outage, fork, upgrade, weak liquidity. Maintain protocol, clients, validators, membership and ecosystem indefinitely	Must benchmark whole network, fault tolerance and recovery; DX/support/security ecosystem must be created or inherited; portability is lowest and licensing spans all chosen components	rejected for initial pilot ; burden dominates learning. Owner: executive architecture/risk

Table 50: Build/buy decisions for high-specialization boundaries.

Choice	Initial posture	Risks to control	Revisit trigger
Custody/signing	Buy qualified institutional custody/HSM/MPC service; retain platform authorization and reconciliation	Concentration, outage, policy fidelity, data/export, key governance, exit, incident cooperation	Volume/economics, control gaps, recovery/data needs, approved in-house custody capability
Node access	Managed provider for submit plus administratively independent read/observation path	Correlated cloud/provider failure, stale/false responses, rate/retention limits, data jurisdiction	Integrity, latency, cost, concentration or archival needs justify self-operated node
Chain analytics	Integrate one or more approved providers as risk signal	False positives/negatives, opaque model, privacy, vendor concentration	Model evidence or legal policy requires second source/in-house analysis
Workflow engine	Evaluate against durable state contract; prefer existing approved platform if fit	Lock-in, nondeterministic upgrade, hidden retry, history growth	Scale/visibility/operating proof or a simpler database scheduler is sufficient
Reconciliation	Own comparison semantics and cases; may use approved data platform	Outsourced black box clearing discrepancies, weak lineage	Existing finance platform proves replay, evidence, segregation, and export

16.4. Custody, nodes, and operating ownership

ASSUMPTION Managed node access plus an independent read path is the pilot posture, subject to provider failure-domain, data, integrity, and recovery tests. ^{CLM-OPS-002} The platform records both observations and their provenance; “two providers agree” is not independent if both proxy the same upstream or share administration.

Consequence. The recommended stack is intentionally incomplete until Phase 0 selects legal entities, route, asset, network, custody, node, FX, and payout contracts. Architecture contracts make those decisions substitutable; they do not pretend substitution is free.

17. Stablecoin and settlement-asset options

Decision resolved. Build issuer/network portability into contracts, but launch one approved combination; do not assume an announced proprietary asset is pilot-ready. **PROPOSED**

Asset selection is not a ticker-symbol choice. It selects an issuer and legal claim; reserve and redemption model; custody and contract; mint, burn, freeze, denylist, pause, and upgrade authorities; supported networks; liquidity and market makers; accounting; privacy; incident response; and an exit path.

Circle describes USDC as multichain, backed by highly liquid assets, and redeemable 1:1 under its product terms; Paxos publishes issuer and reserve information for PYUSD (Circle 2026b; Paxos 2026). **DOC** ^{CLM-DA-001,CLM-DA-002} These facts make external regulated assets diligence candidates, not automatic selections. Eligibility, direct redemption, legal rights, freezes, supported networks, custody, liquidity, jurisdiction, and contract/version still require approval.

PUBLIC EWS has announced ZelleUSD for potential future capabilities in markets other than the first announced India route (Early Warning Services 2026a). ^{CLM-ZEL-013,CLM-ZEL-014} **UNKNOWN** Public evidence does not establish the asset’s issuer entity, authorization, reserves, redemption, network, contracts, custody, keys, controls, partners, circulation, or operating schedule. Accordingly the paper evaluates “future ZelleUSD” as a strategic option whose diligence cells remain open; it neither rejects nor recommends it for the initial pilot.

Table 51. Settlement-asset/control-model options. Every row requires legal, issuer, custody, liquidity, accounting, and operating diligence.

Option	Strategic benefit	Principal obligations/risks	Time/complexity	Exit/migration	Initial recommendation
One external regulated stablecoin	Fastest path to test a liquid institutional rail; issuer handles token/reserve operation	Issuer/redemption/concentration/admission power, network/contract/custody, eligibility	Lowest relative if approvals and corridor liquidity exist	Stop new use; redeem/transfer inventory; introduce new route version	recommended conditionally through ADR
Multiple external stablecoins	Counterparty/network choice and broader reach	Multiplies contracts, decimals, controls, liquidity, reconciliation, accounting and support	High before one route is proven	Per-asset kill switch and inventory unwind, but routing behavior complex	not recommended for initial pilot
Bank-issued tokenized deposits	Direct bank liability; potential institutional identity/single-ness benefits	Issuing-bank claim, convertibility, network governance, interbank liquidity, legal/accounting interoperability	Corridor/consortium dependent	Convert/redeem under bank terms; adapter/network exit required	evaluate as interoperability path
Future ZelleUSD	Strategic control, economics and product alignment may be possible	Every issuer/reserve/redemption/network/custody/admin/liquidity/legal fact currently unspecified publicly	Unknown; issuance is a distinct regulated operating program	Must design redemption, migration, contract upgrade and wind-down before launch	unknown; Phase 0/strategic ADR only

Option	Strategic benefit	Principal obligations/risks	Time/complexity	Exit/migration	Initial recommendation
Permissioned settlement token/network	Membership, privacy and governance can be tailored	New governance, validator/member operations, liquidity, interoperability, exit and shared incident response	Highest unless an existing approved consortium fits	Consortium and token redemption/migration terms critical	not recommended for initial pilot
Stablecoin only as invisible rail	Preserves bank channel and avoids consumer custody/gas while testing settlement	Operator/participants still bear treasury, provider, legal, customer and recovery obligations	Lowest product disruption; substantial back-office rigor remains	Replace route behind stable participant contract	recommended product posture

One external asset concentrates issuer risk but keeps the pilot’s state, accounting, custody, reconciliation, and incident surface testable. Multi-asset routing does not simply add a list: it adds route-dependent decimals, contract powers, networks, price/liquidity, eligibility, disclosure, statements, and loss allocation. Generalization is earned after one asset’s controls and exit drill pass.

Stablecoins and tokenized deposits must remain distinct in the data model. A tokenized deposit can represent a commercial-bank liability subject to that bank/network arrangement; a stablecoin represents a claim and control model defined by its issuer and applicable framework. Both may share an adapter vocabulary, but their legal owner, redemption, insolvency, reserve, liquidity, and finality policy differ.

17.1. Per-option control dossiers

The summary recommendation is not a substitute for diligence. Table 52 exposes every required decision dimension for all six options. “Unknown” is an intentional stop condition, especially for a future ZelleUSD; it is not a neutral score.

Table 52. Full settlement-asset and product-control option dossiers.

Option	Issuer / authorization / reserves / insolvency	Custody / redemption / admin powers / chain-contract	Liquidity / interoperability / accounting / privacy	Complexity / time / strategic control / concentration / customer proposition	Exit / migration / posture
One external regulated stablecoin	Issuer/counterparty: approved external legal issuer. Authorization: issuer/operator/participant roles by jurisdiction. Reserves: issuer evidence and terms. Insolvency: direct and indirect holder treatment requires counsel	Custody: qualified supported custodian. Redemption: institutional access, limits, hours, fees and suspension proven. Powers: mint, burn, freeze, denylist, pause and upgrade authority mapped. Chain: one approved contract/network/version	Liquidity: corridor capacity and rebalance stress. Interop: issuer-supported networks only; no generic bridge. Accounting: asset/valuation/impairment/fees approved. Privacy: public metadata minimized and assessed	Complexity/time: lowest relative after approval. Control: low over issuer/network. Concentration: deliberately bounded to one. Customer: invisible familiar payment	Re-deem/transfer inventory, disable route version, migrate adapter and disclosures. Conditionally recommended through ADR
Multiple external stablecoins	Issuer/counterparty: several independent issuers. Authorization/reserves/insolvency: separately approved; no equivalence inferred	Custody/redemption/powers: per asset and provider. Chain-contract: decimals, addresses, upgrade and network combinations versioned; bridge use is separate ADR	Liquidity: fragmented and route-dependent. Interop: conversion creates new counterparties. Accounting: per-asset classification/valuation. Privacy: multiplied network footprints	Complexity/time: high. Control: routing optionality. Concentration: lower issuer concentration but higher integration/provider exposure. Customer: little benefit if rail remains invisible	Per-asset kill switch, inventory unwind and route migration; preserve operation history. Not recommended initially

Option	Issuer / authorization / reserves / insolvency	Custody / redemption / admin powers / chain-contract	Liquidity / interoperability / accounting / privacy	Complexity / time / strategic control / concentration / customer proposition	Exit / migration / posture
Bank-issued tokenized deposits	Issuer/counterparty: issuing commercial bank/depository bank/network membership and corridor roles. Reserves: deposit balance-sheet/ regime, not stablecoin reserve short-hand. Insolvency: account/deposit treatment requires exact terms	Custody: bank/network account or token control. Redemption: convertibility to bank money under scheme rules. Powers: bank/network issuance, transfer and restriction controls. Chain: permissioned or public implementation-specific	Liquidity: issuing-bank and interbank reach. Interop: bank/network agreements dominate. Accounting: deposit/interbank treatment. Privacy: membership may reduce public exposure but increases shared-data governance	Complexity/time: consortium/corridor dependent. Control: shared with banks. Concentration: issuer-bank and network. Customer: potentially institutional singleness; bank channel can remain unchanged	Convert/redeem under scheme, migrate participant/net-work adapters and settle residual claims. Evaluate as interoperability path
Future ZelleUSD	Issuer/counterparty: unknown. Authorization: unknown. Reserves: unknown. Insolvency: unknown. Public announcement establishes none of these operating facts	Custody/redemption/admin powers/chain-contract: all unknown. Issuance, key authority, contract governance and holder eligibility require primary documents and approvals	Liquidity/interoperability/accounting/privacy: all unknown. No corridor, market-maker, redemption, network, statement or metadata posture is assumed	Complexity/time: unknown; issuance is a separate regulated operating program. Strategic control: potential, not proven. Concentration/customer proposition: unknown	Wind-down, redemption, contract/network migration and holder treatment must be designed before launch. Strategic ADR only; not pilot-ready on public evidence

Option	Issuer / authorization / reserves / insolvency	Custody / redemption / admin powers / chain-contract	Liquidity / interoperability / accounting / privacy	Complexity / time / strategic control / concentration / customer proposition	Exit / migration / posture
Permissioned settlement token/network	Issuer/counter-party: consortium or designated issuer. Authorization: membership/governance. Reserves/insolvency: token and member rules must establish claim	Custody: member/key model. Redemption: scheme-specific. Powers: governance/validator/member controls. Chain-contract: new or existing permissioned protocol with upgrade/fork rules	Liquidity: must be created or proven. Interop: gateways add risk. Accounting: scheme-specific. Privacy: configurable but requires data-sharing, regulator and operator governance	Complexity/time: highest unless approved consortium exists. Control: shared but tailorable. Concentration: governance/operators. Customer: invisible or institutional, not inherently better money	Consortium exit, redeem token, export evidence, remove validator/member keys, migrate obligations. Not recommended initially
Stablecoin only as invisible rail	Issuer/counter-party/authorization/reserves/insolvency: inherited from the separately selected asset; product does not erase them	Custody/redemption/powers/chain-contract: institutional and hidden from customer; still fully controlled, disclosed where required and reconciled	Liquidity/interoperability/accounting/privacy: owned in treasury/back office; no customer wallet, arbitrary address or gas. Domestic route remains independent	Complexity/time: lowest customer disruption but substantial operations. Control: participant contract stable. Concentration: selected rail. Customer: bank account, recipient and status experience	Replace or disable settlement adapter behind stable participant API; reconcile/unwind inventory. Recommended product posture

17.2. Asset due-diligence gate

Table 53. No asset may enter a production route until every mandatory diligence question has approved evidence or an explicit stop decision.

Domain	Required evidence/question	Reject or pause signal
Issuer and authorization	Legal issuer/entity, applicable permission, jurisdictions, audited financial/operating responsibilities	Unclear issuer/role, missing required authorization, unacceptable jurisdiction
Reserve and redemption	Composition, segregation, attestations/audits, direct/indirect redemption rights, timing, fees, minimums, suspension, insolvency treatment	No viable institutional exit or reserve evidence/control below policy
Contract and admin powers	Network, addresses, decimals, proxy/code, mint/burn/freeze/denylist/pause/upgrade roles, delays, events, audits, incident contacts	Undisclosed or unacceptable privileged path; unmonitored upgrade
Custody and keys	Supported custody, address ownership, key policy, insurance/terms, recovery, state-ment/API/export and exit	Keys cannot meet segregation/recovery/audit or provider exit requirements
Liquidity and treasury	Corridor liquidity, market makers, redemption capacity, hours, spread/slippage, concentration, prefunding and stress behavior	Capacity below exposure; no credible rebalance/redemption under stress
Network and nodes	Finality/reorg model, fees, congestion/censorship, governance/upgrades, node diversity, data/analytics, contract/tool support	No independent observation or unacceptable finality/governance risk
Compliance/privacy	Address/party controls, sanctions/freeze cooperation, information exchange, data location, public-chain metadata exposure	Required controls/roles unresolved or data exposure impermissible
Accounting/tax/reporting	Asset classification, valuation, subledger/GL mapping, statements, fees, close and impairment treatment	Cannot close/reconcile or meet reporting policy
Operations and exit	SLAs, incident/change notice, kill switch, contact escalation, test environment, export, wind-down/migration rehearsal	Unbounded incident dependency or no tested exit

18. Tradeoffs and alternatives

Decision resolved. Accept explicit coordination and reconciliation complexity to retain accountable financial control across heterogeneous rails. **PROPOSED**

Table 54. Tradeoff ledger. Optionality has operating cost and every posture has a revisit signal.

Decision	Chosen posture / benefit	Cost and failure exposure	Revisit signal	Owner
Operational truth	Internal ledger/state; represents all obligations	More systems, journals, reconciliation and repair; drift risk	External shared ledger proves every required obligation/control and recovery	Accounting/architecture

Decision	Chosen posture / benefit	Cost and failure exposure	Revisit signal	Owner
Customer instrument	Invisible settlement; familiar channel and support	Operator absorbs treasury/integration complexity; limited token proposition	Approved strategy and controls for customer-held token	Product/risk
Network model	Evaluate public network for pilot	Public fees, privacy, governance, reorg/censorship risk	Existing permissioned consortium proves liquidity, governance, exit and corridor fit	Architecture/legal/treasury
Issuer	One external regulated asset selected by ADR	Concentration and issuer/admin/redemption dependency	Proprietary asset or alternate issuer proves legal, liquidity, control and cost advantage	Treasury/legal/product
Custody/nodes	Managed custody and node plus independent read path	Concentration, contracts, export and incident dependency	Scale, integrity, economics or control justify self-operation	Security/platform
Generality	One asset/network/corridor	Later adapters/migrations; initial concentration	One route passes controls and demand supports another dimension	Product/architecture
Liquidity	Prefunded bounded inventory	Capital inefficiency and rebalance burden	Measured flows and credit/legal approval support just-in-time/credit model	Treasury/risk
Customer speed	Evidence-gated completion; early credit only with explicit risk owner	May feel slower; or creates loss exposure when accelerated	Loss/latency data and disclosure support threshold change	Product/risk/legal
Provider abstraction	Stable operation/evidence contract plus capability-specific branches	More adapter design and tests	Common semantics demonstrably mature without hiding risk	Architecture/integration
Workflow	Durable asynchronous state	Operational engine/history/queue complexity	Route is provably local/atomic or simpler scheduler meets timers/recovery	Platform
Regional availability	Fenced single-writer ownership per ledger partition	Failover may pause writes; operational complexity	Proven multi-region consistency model maintains invariants and recovery objectives	Ledger/SRE
Build versus buy	Build financial authority; buy specialist custody/node capabilities	Vendor dependency plus integration; owned core maintenance	Provider gaps or scale make internal specialization justified	Engineering/security/procurement

Table 55. Operating burden and reversibility companion to the tradeoff ledger.

Decision	Explicit operating burden	Reversibility / exit mechanism
Operational truth	Journal/schema governance, partition ownership, reconciliation, correction, close, recovery and finance/SRE on-call	Export immutable history and migrate opening balances only through controlled parallel reconciliation; no chain switch alone replaces obligations
Customer instrument	Treasury, liquidity, disclosures, support and provider incidents remain behind the bank channel	Disable settlement route while retaining participant API and domestic product; unwind institutional inventory
Network model	Finality policy, node/provider provenance, fees, upgrades, reorg drills, chain analytics and incident response	Stop new submission, observe/settle in-flight work, move future route version to another approved network; assets may require redemption/transfer
Issuer	Reserve/redemption monitoring, admin-power changes, eligibility, concentration, statements and issuer incident escalation	Kill new use, redeem or transfer inventory, close breaks and introduce a separately approved issuer/asset version
Custody/nodes	Key ceremonies/policy, provider due diligence, statement export, RPC integrity, recovery and exit drills	Transfer assets/roles under quorum, preserve signed evidence, switch endpoints only after observation parity and recovery proof
Generality	Adapter capability registry, route policy, migrations and regression matrix even when only one route runs	Add one dimension behind a new ADR; remove it with a scoped kill switch and inventory/case closeout
Liquidity	Forecasting, prefund monitoring, rebalance, limits, stress tests, valuation and concentration reporting	Reduce/close inventory and route limits; adopt credit or just-in-time funding only under new legal/risk approval and journals
Customer speed	Threshold telemetry, interval-risk measurement, disclosure/status policy, loss allocation and complaints	Tighten completion gates immediately for new work; loosening requires versioned policy and explicit risk acceptance
Provider abstraction	Capability-specific adapter tests, version mapping, inquiry, error taxonomy, provider certification and provenance	Replace adapter by route version after conformance, dual observation and in-flight operation closure; contract is stable but migration is not free
Workflow	History/version management, timers, worker capacity, replay, cases, schema evolution and engine disaster recovery	Export domain events and rebuild projection; replace engine only after deterministic replay and active-work migration
Regional availability	Partition fencing, failover runbooks, replication/backup, pause semantics and recovery drills	Move ownership through fenced failover; adopt another consistency model only after invariant and split-brain testing
Build versus buy	Core engineering/SRE plus vendor due diligence, contract/SLA, integration, incident coordination, data export and procurement	Specialist provider can be replaced through tested export/transfer; owned financial authority migrates only through reconciled cutover

A chain-as-source-of-truth advocate correctly notes that public ledgers provide shared, independently observable transaction history. The chosen posture retains that evidence and adds internal obligation truth rather than discarding it. A permissioned-network advocate correctly values identity, controlled membership, and privacy; the countervailing cost is governance, operation, liquidity, and interoperability. A proprietary-asset advocate correctly values strategic

control; the countervailing fact is that issuance creates a regulated product and operating system, not merely a contract deployment.

Similarly, managed services reduce time to first learning but create concentration and exit work. Self-operation is not inherently safer: it transfers patching, key recovery, node integrity, incident response, and staffing to the operator. The decision changes only when those obligations can be performed better and proven, not when ownership feels architecturally pure.

Consequence. The initial posture is intentionally bounded and reversible. It optimizes learning about controlled settlement, not breadth of asset marketing or infrastructure ownership.

19. Evidence-gated roadmap

Decision resolved. Increase real-money exposure only after the preceding phase proves ownership, invariants, reconciliation, recovery, and operational control. **PROPOSED**

The roadmap has exactly seven phases, numbered 0 through 6. It is not a calendar. A phase can stop, repeat, or reduce scope; elapsed time never substitutes for exit evidence. Numeric limits, recovery objectives, and service measures are **ASSUMPTION** values to approve from workload, loss, liquidity, customer, and regulatory data. ^{CLM-OPS-001}

Table 56. Seven-phase delivery and control gates. The phase/gate visual is the lower panel of figure 17.

Phase	Objective	Capabilities/deliverables	Measurable exit evidence	Exposure / readiness	Stop conditions
0 Discover and decide	Close facts that materially shape the system	Legal/entity/role map; participant contract baseline; corridor, asset, network, custody, node, FX, payout and accounting ADRs; threat model; finality policy; success/loss measures	Mandatory unknowns have accountable disposition; source/contract diligence complete; failure and exit obligations accepted	No customer funds; tabletop only	Issuer/route/legal role unresolved; no viable redemption/exit; controls or economics unacceptable
1 Control-plane and ledger foundation	Prove owned financial truth independent of blockchain	Intent/idempotency; ledger/journals/reservations; state machine; policy/config versions; outbox/inbox; audit; cases; domestic regression isolation	Balance/property/concurrency tests pass; duplicate/replay produce one effect; point-in-time recovery and audit reconstruction demonstrated	Synthetic funds; production-like security; on-call/run-books drafted	Invariant, access, recovery, or participant compatibility failure
2 Sandbox adapters	Prove provider contracts and ambiguity handling	Custody/signing, node/observer, asset, FX, payout adapters; simulation; capability registry; reconciliation sources; kill switches	Golden transaction vectors; duplicate/timeout/conflict/reorg simulations; provider export and inquiry proven; all external accounts covered by reconciliation	Test networks/provider sandboxes only; no customer promise	Provider cannot supply stable identity, inquiry, evidence, segregation, or exit
3 Bounded end-to-end proof	Exercise full lifecycle without open customer exposure	Production-shaped route, treasury operations, policy/case integration, monitoring, close, incident and disaster drills	Complete 13-flow suite; reconciliation closes; signing/key/node/payout drills meet approved measures; no critical/high review findings	Internal/tightly bounded value under explicit loss cap	Unexplained break, unsafe retry, drill failure, policy bypass, material security finding
4 Production pilot	Validate one corridor/asset/network with selected institutions	Participant onboarding, customer disclosures/support, prefunding, limits, staffed operations, regulatory reporting as applicable	Approved reliability, reconciliation aging, customer outcomes, loss/exposure, cost and support measures across required observation window	One corridor, one asset, one network, bounded participants/value; no wallet/bridge	Limit breach, unresolved material break, provider/issuer incident, legal/control change, adverse customer/loss threshold
5 Controlled expansion	Expand one dimension at a time	Additional participants or value bands, then corridor/provider capacity; automation justified by evidence	Each expansion cohort passes Phase 4 measures; concentration and liquidity stress tests; recovery remains within limits	Incremental approved exposure; rollback to prior route/version	Metrics degrade, operations saturation, concentration/credit exceeds policy
6 Strategic options	Evaluate new assets/networks or proprietary capability	Multi-asset only if needed; tokenized-deposit interoperability; possible ZelleUSD/permissioned decisions; self-operated node/custody only if justified	Separate business, legal, technology, governance, security, liquidity, migration and exit ADRs; independent pilot gates	No automatic inheritance from Phase 5; each option bounded anew	Optionality cost exceeds value; issuance/governance/exit unresolved; control regression

19.1. Decision records and risk burn-down

The program maintains at least: ADR-001 customer/participant boundary; ADR-002 ledger authority and accounting model; ADR-003 state vocabulary and consistency model; ADR-004 corridor/legal/finality policy; ADR-005 asset/issuer; ADR-006 network/finality; ADR-007 custody/signing/key roles; ADR-008 node/observation; ADR-009 FX/liquidity; ADR-010 payout; ADR-011 data/privacy/compliance roles; ADR-012 Java/module/deployment; ADR-013 messaging/workflow; ADR-014 reconciliation/close; ADR-015 limits/degraded modes; ADR-016 migration/exit. An ADR records owner, evidence, options, decision, dissent, assumptions, operational obligations, test, review date, and change trigger.

Route disable never deletes obligations. It blocks new exposure, permits inquiry/observation/reconciliation, and allows only explicitly authorized recovery effects. Promotion and rollback preserve the route, policy, schema, contract, and configuration versions that governed every existing payment.

20. Conclusion

The central architecture choice is not a blockchain or token. It is the decision to preserve a regulated payment system's own accountabilities while adding another settlement rail. Participant institutions retain their customer channel and account relationship. The proposed control plane retains intent, policy, internal ledger, durable payment state, participant truth, recovery, and reconciliation. Issuers, custody, signing, chains, FX, and payout providers contribute bounded effects and evidence through explicit trust boundaries.

That allocation makes hard cases representable. A blockchain can succeed while a database write fails; a database can commit while chain submission fails; a transaction can reorganize or remain unfinalized; a finalized token movement can be followed by payout failure; a sanctions fact can change after payment; delivery can duplicate; and external records can disagree. None is repaired by changing a status field or "rolling back the chain." Stable operation requires inquiry, immutable journals, versioned states, compensation, independent observation, cases, exposure limits, and reconciliation.

Java and Spring are strong choices for enterprise APIs, orchestration, ledger-facing services, policy integration, messaging, reconciliation, and operations. They are not a mandate to select a weaker node, reproduce custody cryptography, or move smart contracts out of their native ecosystem. Besu and other DLT technologies remain evaluated options, not defaults. The pilot selects one approved asset and public network through separate decisions, with qualified custody, managed node access plus independent observation, one corridor, bounded participants, prefunding, no consumer wallet, and no bridge.

The unresolved facts are material: private EWS topology and contracts; ZelleUSD issuer, reserves, redemption, network and controls; corridor legal roles and finality; selected providers; customer remedies; workload and recovery objectives; liquidity and loss thresholds. They belong in Phase 0 and governed ADRs, not in plausible-sounding prose.

The general lesson extends beyond Zelle: successful stablecoin integration is disciplined payment infrastructure with an additional asynchronous rail. It succeeds when every obligation has an owner, every transition has evidence, every external effect can be inquired and reconciled, and every increase in exposure is earned.

21. Technical references and annotated bibliography

The bibliography is generated from the canonical references.bib. Each entry identifies its category in keywords and includes an annotation stating relevance and limitation; the canonical evidence CSV maps source keys to related claim IDs and paper locations. Official statements validate public product facts; standards and regulations validate their text; documentation validates described technology behavior; source repositories validate code and release behavior. None proves private EWS deployment or final legal applicability.

The catalog below follows the thirteen source categories required by the research protocol. A source may appear in more than one category when it performs distinct evidentiary work. The visible entry type distinguishes organizational statements and documentation (online), normative or analytical reports (report), standards (where separately typed), and immutable software releases (software). The evidence key joins each entry to the generated claim index in section E.

Table 57: Dependency and risk burn-down. Learning without customer funds precedes exposure.

Risk	Earliest learning	Evidence before customer funds	Evidence before expansion
Ledger/state correctness	Phase 1	Property, contention, duplicate, recovery and audit reconstruction	Production invariant and close history
Provider ambiguity	Phase 2	Timeout/lost-response/inquiry and duplicate test vectors	Measured ambiguity aging and successful cases
Chain/reorg/finality	Phase 2	Historical/simulated reorg, node disagreement and threshold evaluation	Network-health history, drills and loss/exposure performance
Key/custody incident	Phase 2–3	Signer refusal, outage, recovery and migration drill with synthetic assets	Contractual incident/export proof and periodic production drill
Liquidity/FX/payout	Phase 2–3	Capacity, quote expiry, shortfall, failed payout and reconciliation tests	Stress/concentration measures and operating history
Legal/compliance/customer	Phase 0–3	Approved roles, terms, decision evidence, case and disclosure walkthrough	Pilot cases, regulatory change review, customer outcome measures
Operational load	Phase 3	On-call, dashboards, runbooks, chaos/tabletop and disaster recovery	Capacity/incident trend and staffing readiness
Exit/concentration	Phase 2–3	Data/key/record export, route disable, inventory unwind and alternate observation	Periodic provider/asset/network exit rehearsal

Payment networks

Source type: report; evidence key: bis-tokenised-ledger-2023.

Relevance and limitation: Policy analysis comparing tokenized deposits, stablecoins, and unified-ledger concepts. It is not a mandate or deployment fact. Bank for International Settlements (2023). *Blueprint for the Future Monetary System: Improving the Old, Enabling the New*. BIS Bulletin 73. URL: <https://www.bis.org/publ/bisbull73.htm> (visited on 07/13/2026).

Source type: report; evidence key: bis-annual-2026.

Relevance and limitation: Current BIS analysis of tokenisation and monetary-system design; useful for tradeoffs, not evidence of EWS choices.– (2026). *Annual Economic Report 2026, Chapter III*. URL: <https://www.bis.org/publ/arpdf/ar2026e3.htm> (visited on 07/13/2026).

Source type: report; evidence key: ews-fraud-rfi-2025.

Relevance and limitation: Primary EWS description of fraud signals, warnings, enrollment controls, and participant responsibilities. Confidential rules and models remain undisclosed. Early Warning Services (Sept. 18, 2025a). *Actions to Address Payments Fraud RFI Response*. URL: <https://www.earlywarning.com/sites/default/files/2025-09/EWS%20Actions%20to%20Address%20Payments%20Fraud%20RFI%20%28September%2018%202025%29.pdf> (visited on 07/13/2026).

Source type: online; evidence key: ews-international-2025.

Relevance and limitation: Primary announcement of EWS's cross-border stablecoin initiative. It establishes direction, not chain, issuer, custody, reserve, contract, ledger, or API design.– (Oct. 24, 2025b). *Zelle Goes International: Early Warning Expands One-Trillion-Dollar Payments Network with Stablecoin Initiative*. URL: <https://www.earlywarning.com/press-release/zelle-goes-international-early-warning-expands-1t-payments-network-stablecoin> (visited on 07/13/2026).

Source type: report; evidence key: ews-zelle-p2p-2025.

Relevance and limitation: Official product brief depicting participating financial institutions and ACH or RTP settlement context. It does not prove one universal route.– (Oct. 2025c). *Zelle P2P Product Brief, Q4 2025*. URL: <https://www.earlywarning.com/sites/default/files/2025-10/Zelle-P2P-Product-Brief-Q425.pdf> (visited on 07/13/2026).

Source type: online; evidence key: ews-india-zelleusd-2026.

Relevance and limitation: Primary source for India as the first announced destination and ZelleUSD as a future option in other markets. It does not state that India uses ZelleUSD or disclose implementation details.– (June 11, 2026a). *Zelle Heads to India, Unveils ZelleUSD Stablecoin for Other Markets*. URL: <https://www.earlywarning.com/press-release/zelle-heads-india-unveils-zelleusd-stablecoin-other-markets> (visited on 07/13/2026).

Source type: online; evidence key: ews-zelle-results-2026.

Relevance and limitation: Primary source for EWS-reported 2025 volume, transaction count, participant scale, and network ownership. Annual totals do not establish peak load or SLOs.– (Feb. 11, 2026b). *Zelle Posts 20 Percent Growth with 1.2 Trillion Dollars Sent*. URL: <https://www.earlywarning.com/press-release/zelle-posts-20-growth-12-trillion-sent-far-outpacing-consumer-spending-and-cementing> (visited on 07/13/2026).

Source type: online; evidence key: ews-community-2026.

Relevance and limitation: Primary source for participant reach and the variety of financial-institution integration contexts. It does not enumerate current interfaces or versions.– (Jan. 27, 2026c). *Zelle Widens the On-Ramp for Community Banks and Credit Unions*. URL: <https://www.earlywarning.com/press-release/zelle-widens-ramp-community-banks-and-credit-unions-337-new-bank-and-credit-unions> (visited on 07/13/2026).

Source type: online; evidence key: ews-brands.

Relevance and limitation: Official organizational description of Zelle. It provides product context rather than implementation details.– (2026d). *Brands: Zelle*. URL: <https://www.earlywarning.com/brands> (visited on 07/13/2026).

Source type: online; evidence key: fednow-about.

Relevance and limitation: Official overview of continuous instant-payment processing and eligible participants. Institution-specific product behavior remains outside its scope. Federal Reserve Financial Services (2026a). *About the FedNow Service*. URL: <https://www.frb services.org/financial-services/fednow/about.html> (visited on 07/13/2026).

Source type: report; evidence key: fednow-product-sheet.

Relevance and limitation: Official feature and settlement description used to distinguish rail finality from customer and legal outcomes.– (2026b). *FedNow Service Product Sheet*. URL: <https://www.frb.services.org/binaries/content/assets/crsocms/financial-services/fednow/fednow-product-sheet.pdf> (visited on 07/13/2026).

Source type: online; *evidence key:* nacha-reversals.

Relevance and limitation: Official summary of defined ACH reversal reasons and timing. Current rules and counsel govern specific cases. NACHA (2026). *Reversals and Enforcement*. URL: <https://www.nacha.org/rules/reversals-and-enforcement> (visited on 07/13/2026).

Source type: online; *evidence key:* tch-rtp.

Relevance and limitation: Official RTP description of real-time final settlement and irrevocable credit transfers. Product recourse remains separate. The Clearing House (2026). *RTP for Financial Institutions*. URL: <https://www.theclearinghouse.org/payment-systems/rtp/institution> (visited on 07/13/2026).

Source type: online; *evidence key:* zelle-support.

Relevance and limitation: Official support routing that preserves the participating institution's customer-support role. It does not define loss allocation. Zelle (2026a). *Contact Support*. URL: <https://www.zellepay.com/support/contact-support> (visited on 07/13/2026).

Source type: online; *evidence key:* zelle-how-it-works.

Relevance and limitation: Public product description of bank-channel access and email or mobile-number enrollment. It is not topology documentation.– (2026b). *How It Works*. URL: <https://www.zellepay.com/how-it-works> (visited on 07/13/2026).

Source type: online; *evidence key:* zelle-security.

Relevance and limitation: Official customer guidance on trusted recipients and cancellation limits. It is not a complete legal or dispute rulebook.– (2026c). *Security*. URL: <https://www.zellepay.com/security> (visited on 07/13/2026).

Stablecoins

Source type: report; *evidence key:* bis-tokenised-ledger-2023.

Relevance and limitation: Policy analysis comparing tokenized deposits, stablecoins, and unified-ledger concepts. It is not a mandate or deployment fact. Bank for International Settlements (2023). *Blueprint for the Future Monetary System: Improving the Old, Enabling the New*. BIS Bulletin 73. URL: <https://www.bis.org/publ/bisbull73.htm> (visited on 07/13/2026).

Source type: report; *evidence key:* bis-annual-2026.

Relevance and limitation: Current BIS analysis of tokenisation and monetary-system design; useful for tradeoffs, not evidence of EWS choices.– (2026). *Annual Economic Report 2026, Chapter III*. URL: <https://www.bis.org/publ/arpdf/ar2026e3.htm> (visited on 07/13/2026).

Source type: online; *evidence key:* circle-ccfp.

Relevance and limitation: Official documentation for issuer-mediated burn, attestation, and mint flows. It is specific to Circle and USDC. Circle (2026a). *Cross-Chain Transfer Protocol Technical Guide*. URL: <https://developers.circle.com/ccfp/references/technical-guide> (visited on 07/13/2026).

Source type: online; *evidence key:* circle-usdc.

Relevance and limitation: Issuer documentation for reserves, redemption framing, and multichain availability. Contractual rights and approved access require diligence.– (2026b). *What Is USDC?* URL: <https://developers.circle.com/stablecoins/what-is-usdc> (visited on 07/13/2026).

Source type: online; *evidence key:* ews-international-2025.

Relevance and limitation: Primary announcement of EWS's cross-border stablecoin initiative. It establishes direction, not chain, issuer, custody, reserve, contract, ledger, or API design. Early Warning Services (Oct. 24, 2025b). *Zelle Goes International: Early Warning Expands One-Trillion-Dollar Payments Network with Stablecoin Initiative*. URL: <https://www.earlywarning.com/news/ews-expands-one-trillion-dollar-payments-network-with-stablecoin-initiative>

[//www.earlywarning.com/press-release/zelle-goes-international-early-warning-expands-1t-payments-network-stablecoin](https://www.earlywarning.com/press-release/zelle-goes-international-early-warning-expands-1t-payments-network-stablecoin) (visited on 07/13/2026).

Source type: online; evidence key: ewe-india-zelleusd-2026.

Relevance and limitation: Primary source for India as the first announced destination and ZelleUSD as a future option in other markets. It does not state that India uses ZelleUSD or disclose implementation details. Early Warning Services (June 11, 2026a). *Zelle Heads to India, Unveils ZelleUSD Stablecoin for Other Markets*. URL: <https://www.earlywarning.com/press-release/zelle-heads-india-unveils-zelleusd-stablecoin-other-markets> (visited on 07/13/2026).

Source type: online; evidence key: paxos-pyusd.

Relevance and limitation: Issuer documentation for PYUSD and its reserve materials. Product, chain, redemption, and jurisdictional details remain decision inputs. Paxos (2026). *PayPal USD*. URL: <https://docs.paxos.com/guides/stablecoin/pyusd> (visited on 07/13/2026).

Source type: online; evidence key: genius-act-2025.

Relevance and limitation: Authoritative legislative record for the United States payment-stablecoin issuer framework. Application and implementing rules require legal review. United States Congress (July 18, 2025). *S.1582 - GENIUS Act, Public Law 119-27*. URL: <https://www.congress.gov/bill/119th-congress/senate-bill/1582> (visited on 07/13/2026).

Blockchain protocols

Source type: software; evidence key: besu-source.

Relevance and limitation: Canonical source repository and release history. Source language and activity do not prove deployment or fit. Besu Project (July 13, 2026a). *Besu Ethereum Client Source*. Version 26.7.0. URL: <https://github.com/besu-eth/besu/releases/tag/26.7.0> (visited on 07/13/2026).

Source type: online; evidence key: besu-docs.

Relevance and limitation: Official documentation for Besu's execution-client role, public/private operation, JSON-RPC, and key boundary. It does not establish EWS use.– (2026b). *Besu Documentation*. URL: <https://docs.besu-eth.org/> (visited on 07/13/2026).

Source type: software; evidence key: bitcoinj-source.

Relevance and limitation: Canonical Java library source and releases. Bitcoin support does not supply issuer or EVM functionality. bitcoinj Project (May 4, 2026a). *bitcoinj Source*. Version 0.17.1. URL: <https://github.com/bitcoinj/bitcoinj/releases/tag/v0.17.1> (visited on 07/13/2026).

Source type: online; evidence key: bitcoinj-docs.

Relevance and limitation: Official library documentation used to distinguish a protocol integration library from a general blockchain platform.– (2026b). *bitcoinj*. URL: <https://bitcoinj.org/> (visited on 07/13/2026).

Source type: online; evidence key: circle-cctp.

Relevance and limitation: Official documentation for issuer-mediated burn, attestation, and mint flows. It is specific to Circle and USDC. Circle (2026a). *Cross-Chain Transfer Protocol Technical Guide*. URL: <https://developers.circle.com/cctp/references/technical-guide> (visited on 07/13/2026).

Source type: online; evidence key: ethereum-pos.

Relevance and limitation: Official educational documentation for validator voting and finality behavior. Route policy still requires network-specific risk decisions. Ethereum Foundation (2026). *Proof-of-Stake*. URL: <https://ethereum.org/developers/docs/consensus-mechanisms/pos/> (visited on 07/13/2026).

Source type: online; evidence key: hedera-docs.

Relevance and limitation: Official SDK, consensus, mirror-node, and smart-contract documentation. A Java SDK does not resolve governance or network fit. Hedera (2026). *Hedera Documentation*. URL: <https://docs.hedera.com/> (visited on 07/13/2026).

Source type: software; evidence key: hiero-source.

Relevance and limitation: Canonical Java consensus-node source. Source language does not establish operating-model suitability. Hiero Project (May 21, 2026). *Hiero Consensus Node Source*. Version 0.74.0. URL: <https://github.com/hiero-ledger/hiero-consensus-node/releases/tag/v0.74.0> (visited on 07/13/2026).

Source type: software; *evidence key:* web3j-source.

Relevance and limitation: Canonical source repository identifying the evaluated release. Version support must be pinned during implementation. LF Decentralized Trust Web3j Project (June 29, 2026a). *Web3j Source and Releases*. Version 6.0.0. URL: <https://github.com/LFDT-web3j/web3j/releases/tag/v6.0.0> (visited on 07/13/2026).

Source type: online; *evidence key:* web3j-docs.

Relevance and limitation: Official Java integration-library documentation for RPC, contract wrappers, events, and HSM support. It is not a node or custody policy.– (2026b). *Web3j Documentation*. URL: <https://docs.web3j.io/latest/> (visited on 07/13/2026).

Source type: software; *evidence key:* corda-source.

Relevance and limitation: Canonical Corda runtime source. Licensing, support, and application-network governance remain separate decisions. R3 (Nov. 28, 2024). *Corda Runtime Open Source*. Version 5.2.2. URL: <https://github.com/corda/corda-runtime-os/releases/tag/release-5.2.2.0> (visited on 07/13/2026).

Source type: online; *evidence key:* corda-docs.

Relevance and limitation: Official documentation for permissioned application networks, flows, UTXO ledgers, identities, and notaries. It is not a public stablecoin chain.– (2026). *Corda 5.2 Documentation*. Version 5.2. URL: <https://docs.r3.com/en/platform/corda/5.2.html> (visited on 07/13/2026).

Java

Source type: software; *evidence key:* besu-source.

Relevance and limitation: Canonical source repository and release history. Source language and activity do not prove deployment or fit. Besu Project (July 13, 2026a). *Besu Ethereum Client Source*. Version 26.7.0. URL: <https://github.com/besu-eth/besu/releases/tag/26.7.0> (visited on 07/13/2026).

Source type: online; *evidence key:* besu-docs.

Relevance and limitation: Official documentation for Besu's execution-client role, public/private operation, JSON-RPC, and key boundary. It does not establish EWS use.– (2026b). *Besu Documentation*. URL: <https://docs.besu-eth.org/> (visited on 07/13/2026).

Source type: software; *evidence key:* bitcoinj-source.

Relevance and limitation: Canonical Java library source and releases. Bitcoin support does not supply issuer or EVM functionality. bitcoinj Project (May 4, 2026a). *bitcoinj Source*. Version 0.17.1. URL: <https://github.com/bitcoinj/bitcoinj/releases/tag/v0.17.1> (visited on 07/13/2026).

Source type: online; *evidence key:* bitcoinj-docs.

Relevance and limitation: Official library documentation used to distinguish a protocol integration library from a general blockchain platform.– (2026b). *bitcoinj*. URL: <https://bitcoinj.org/> (visited on 07/13/2026).

Source type: online; *evidence key:* hedera-docs.

Relevance and limitation: Official SDK, consensus, mirror-node, and smart-contract documentation. A Java SDK does not resolve governance or network fit. Hedera (2026). *Hedera Documentation*. URL: <https://docs.hedera.com/> (visited on 07/13/2026).

Source type: software; *evidence key:* hiero-source.

Relevance and limitation: Canonical Java consensus-node source. Source language does not establish operating-model suitability. Hiero Project (May 21, 2026). *Hiero Consensus Node Source*. Version 0.74.0. URL: <https://github.com/hiero-ledger/hiero-consensus-node/releases/tag/v0.74.0> (visited on 07/13/2026).

Source type: software; *evidence key:* web3j-source.

Relevance and limitation: Canonical source repository identifying the evaluated release. Version support must be pinned during implementation.LF Decentralized Trust Web3j Project (June 29, 2026a). *Web3j Source and Releases*. Version 6.0.0. URL: <https://github.com/LFDT-web3j/web3j/releases/tag/v6.0.0> (visited on 07/13/2026).

Source type: online; *evidence key:* web3j-docs.

Relevance and limitation: Official Java integration-library documentation for RPC, contract wrappers, events, and HSM support. It is not a node or custody policy.– (2026b). *Web3j Documentation*. URL: <https://docs.web3j.io/latest/> (visited on 07/13/2026).

Source type: online; *evidence key:* java-se-25-spec.

Relevance and limitation: Normative platform specifications for the evaluated Java 25 runtime. They establish language and VM behavior, not organizational supportability or workload performance.Oracle (2025). *Java Platform, Standard Edition 25 Specifications*. URL: <https://docs.oracle.com/javase/specs/> (visited on 07/13/2026).

Source type: online; *evidence key:* jep-444.

Relevance and limitation: Official Java platform proposal for virtual threads. It is relevant only when workload and library behavior support the model.Pressler, R. and A. Bateman (2023). *JEP 444: Virtual Threads*. URL: <https://openjdk.org/jeps/444> (visited on 07/13/2026).

Source type: software; *evidence key:* corda-source.

Relevance and limitation: Canonical Corda runtime source. Licensing, support, and application-network governance remain separate decisions.R3 (Nov. 28, 2024). *Corda Runtime Open Source*. Version 5.2.2. URL: <https://github.com/corda/corda-runtime-os/releases/tag/release-5.2.2.0> (visited on 07/13/2026).

Source type: online; *evidence key:* corda-docs.

Relevance and limitation: Official documentation for permissioned application networks, flows, UTXO ledgers, identities, and notaries. It is not a public stablecoin chain.– (2026). *Corda 5.2 Documentation*. Version 5.2. URL: <https://docs.r3.com/en/platform/corda/5.2.html> (visited on 07/13/2026).

Source type: online; *evidence key:* spring-kafka-eos.

Relevance and limitation: Official scope for Kafka read-process-write transaction semantics. It does not extend atomicity to external financial systems.Spring (2026). *Spring for Apache Kafka: Exactly Once Semantics*. URL: <https://docs.spring.io/spring-kafka/reference/kafka/exactly-once.html> (visited on 07/13/2026).

Source type: software; *evidence key:* spring-boot-source.

Relevance and limitation: Immutable release page for the evaluated Spring Boot baseline. It documents a framework release and dependencies, not fitness for this architecture without compatibility and workload testing.Spring Project (Apr. 23, 2026). *Spring Boot 4.0.6*. Version 4.0.6. URL: <https://github.com/spring-projects/spring-boot/releases/tag/v4.0.6> (visited on 07/13/2026).

Enterprise integration

Source type: online; *evidence key:* kafka-41-docs.

Relevance and limitation: Official documentation for the event-streaming reference implementation used in the broker-class evaluation. Kafka delivery and transactions do not make external financial effects atomic.Apache Software Foundation (2026). *Apache Kafka 4.1 Documentation*. URL: <https://kafka.apache.org/41/documentation.html> (visited on 07/13/2026).

Source type: online; *evidence key:* cloudevents-spec.

Relevance and limitation: Open event-envelope specification used for metadata vocabulary. It does not define business authority or delivery guarantees.Cloud Native Computing Foundation (2026). *CloudEvents Specification*. URL: <https://github.com/cloudevents/spec/blob/main/cloudevents/spec.md> (visited on 07/13/2026).

Source type: online; *evidence key:* debezium-outbox.

Relevance and limitation: Official implementation documentation for the transactional outbox pattern. Publication and consumption remain asynchronous. Debezium (2026). *Outbox Event Router*. URL: <https://debezium.io/documentation/reference/stable/transformations/outbox-event-router.html> (visited on 07/13/2026).

Source type: online; *evidence key:* aws-idempotent-apis.

Relevance and limitation: Primary AWS engineering guidance for idempotency-token contracts and retry ambiguity. It does not replace financial reconciliation. Featonby, M. (2026). *Making Retries Safe with Idempotent APIs*. Amazon Web Services. URL: <https://aws.amazon.com/builders-library/making-retries-safe-with-idempotent-APIs/> (visited on 07/13/2026).

Source type: online; *evidence key:* spring-kafka-eos.

Relevance and limitation: Official scope for Kafka read-process-write transaction semantics. It does not extend atomicity to external financial systems. Spring (2026). *Spring for Apache Kafka: Exactly Once Semantics*. URL: <https://docs.spring.io/spring-kafka/reference/kafka/exactly-once.html> (visited on 07/13/2026).

Source type: online; *evidence key:* temporal-docs.

Relevance and limitation: Official documentation for a representative durable-workflow product considered during class evaluation. It does not preselect Temporal or make workflow history a financial ledger. Temporal Technologies (2026). *Temporal Platform Documentation*. URL: <https://docs.temporal.io/> (visited on 07/13/2026).

Security

Source type: report; *evidence key:* ewe-fraud-rfi-2025.

Relevance and limitation: Primary EWS description of fraud signals, warnings, enrollment controls, and participant responsibilities. Confidential rules and models remain undisclosed. Early Warning Services (Sept. 18, 2025a). *Actions to Address Payments Fraud RFI Response*. URL: <https://www.earlywarning.com/sites/default/files/2025-09/EWS%20Actions%20to%20Address%20Payments%20Fraud%20RFI%20%28September%2018%202025%29.pdf> (visited on 07/13/2026).

Source type: online; *evidence key:* fireblocks-recovery.

Relevance and limitation: Provider documentation for its key-share and recovery model. Contractual availability, implementation quality, export, jurisdiction, and tested recovery still require independent diligence. Fireblocks (2026a). *Disaster Recovery Service Process Overview*. URL: <https://developers.fireblocks.com/docs/perform-drs-process> (visited on 07/13/2026).

Source type: online; *evidence key:* fireblocks-capabilities.

Relevance and limitation: Provider documentation for roles, administrative quorum, policy controls, vaults, MPC key shares, and transaction authorization. It is a diligence example, not evidence of EWS use, independent assurance, or selection.– (2026b). *Fireblocks Key Features and Capabilities*. URL: <https://developers.fireblocks.com/docs/capabilities> (visited on 07/13/2026).

Source type: report; *evidence key:* nist-fips-140-3.

Relevance and limitation: Normative cryptographic-module security baseline. Validation alone does not define custody or authorization policy. National Institute of Standards and Technology (Mar. 22, 2019). *Security Requirements for Cryptographic Modules*. FIPS 140-3. URL: <https://csrc.nist.gov/pubs/fips/140-3/final> (visited on 07/13/2026).

Source type: report; *evidence key:* nist-sp-800-57.

Relevance and limitation: Authoritative key-management lifecycle guidance used for custody, compromise, recovery, and cryptoperiod design.– (May 2020). *Recommendation for Key Management: Part 1, General*. NIST SP 800-57 Part 1 Revision 5. URL: <https://csrc.nist.gov/pubs/sp/800/57/pt1/r5/final> (visited on 07/13/2026).

Source type: report; *evidence key:* nist-ssdf.

Relevance and limitation: Authoritative software-development control framework. It complements rather than replaces system threat modeling.– (Feb. 2022). *Secure Software Development Framework Version 1.1*. NIST SP 800-218. URL: <https://csrc.nist.gov/pubs/sp/800/218/final> (visited on 07/13/2026).

Source type: report; *evidence key:* ofac-virtual-currency-2021.

Relevance and limitation: Authoritative sanctions guidance establishing that virtual-currency activity remains subject to sanctions obligations. Office of Foreign Assets Control (Oct. 2021). *Sanctions Compliance Guidance for the Virtual Currency Industry*. URL: https://ofac.treasury.gov/system/files/126/virtual_currency_guidance_brochure.pdf (visited on 07/13/2026).

Source type: online; *evidence key:* openzeppelin-access-control.

Relevance and limitation: Official smart-contract library documentation for roles and delayed administration. Governance and monitoring remain design obligations. OpenZeppelin (2026). *Access Control*. URL: <https://docs.openzeppelin.com/contracts/access-control> (visited on 07/13/2026).

Source type: online; *evidence key:* zelle-security.

Relevance and limitation: Official customer guidance on trusted recipients and cancellation limits. It is not a complete legal or dispute rulebook. Zelle (2026c). *Security*. URL: <https://www.zellepay.com/security> (visited on 07/13/2026).

Custody and signing

Source type: online; *evidence key:* fireblocks-recovery.

Relevance and limitation: Provider documentation for its key-share and recovery model. Contractual availability, implementation quality, export, jurisdiction, and tested recovery still require independent diligence. Fireblocks (2026a). *Disaster Recovery Service Process Overview*. URL: <https://developers.fireblocks.com/docs/perform-drs-process> (visited on 07/13/2026).

Source type: online; *evidence key:* fireblocks-capabilities.

Relevance and limitation: Provider documentation for roles, administrative quorum, policy controls, vaults, MPC key shares, and transaction authorization. It is a diligence example, not evidence of EWS use, independent assurance, or selection. – (2026b). *Fireblocks Key Features and Capabilities*. URL: <https://developers.fireblocks.com/docs/capabilities> (visited on 07/13/2026).

Compliance

Source type: online; *evidence key:* ecf-ecfr-reg-e.

Relevance and limitation: Authoritative electronic-fund-transfer regulation. Coverage of a specific stablecoin or cross-border product requires legal analysis. Consumer Financial Protection Bureau (2026). *12 CFR Part 1005 - Electronic Fund Transfers (Regulation E)*. URL: <https://www.ecfr.gov/current/title-12/chapter-X/part-1005> (visited on 07/13/2026).

Source type: report; *evidence key:* ews-fraud-rfi-2025.

Relevance and limitation: Primary EWS description of fraud signals, warnings, enrollment controls, and participant responsibilities. Confidential rules and models remain undisclosed. Early Warning Services (Sept. 18, 2025a). *Actions to Address Payments Fraud RFI Response*. URL: <https://www.earlywarning.com/sites/default/files/2025-09/EWS%20Actions%20to%20Address%20Payments%20Fraud%20RFI%20%28September%2018%202025%29.pdf> (visited on 07/13/2026).

Source type: report; *evidence key:* fatf-virtual-assets-2025.

Relevance and limitation: Primary international policy source for risk-based AML/CFT and transfer-information expectations. Jurisdictional implementation varies. Financial Action Task Force (2025). *Targeted Update on Implementation of the FATF Standards on Virtual Assets and VASPs*. URL: <https://www.fatf-gafi.org/en/publications/Fatfrecommendations/targeted-update-virtual-assets-vasps-2025.html> (visited on 07/13/2026).

Source type: report; *evidence key:* fincen-cvc-2019.

Relevance and limitation: Primary guidance for role- and facts-based money-transmission analysis. It does not decide the proposed product's legal status. Financial Crimes Enforcement Network (May 9, 2019). *Application of FinCEN's Regulations to Certain Business Models Involving Convertible Virtual Currencies*. FIN-2019-G001. URL: <https://www.fincen.gov/resources/statutes-regulations/guidance/application-fincens-regulations-certain-business-models> (visited on 07/13/2026).

Source type: report; evidence key: ofac-virtual-currency-2021.

Relevance and limitation: Authoritative sanctions guidance establishing that virtual-currency activity remains subject to sanctions obligations. Office of Foreign Assets Control (Oct. 2021). *Sanctions Compliance Guidance for the Virtual Currency Industry*. URL: https://ofac.treasury.gov/system/files/126/virtual_currency_guidance_brochure.pdf (visited on 07/13/2026).

Source type: online; evidence key: genius-act-2025.

Relevance and limitation: Authoritative legislative record for the United States payment-stablecoin issuer framework. Application and implementing rules require legal review. United States Congress (July 18, 2025). *S.1582 - GENIUS Act, Public Law 119-27*. URL: <https://www.congress.gov/bill/119th-congress/senate-bill/1582> (visited on 07/13/2026).

Smart contracts

Source type: online; evidence key: openzeppelin-access-control.

Relevance and limitation: Official smart-contract library documentation for roles and delayed administration. Governance and monitoring remain design obligations. OpenZeppelin (2026). *Access Control*. URL: <https://docs.openzeppelin.com/contracts/access-control> (visited on 07/13/2026).

Source type: online; evidence key: eip-20.

Relevance and limitation: Normative token interface used to explain why interface compatibility does not establish economic equivalence. Vogelsteller, F. and V. Buterin (Nov. 19, 2015). *ERC-20: Token Standard*. URL: <https://eips.ethereum.org/EIPS/eip-20> (visited on 07/13/2026).

Source type: online; evidence key: eip-1967.

Relevance and limitation: Normative proxy storage-slot standard used to identify upgrade and admin-control monitoring requirements. Worthington, S. et al. (Apr. 24, 2019). *ERC-1967: Proxy Storage Slots*. URL: <https://eips.ethereum.org/EIPS/eip-1967> (visited on 07/13/2026).

Standards

Source type: online; evidence key: cloudevents-spec.

Relevance and limitation: Open event-envelope specification used for metadata vocabulary. It does not define business authority or delivery guarantees. Cloud Native Computing Foundation (2026). *CloudEvents Specification*. URL: <https://github.com/cloudevents/spec/blob/main/cloudevents/spec.md> (visited on 07/13/2026).

Source type: online; evidence key: ecfr-reg-e.

Relevance and limitation: Authoritative electronic-fund-transfer regulation. Coverage of a specific stablecoin or cross-border product requires legal analysis. Consumer Financial Protection Bureau (2026). *12 CFR Part 1005 - Electronic Fund Transfers (Regulation E)*. URL: <https://www.ecfr.gov/current/title-12/chapter-X/part-1005> (visited on 07/13/2026).

Source type: report; evidence key: fatf-virtual-assets-2025.

Relevance and limitation: Primary international policy source for risk-based AML/CFT and transfer-information expectations. Jurisdictional implementation varies. Financial Action Task Force (2025). *Targeted Update on Implementation of the FATF Standards on Virtual Assets and VASPs*. URL: <https://www.fatf-gafi.org/en/publications/Fatfrecommendations/targeted-update-virtual-assets-vasps-2025.html> (visited on 07/13/2026).

Source type: report; evidence key: nist-fips-140-3.

Relevance and limitation: Normative cryptographic-module security baseline. Validation alone does not define custody or authorization policy. National Institute of Standards and Technology (Mar. 22, 2019). *Security Requirements for Cryptographic Modules*. FIPS 140-3. URL: <https://csrc.nist.gov/pubs/fips/140-3/final> (visited on 07/13/2026).

Source type: report; evidence key: nist-sp-800-57.

Relevance and limitation: Authoritative key-management lifecycle guidance used for custody, compromise, recovery, and cryptoperiod design. National Institute of Standards and Technology (May 2020). *Recommendation for Key Management: Part 1, General*. NIST SP 800-57 Part 1 Revision 5. URL: <https://csrc.nist.gov/pubs/sp/800/57/pt1/r5/final> (visited on 07/13/2026).

Source type: report; *evidence key:* nist-ssdf.

Relevance and limitation: Authoritative software-development control framework. It complements rather than replaces system threat modeling.– (Feb. 2022). *Secure Software Development Framework Version 1.1*. NIST SP 800-218. URL: <https://csrc.nist.gov/pubs/sp/800/218/final> (visited on 07/13/2026).

Source type: online; *evidence key:* opentelemetry-spec.

Relevance and limitation: Official telemetry standard for traces, metrics, and logs. Audit and business evidence remain application responsibilities. OpenTelemetry (2026). *OpenTelemetry Specification*. URL: <https://opentelemetry.io/docs/specs/otel/> (visited on 07/13/2026).

Source type: online; *evidence key:* java-se-25-spec.

Relevance and limitation: Normative platform specifications for the evaluated Java 25 runtime. They establish language and VM behavior, not organizational supportability or workload performance. Oracle (2025). *Java Platform, Standard Edition 25 Specifications*. URL: <https://docs.oracle.com/javase/specs/> (visited on 07/13/2026).

Source type: online; *evidence key:* eip-20.

Relevance and limitation: Normative token interface used to explain why interface compatibility does not establish economic equivalence. Vogelsteller, F. and V. Buterin (Nov. 19, 2015). *ERC-20: Token Standard*. URL: <https://eips.ethereum.org/EIPS/eip-20> (visited on 07/13/2026).

Source type: online; *evidence key:* eip-1967.

Relevance and limitation: Normative proxy storage-slot standard used to identify upgrade and admin-control monitoring requirements. Worthington, S. et al. (Apr. 24, 2019). *ERC-1967: Proxy Storage Slots*. URL: <https://eips.ethereum.org/EIPS/eip-1967> (visited on 07/13/2026).

Infrastructure

Source type: online; *evidence key:* kafka-41-docs.

Relevance and limitation: Official documentation for the event-streaming reference implementation used in the broker-class evaluation. Kafka delivery and transactions do not make external financial effects atomic. Apache Software Foundation (2026). *Apache Kafka 4.1 Documentation*. URL: <https://kafka.apache.org/41/documentation.html> (visited on 07/13/2026).

Source type: software; *evidence key:* besu-source.

Relevance and limitation: Canonical source repository and release history. Source language and activity do not prove deployment or fit. Besu Project (July 13, 2026a). *Besu Ethereum Client Source*. Version 26.7.0. URL: <https://github.com/besu-eth/besu/releases/tag/26.7.0> (visited on 07/13/2026).

Source type: online; *evidence key:* besu-docs.

Relevance and limitation: Official documentation for Besu's execution-client role, public/private operation, JSON-RPC, and key boundary. It does not establish EWS use.– (2026b). *Besu Documentation*. URL: <https://docs.besu-eth.org/> (visited on 07/13/2026).

Source type: online; *evidence key:* fireblocks-capabilities.

Relevance and limitation: Provider documentation for roles, administrative quorum, policy controls, vaults, MPC key shares, and transaction authorization. It is a diligence example, not evidence of EWS use, independent assurance, or selection. Fireblocks (2026b). *Fireblocks Key Features and Capabilities*. URL: <https://developers.fireblocks.com/docs/capabilities> (visited on 07/13/2026).

Source type: online; *evidence key:* postgresql-transactions.

Relevance and limitation: Official relational-database transaction and isolation documentation. Its guarantees stop at the database boundary. PostgreSQL Global Development Group (2026). *Transaction Isolation*. URL: <https://www.postgresql.org/docs/current/transaction-iso.html> (visited on 07/13/2026).

Source type: software; *evidence key:* spring-boot-source.

Relevance and limitation: Immutable release page for the evaluated Spring Boot baseline. It documents a framework release and dependencies, not fitness for this architecture without compatibility and workload testing. Spring Project (Apr. 23, 2026). *Spring Boot 4.0.6*. Version 4.0.6. URL: <https://github.com/spring-projects/spring-boot/releases/tag/v4.0.6> (visited on 07/13/2026).

Source type: online; *evidence key:* temporal-docs.

Relevance and limitation: Official documentation for a representative durable-workflow product considered during class evaluation. It does not preselect Temporal or make workflow history a financial ledger. Temporal Technologies (2026). *Temporal Platform Documentation*. URL: <https://docs.temporal.io/> (visited on 07/13/2026).

Observability

Source type: online; *evidence key:* opentelemetry-spec.

Relevance and limitation: Official telemetry standard for traces, metrics, and logs. Audit and business evidence remain application responsibilities. OpenTelemetry (2026). *OpenTelemetry Specification*. URL: <https://opentelemetry.io/docs/specs/otel/> (visited on 07/13/2026).

Reliability

Source type: online; *evidence key:* kafka-41-docs.

Relevance and limitation: Official documentation for the event-streaming reference implementation used in the broker-class evaluation. Kafka delivery and transactions do not make external financial effects atomic. Apache Software Foundation (2026). *Apache Kafka 4.1 Documentation*. URL: <https://kafka.apache.org/41/documentation.html> (visited on 07/13/2026).

Source type: online; *evidence key:* debezium-outbox.

Relevance and limitation: Official implementation documentation for the transactional outbox pattern. Publication and consumption remain asynchronous. Debezium (2026). *Outbox Event Router*. URL: <https://debezium.io/documentation/reference/stable/transformations/outbox-event-router.html> (visited on 07/13/2026).

Source type: online; *evidence key:* ethereum-pos.

Relevance and limitation: Official educational documentation for validator voting and finality behavior. Route policy still requires network-specific risk decisions. Ethereum Foundation (2026). *Proof-of-Stake*. URL: <https://ethereum.org/developers/docs/consensus-mechanisms/pos/> (visited on 07/13/2026).

Source type: online; *evidence key:* aws-idempotent-apis.

Relevance and limitation: Primary AWS engineering guidance for idempotency-token contracts and retry ambiguity. It does not replace financial reconciliation. Featonby, M. (2026). *Making Retries Safe with Idempotent APIs*. Amazon Web Services. URL: <https://aws.amazon.com/builders-library/making-retries-safe-with-idempotent-APIs/> (visited on 07/13/2026).

Source type: online; *evidence key:* fireblocks-recovery.

Relevance and limitation: Provider documentation for its key-share and recovery model. Contractual availability, implementation quality, export, jurisdiction, and tested recovery still require independent diligence. Fireblocks (2026a). *Disaster Recovery Service Process Overview*. URL: <https://developers.fireblocks.com/docs/perform-drs-process> (visited on 07/13/2026).

Source type: online; *evidence key:* postgresql-transactions.

Relevance and limitation: Official relational-database transaction and isolation documentation. Its guarantees stop at the database boundary. PostgreSQL Global Development Group (2026). *Transaction Isolation*. URL: <https://www.postgresql.org/docs/current/transaction-iso.html> (visited on 07/13/2026).

Source type: online; *evidence key:* temporal-docs.

Relevance and limitation: Official documentation for a representative durable-workflow product considered during class evaluation. It does not preselect Temporal or make workflow history a financial ledger. Temporal Technologies (2026). *Temporal Platform Documentation*. URL: <https://docs.temporal.io/> (visited on 07/13/2026).

A. Detailed state transitions

This appendix expands table 26. Routes may omit inapplicable states, but may not bypass their equivalent guards. Every transition appends an event with previous/new state, expected/new aggregate version, business operation ID where applicable, actor/workload, time, reason, evidence references, and policy/configuration versions.

Table 58. Full canonical transition matrix.

From	To	Guard/evidence	Side effect and timeout
New	RECEIVED	Participant authenticated/authorized; key+hash accepted; intent committed	Publish validation command; ingress budget
RECEIVED	VALIDATED	Schema/contract, participant, amount/asset/corridor and identity prerequisites valid	Request policy facts/decisions
RECEIVED	REJECTED	Deterministic validation or authorization failure	Publish reason-safe participant status; terminal
VALIDATED	POLICY_PENDING	Required policy set and versions recorded	Start decision SLA; no financial effect
POLICY_PENDING	REJECTED	Binding reject with evidence	Release any preauthorization if present; terminal
POLICY_PENDING	MANUAL_REVIEW	Hold/enhanced-review obligation or conflicting facts	Open assigned case and preserve restricted posture
POLICY_PENDING	FUNDS_RESERVED	All decisions allow; unique reservation operation and obligations committed	Request quote; reservation-aging timer
FUNDS_RESERVED	QUOTED	Required inventory/capacity present; quote and route stored	Start quote-expiry timer
FUNDS_RESERVED	FAILED_RETRYABLE	Quote/liquidity transient with no external effect	Bounded re-quote/rebalance; reservation aging policy
QUOTED	LEDGER_POSTED	Quote unexpired; capacity and policy rechecked; authorization journal and outbox committed atomically	Dispatch signing or route command from outbox
LEDGER_POSTED	SIGNING	Route requires chain; simulation and signing authorization pass	Send canonical bytes/hash using sign operation ID
LEDGER_POSTED	Route-specific settlement/payout	Non-chain route; authorized provider operation recorded	Execute through route adapter
SIGNING	SUBMITTED	Signature and transaction identity durably recorded; submission evidence adequate	Submit signed bytes once; ambiguity timer

From	To	Guard/evidence	Side effect and timeout
SIGNING	FAILED_RETRYABLE	Signer proves no signature/effect and outage/refusal is retryable	Retry same canonical operation within policy
SIGNING	MANUAL_REVIEW	Signature existence/authorization uncertain or security concern	Disable scope as required; case/security response
SUBMITTED	CONFIRMING	Independent node observes exact transaction and expected effects	Poll/subscribe through finality policy
SUBMITTED	MANUAL_REVIEW	Inquiry cannot prove absence or presence within ambiguity policy	No resubmission; exposure/case timer
CONFIRMING	ECONOMICALLY_FINAL	Canonical lineage, independent sources, threshold and network health pass	Apply chain-final journal classification; evaluate payout
CONFIRMING	FAILED_RETRYABLE	Transaction proven reverted/absent and retry policy authorizes correction	New authorized operation or compensation
CONFIRMING	MANUAL_REVIEW	Reorg/delay/conflicting observation exceeds policy	Contain affected route/value; case
ECONOMICALLY_FINAL	PAYOUT_PENDING	Payout obligation, policy approval, capacity, and unique payout operation are committed	Dispatch payout request with recorded operation ID
ECONOMICALLY_FINAL	RECIPIENT_CREDITED	Token-to-token route has sufficient destination ownership evidence	Publish eligible participant status
PAYOUT_PENDING	RECIPIENT_CREDITED	Contractually sufficient destination-credit evidence	Set customer completion gate if product policy allows
PAYOUT_PENDING	MANUAL_REVIEW	Timeout/terminal error/conflicting payout evidence	Retain payable; inquiry and corridor remedy
RECIPIENT_CREDITED	RECONCILING	All route effects have comparison coverage	Launch/associate reconciliation run
RECONCILING	RECONCILED	All required sources match within approved tolerance or disposition	Record close evidence; terminal route state
RECONCILING	MANUAL_REVIEW	Material, aged, or integrity break	Open break/case; narrow kill switch if needed
FAILED_RETRYABLE	Prior safe execution state	Absence/no-effect proven; attempts/time/policy valid	Reissue same effect or create approved new attempt
Any recoverable state	COMPENSATING	Original effects known; compensation plan/legal/ledger authority approved	Create separate idempotent recovery operations
COMPENSATING	COMPENSATED	Every recovery effect and journal proven/reconciled as required	Participant notification and terminal evidence
Any nonrecoverable state	FAILED_FINAL	Permanent result; no outstanding effect or recovery obligation	Terminal notification; retain audit and reconciliation duties

B. Financial and flow exactness

The complete illustrative chart of accounts is table 10; exact journals are tables 11 to 21 and 23; the successful closeout is table 22; invariants are table 24; and external-source coverage is table 25. Those tables remain in the main argument because ledger authority cannot be delegated to an appendix.

The complete normal/failure flow catalog is tables 28 and 29; the concern map is table 30; and the local/external success-failure quadrants are table 35. Together they form a release consistency check: a route may add a provider-specific

branch, but every mutation must map to an operation ID, ledger posture, state guard, ambiguity rule, customer status, audit evidence, and reconciliation source.

B.1. Illustrative participant intent and event

Illustrative participant intent. Fields are schematic, not a production API.

```
POST /v1/payment-intents
Idempotency-Key: participant-generated-opaque-key
{
  "contractVersion": "1.0",
  "source": {"participantAccountRef": "opaque", "currency": "USD"},
  "amount": "100.00",
  "recipient": {"participantAliasRef": "opaque"},
  "destinationCurrency": "INR",
  "product": "INVISIBLE_CROSS_BORDER",
  "termsAcceptanceRef": "opaque"
}
```

The platform derives asset, chain, contract, custody address, FX/payout providers, and finality policy from an approved route. A participant cannot inject them through the generic intent.

Illustrative durable payment event envelope.

```
{
  "eventId": "01...",
  "type": "payment.transaction-observed.v1",
  "aggregateId": "PAY-123",
  "aggregateVersion": 11,
  "businessOperationId": "PAY-123/CHAIN-SUBMIT/1",
  "causationId": "...",
  "correlationId": "PAY-123",
  "occurredAt": "2026-07-13T17:00:00Z",
  "producer": "chain-observer",
  "policyVersion": "finality-policy-7",
  "configurationVersion": "route-12",
  "evidenceRefs": ["evidence://opaque/..."]
}
```

C. Evaluation and decision artifacts

Technology decisions are summarized in tables 47, 48 and 50; asset/control options and diligence are tables 51 and 53; and major architecture costs/change triggers are table 54. A production decision record must add evaluated product version/release/commit, deployment model, contract terms, benchmark plan/results, security assessment, support ownership, total operating cost, recovery/export test, and sign-offs. A numeric score may supplement but never override a mandatory constraint.

C.1. Required organization-specific validation questions

Table 59. Questions that cannot be answered from public evidence.

Domain	Question	Closure / accountable roles
Current platform	Which contracts and systems currently own participant intent, alias, account authorization, risk, settlement status, notification, case, reconciliation, and audit?	Internal discovery; product/platform/accounting
Product	Is stablecoin visible or entirely institutional; who is sender/recipient; what status, disclosure, error and remedy promises apply?	Product/legal/participant agreement
Legal entities/roles	Who issues, transmits, custodies, converts, pays out, guarantees, screens, reports, supports and absorbs loss in each jurisdiction?	Counsel/compliance/entity RACI
Asset	Who is issuer; what are reserve, redemption, eligibility, insolvency and admin-control terms; is ZelleUSD relevant and when?	Asset ADR; treasury/legal/compliance
Network/contract	Which network/contract versions; what finality, fees, governance, upgrade, privacy, node and incident risks are approved?	Network ADR; chain risk/security
Custody/signing	Who owns keys/addresses; what quorum, value bands, recovery, export, incident and exit model applies?	Custody ADR; security/treasury
FX/liquidity	Who provides quotes and inventory; what prefunding, credit, slippage, concentration, stress and rebalance limits apply?	Treasury/risk/finance
Payout	What proves recipient credit; is retry idempotent; what cancellation/return/remedy and statement evidence exists?	Payout ADR; operations/legal
Finality	What evidence separately establishes blockchain, legal, customer and accounting finality per route?	Four-owner policy template
Workload/reliability	Peak shape, value bands, latency/status expectations, availability, RPO/RTO, backlog, dispute and retention windows?	Product/SRE/accounting evidence
Accounting	What is the legal/accounting system of record; chart, valuation, revenue, loss, suspense, close and auditor evidence?	Controller-approved accounting design
Data/privacy	Which personal/transaction data may cross institutions, providers, chains and regions; retention/deletion/access rules?	Privacy/security/data governance
Operations	Staffing, on-call, case SLAs, four-eyes actions, limit/kill-switch ownership, restart authority, drills and evidence retention?	Operations/SRE/security

D. Evidence taxonomy and claim ledger

E. Generated evidence matrix

The following view is generated deterministically from the canonical CSV. Source keys resolve in references.bib; locations use LaTeX labels. Confidence describes support for the narrowly written claim, not implementation suitability.

Table 61: Evidence claim ledger. The canonical CSV generates this readable view.

Table 60: Evidence badges map exactly to the approved taxonomy.

Badge	Class	Permitted inference
PUBLIC	Validated from public source	Exact fact established by authoritative public source; no private topology inference
CODE	Validated from source code	Official repository behavior at cited release/commit; no deployment/suitability inference
DOC	Validated from documentation	Responsible organization's documented behavior; no production-use inference
INFERRED	Inferred architecture	Logical decomposition used to reason; explicitly not deployed topology
PROPOSED	Proposed target architecture	Recommendation and rationale owned by this paper
ASSUMPTION	Operational assumption	Planning posture requiring measurement, owner and approval
UNKNOWN	Unknown / needs validation	Material fact not established and retained as a gate

Claim	Evidence record
CLM-CUST-001 Confidence: Medium-high Accessed: 2026-07-13	Fireblocks documents role-based access, administrative quorum, policy controls, vaults, MPC key shares, and transaction authorization capabilities. <i>Class:</i> Validated from documentation <i>Sources:</i> fireblocks-capabilities <i>Locations:</i> sec:security <i>Caveat:</i> This validates provider documentation only, not independent assurance, contractual terms, EWS use, or selection.
CLM-CUST-002 Confidence: Medium-high Accessed: 2026-07-13	Fireblocks documents a disaster-recovery process and its described distribution of MPC key shares. <i>Class:</i> Validated from documentation <i>Sources:</i> fireblocks-recovery <i>Locations:</i> sec:security <i>Caveat:</i> Recovery availability, export, contractual rights, jurisdiction, and operational effectiveness require independent testing and diligence.
CLM-DA-001 Confidence: High Accessed: 2026-07-13	Circle describes USDC as a multichain dollar stablecoin backed by highly liquid assets with 1:1 redemption and published reserve assurance. <i>Class:</i> Validated from documentation <i>Sources:</i> circle-usdc <i>Locations:</i> sec:stablecoin-options;sec:technology-evaluation <i>Caveat:</i> Issuer terms, eligibility, freeze powers, chain support, and redemption access still require diligence.
CLM-DA-002 Confidence: High	

Continued on next page

Table 62 – continued

Claim	Evidence record
Accessed: 2026-07-13	Paxos identifies itself as the issuer of PYUSD and publishes product and reserve information for the asset. <i>Class:</i> Validated from documentation <i>Sources:</i> paxos-pyusd <i>Locations:</i> sec:stablecoin-options <i>Caveat:</i> Availability, redemption, custody, contract, and jurisdictional rights depend on the approved product structure.
CLM-DA-003	
Confidence: Medium-high	
Accessed: 2026-07-13	BIS analysis distinguishes stablecoins from tokenized commercial-bank deposits and identifies different singleness, liquidity, and settlement implications. <i>Class:</i> Validated from documentation <i>Sources:</i> bis-tokenised-ledger-2023;bis-annual-2026 <i>Locations:</i> sec:stablecoin-options;sec:problem-context <i>Caveat:</i> BIS analysis is policy research rather than an EWS requirement or implementation fact.
CLM-DA-004	
Confidence: High	
Accessed: 2026-07-13	Circle CCTP documents issuer-mediated burn and mint interoperability with off-chain attestation and source/destination finality thresholds. <i>Class:</i> Validated from documentation <i>Sources:</i> circle-cctp <i>Locations:</i> sec:payment-lifecycle;sec:stablecoin-options <i>Caveat:</i> CCTP is a Circle and USDC-specific design example, not a neutral bridge or Zelle architecture fact.
CLM-DA-005	
Confidence: High	
Accessed: 2026-07-13	Ethereum proof-of-stake finality depends on validator votes and can be delayed when a blocking share does not vote. <i>Class:</i> Validated from documentation <i>Sources:</i> ethereum-pos <i>Locations:</i> sec:finality;sec:failure-model <i>Caveat:</i> Confirmation and finality policy must be chain, route, and value specific.
CLM-DA-006	
Confidence: High	
Accessed: 2026-07-13	ERC-20 defines a fungible-token interface and event vocabulary but does not define issuer reserves, redemption, custody, or legal rights. <i>Class:</i> Validated from documentation <i>Sources:</i> eip-20 <i>Locations:</i> sec:technology-evaluation;sec:stablecoin-options <i>Caveat:</i> Interface compatibility does not make two assets economically or operationally equivalent.
CLM-DA-007	
Confidence: High	

Continued on next page

Table 62 – continued

Claim	Evidence record
Accessed: 2026-07-13	EIP-1967 standardizes proxy implementation and administration storage slots used by upgradeable EVM contracts. <i>Class:</i> Validated from documentation <i>Sources:</i> eip-1967 <i>Locations:</i> sec:security;sec:technology-evaluation <i>Caveat:</i> The standard identifies slots; it does not establish that upgrade authority is acceptably governed.
CLM-INF-001	
Confidence: Medium	
Accessed: 2026-07-13	Participant API, alias resolution, policy, routing, notification, exception, and operations capabilities form a useful logical decomposition of publicly observable behavior. <i>Class:</i> Inferred architecture <i>Sources:</i> ews-zelle-p2p-2025;zelle-how-it-works <i>Locations:</i> sec:problem-context <i>Caveat:</i> These capabilities are not asserted as EWS services, deployment units, data stores, or ownership boundaries.
CLM-INF-002	
Confidence: High	
Accessed: 2026-07-13	A stablecoin settlement extension introduces issuer, custody, signer, chain, node, FX, payout, analytics, legal, treasury, and reconciliation roles. <i>Class:</i> Inferred architecture <i>Sources:</i> ews-international-2025;ews-india-zelleusd-2026;ofac-virtual-currency-2021 <i>Locations:</i> sec:problem-context;sec:architecture-overview <i>Caveat:</i> Exact organizations, contracts, and role allocations require product and legal discovery.
CLM-INT-001	
Confidence: High	
Accessed: 2026-07-13	Idempotency tokens make mutating retries safe only when token recording, payload equivalence, and the mutation share a reliable atomic boundary. <i>Class:</i> Validated from documentation <i>Sources:</i> aws-idempotent-apis <i>Locations:</i> sec:consistency;sec:component-design <i>Caveat:</i> Retention, conflict behavior, and external side effects still require explicit contracts.
CLM-INT-002	
Confidence: High	
Accessed: 2026-07-13	CloudEvents defines a common event metadata envelope without defining an application's business authority or delivery guarantee. <i>Class:</i> Validated from documentation <i>Sources:</i> cloudevents-spec <i>Locations:</i> sec:messaging <i>Caveat:</i> A standard envelope does not supply ordering, deduplication, ledger authority, or reconciliation.
CLM-INT-003	
Confidence: High	

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Table 62 – continued

Claim	Evidence record
Accessed: 2026-07-13	OpenTelemetry defines vendor-neutral telemetry signals and semantic conventions for traces, metrics, and logs. <i>Class:</i> Validated from documentation <i>Sources:</i> opentelemetry-spec <i>Locations:</i> sec:component-design;sec:roadmap <i>Caveat:</i> Telemetry standards do not define business audit evidence or operating thresholds.
CLM-INT-004	
Confidence: High	
Accessed: 2026-07-13	PostgreSQL documents transaction isolation and concurrency behavior within one database transaction. <i>Class:</i> Validated from documentation <i>Sources:</i> postgresql-transactions <i>Locations:</i> sec:consistency;sec:internal-ledger <i>Caveat:</i> Database isolation cannot atomically include custody, chain, issuer, FX, payout, or participant systems.
CLM-INT-005	
Confidence: High	
Accessed: 2026-07-13	Debezium documents the transactional outbox pattern for atomically recording business state and an event record in one database. <i>Class:</i> Validated from documentation <i>Sources:</i> debezium-outbox <i>Locations:</i> sec:consistency;sec:messaging <i>Caveat:</i> Outbox publication is still asynchronous and consumers require inbox or idempotent effect handling.
CLM-JAVA-001	
Confidence: High	
Accessed: 2026-07-13	Besu is an Apache-licensed Ethereum execution client written in Java for public and private networks; public Ethereum operation also requires a consensus client. <i>Class:</i> Validated from documentation <i>Sources:</i> besu-docs <i>Locations:</i> sec:java-architecture;sec:technology-evaluation <i>Caveat:</i> A Java implementation does not justify operating a private network or selecting Besu for the pilot.
CLM-JAVA-002	
Confidence: High	
Accessed: 2026-07-13	The official Besu repository contains the execution-client implementation and current release history. <i>Class:</i> Validated from source code <i>Sources:</i> besu-source <i>Locations:</i> sec:technology-evaluation <i>Caveat:</i> Repository activity is not evidence of EWS use, procurement approval, or production suitability.
CLM-JAVA-003	
Confidence: High	

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Table 62 – continued

Claim	Evidence record
Accessed: 2026-07-13	Web3j provides Java and Android APIs for Ethereum JSON-RPC, contract wrappers, events, wallets, and HSM integration. <i>Class:</i> Validated from documentation <i>Sources:</i> web3j-docs <i>Locations:</i> sec:java-architecture;sec:technology-evaluation <i>Caveat:</i> Web3j is not a node, ledger, custody policy, or consensus implementation.
CLM-JAVA-004	
Confidence: High	
Accessed: 2026-07-13	The official Web3j repository identifies v6.0.0 as the current release at the research date. <i>Class:</i> Validated from source code <i>Sources:</i> web3j-source <i>Locations:</i> sec:technology-evaluation <i>Caveat:</i> Version compatibility and support policy must be pinned during implementation and procurement.
CLM-JAVA-005	
Confidence: High	
Accessed: 2026-07-13	Hedera provides Java SDKs and EVM-compatible contracts while retaining distinct consensus-node and mirror-node responsibilities. <i>Class:</i> Validated from documentation <i>Sources:</i> hedera-docs <i>Locations:</i> sec:technology-evaluation <i>Caveat:</i> A Java SDK does not make network governance, fees, permissioning, or operating ownership equivalent to an EVM chain.
CLM-JAVA-006	
Confidence: High	
Accessed: 2026-07-13	The official Hiero consensus-node repository contains the Java-based node implementation. <i>Class:</i> Validated from source code <i>Sources:</i> hiero-source <i>Locations:</i> sec:technology-evaluation <i>Caveat:</i> Source language does not resolve network, governance, data, custody, or interoperability fit.
CLM-JAVA-007	
Confidence: High	
Accessed: 2026-07-13	Corda 5 documents a permissioned application network with verified identities, flows, UTXO ledger options, and notary-based uniqueness. <i>Class:</i> Validated from documentation <i>Sources:</i> corda-docs <i>Locations:</i> sec:technology-evaluation <i>Caveat:</i> Corda is not a public stablecoin network or EVM client and adds application-network governance.
CLM-JAVA-008	
Confidence: High	
Accessed: 2026-07-13	The official Corda runtime repository implements the Corda 5 platform primarily on the JVM. <i>Class:</i> Validated from source code <i>Sources:</i> corda-source <i>Locations:</i> sec:technology-evaluation <i>Caveat:</i> Licensing, commercial support, version compatibility, and consortium operation require diligence.

Continued on next page

Table 62 – continued

Claim	Evidence record
CLM-JAVA-009	
Confidence: High	
Accessed: 2026-07-13	bitcoinj is a Java library for Bitcoin protocol and wallet integration that can operate without a local Bitcoin Core copy. <i>Class:</i> Validated from documentation <i>Sources:</i> bitcoinj-docs <i>Locations:</i> sec:technology-evaluation <i>Caveat:</i> It is not a general enterprise blockchain node or stablecoin platform.
CLM-JAVA-010	
Confidence: High	
Accessed: 2026-07-13	The official bitcoinj repository contains the Java library implementation and release history. <i>Class:</i> Validated from source code <i>Sources:</i> bitcoinj-source <i>Locations:</i> sec:technology-evaluation <i>Caveat:</i> Bitcoin protocol support does not provide EVM contracts, issuer controls, or bank settlement.
CLM-JAVA-011	
Confidence: High	
Accessed: 2026-07-13	Spring Kafka exactly-once semantics cover Kafka read-process-write transactions while reads and processing remain at least once. <i>Class:</i> Validated from documentation <i>Sources:</i> spring-kafka-eos <i>Locations:</i> sec:java-architecture;sec:consistency <i>Caveat:</i> The guarantee does not atomically include a ledger database, custody, chain, FX, or payout system.
CLM-JAVA-012	
Confidence: High	
Accessed: 2026-07-13	Java virtual threads are designed to support high-throughput thread-per-request code with compatible blocking APIs. <i>Class:</i> Validated from documentation <i>Sources:</i> jep-444 <i>Locations:</i> sec:java-architecture <i>Caveat:</i> They do not remove external latency, database contention, backpressure, or library compatibility constraints.
CLM-OPS-001	
Confidence: Medium	
Accessed: 2026-07-13	Pilot limits, finality thresholds, liquidity buffers, recovery objectives, and manual-approval thresholds should begin conservatively. <i>Class:</i> Operational assumption <i>Sources:</i> author-design <i>Locations:</i> sec:roadmap;sec:finality <i>Caveat:</i> Numeric values require approved corridor, workload, loss, chain, liquidity, and regulatory data.
CLM-OPS-002	
Confidence: Medium	

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Table 62 – continued

Claim	Evidence record
Accessed: 2026-07-13	Managed node access plus an independent read path is a pragmatic initial topology. <i>Class:</i> Operational assumption <i>Sources:</i> author-design <i>Locations:</i> sec:technology-evaluation;sec:roadmap <i>Caveat:</i> Provider concentration, integrity, latency, data, and incident controls require testing and contract review.
CLM-PROP-001	
Confidence: High	
Accessed: 2026-07-13	A regulated control plane should isolate custody, signing, issuers, chains, FX, and payout providers behind explicit versioned adapters. <i>Class:</i> Proposed target architecture <i>Sources:</i> author-design <i>Locations:</i> sec:architecture-overview;sec:component-design <i>Caveat:</i> Provider-specific risks must remain visible rather than erased by a least-common-denominator interface.
CLM-PROP-002	
Confidence: High	
Accessed: 2026-07-13	An internal double-entry ledger should be authoritative for reservations, obligations, fees, FX, payout, suspense, compensation, and reconciliation. <i>Class:</i> Proposed target architecture <i>Sources:</i> author-design <i>Locations:</i> sec:internal-ledger;sec:architecture-overview <i>Caveat:</i> The exact chart of accounts and legal system-of-record boundary require organizational accounting decisions.
CLM-PROP-003	
Confidence: High	
Accessed: 2026-07-13	A durable versioned payment state machine should gate every external effect on recorded evidence. <i>Class:</i> Proposed target architecture <i>Sources:</i> author-design <i>Locations:</i> sec:payment-state-machine;sec:payment-lifecycle <i>Caveat:</i> State vocabulary and route guards require governed versioning and production validation.
CLM-PROP-004	
Confidence: High	
Accessed: 2026-07-13	Blockchain, legal settlement, customer-visible completion, and accounting finality should be modeled as four separate decisions. <i>Class:</i> Proposed target architecture <i>Sources:</i> author-design <i>Locations:</i> sec:finality <i>Caveat:</i> Evidence, thresholds, and order vary by product, corridor, asset, chain, issuer, and payout rail.
CLM-PROP-005	
Confidence: Medium-high	

Continued on next page

Table 62 – continued

Claim	Evidence record
Accessed: 2026-07-13	The initial pilot should use one approved regulated asset, one approved public network, one corridor, bounded participants, prefunded liquidity, no consumer wallet, and no bridge. <i>Class:</i> Proposed target architecture <i>Sources:</i> author-design <i>Locations:</i> sec:non-goals;sec:roadmap;sec:stablecoin-options <i>Caveat:</i> The approved asset, network, corridor, participants, and limits require Phase 0 decisions.
CLM-PROP-006	
Confidence: High	
Accessed: 2026-07-13	Java and Spring should own participant APIs, orchestration, ledger-facing services, policy integration, events, reconciliation, and operations while chain-native components remain in their supported ecosystems. <i>Class:</i> Proposed target architecture <i>Sources:</i> author-design;web3j-docs;besu-docs <i>Locations:</i> sec:java-architecture <i>Caveat:</i> Component boundaries should follow measured workload, support, security, and ownership rather than language preference.
CLM-PROP-007	
Confidence: High	
Accessed: 2026-07-13	Exactly-once business effects should be constructed with idempotency, local ACID transactions, versions, outbox/inbox, inquiry, compensation, and reconciliation rather than claimed as distributed delivery behavior. <i>Class:</i> Proposed target architecture <i>Sources:</i> author-design;aws-idempotent-apis;debezium-outbox;spring-kafka-eos <i>Locations:</i> sec:consistency;sec:messaging <i>Caveat:</i> Every external system remains a separate atomicity and evidence domain.
CLM-RAIL-001	
Confidence: High	
Accessed: 2026-07-13	FedNow supports continuous processing between eligible depository institutions and describes immediate funds availability and final settlement. <i>Class:</i> Validated from documentation <i>Sources:</i> fednow-about;fednow-product-sheet <i>Locations:</i> sec:problem-context;sec:finality <i>Caveat:</i> Participant product behavior, coverage, limits, and customer recourse remain institution specific.
CLM-RAIL-002	
Confidence: High	
Accessed: 2026-07-13	The RTP network describes real-time final interbank settlement and its credit transfers as irrevocable. <i>Class:</i> Validated from documentation <i>Sources:</i> tch-rtsp <i>Locations:</i> sec:problem-context;sec:finality <i>Caveat:</i> Network finality does not by itself resolve customer disputes, fraud, legal obligations, or product disclosures.
CLM-RAIL-003	
Confidence: High	

Continued on next page

Table 62 – continued

Claim	Evidence record
Accessed: 2026-07-13	Nacha rules permit ACH reversals only for defined error conditions and within specified timeframes. <i>Class:</i> Validated from documentation <i>Sources:</i> nacha-reversals <i>Locations:</i> sec:problem-context;sec:failure-model <i>Caveat:</i> Current Nacha rules and participant counsel govern a specific payment and may change.
CLM-REG-001	
Confidence: High	
Accessed: 2026-07-13	The GENIUS Act became Public Law 119-27 on July 18, 2025 and establishes a United States payment-stablecoin issuer framework. <i>Class:</i> Validated from public source <i>Sources:</i> genius-act-2025 <i>Locations:</i> sec:compliance;sec:stablecoin-options <i>Caveat:</i> Effective dates, implementing rules, approvals, and application to a particular structure require current legal review.
CLM-REG-002	
Confidence: High	
Accessed: 2026-07-13	The GENIUS Act includes permitted-issuer, reserve, redemption, disclosure, and Bank Secrecy Act provisions. <i>Class:</i> Validated from public source <i>Sources:</i> genius-act-2025 <i>Locations:</i> sec:compliance;sec:stablecoin-options <i>Caveat:</i> This paper provides architecture context rather than legal advice or an issuer eligibility conclusion.
CLM-REG-003	
Confidence: High	
Accessed: 2026-07-13	FinCEN applies existing money-transmission analysis to convertible virtual-currency business models based on the facts and roles involved. <i>Class:</i> Validated from public source <i>Sources:</i> fincen-cvc-2019 <i>Locations:</i> sec:compliance <i>Caveat:</i> Applicability depends on exact custody, control, flow, counterparty, and jurisdiction facts.
CLM-REG-004	
Confidence: High	
Accessed: 2026-07-13	OFAC states that sanctions obligations apply to virtual-currency transactions as they do to fiat-currency transactions. <i>Class:</i> Validated from public source <i>Sources:</i> ofac-virtual-currency-2021 <i>Locations:</i> sec:compliance;sec:security <i>Caveat:</i> Screening cannot guarantee that facts or designations will not change after submission.
CLM-REG-005	
Confidence: High	

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Table 62 – continued

Claim	Evidence record
Accessed: 2026-07-13	Regulation E establishes United States electronic-fund-transfer disclosure, error-resolution, and liability requirements for covered transfers. <i>Class:</i> Validated from public source <i>Sources:</i> ecf-reg-e <i>Locations:</i> sec:compliance <i>Caveat:</i> Coverage and application to a particular cross-border or stablecoin product require counsel and product facts.
CLM-REG-006	
Confidence: High	
Accessed: 2026-07-13	FATF guidance applies risk-based AML/CFT expectations and information-sharing recommendations to virtual-asset transfers. <i>Class:</i> Validated from documentation <i>Sources:</i> fatf-virtual-assets-2025 <i>Locations:</i> sec:compliance <i>Caveat:</i> Implementation and Travel Rule applicability depend on jurisdiction, role, asset, and transaction facts.
CLM-SEC-001	
Confidence: High	
Accessed: 2026-07-13	FIPS 140-3 specifies security requirements for cryptographic modules including roles, interfaces, key material, physical security, and lifecycle assurance. <i>Class:</i> Validated from documentation <i>Sources:</i> nist-fips-140-3 <i>Locations:</i> sec:security <i>Caveat:</i> Module validation does not define custody governance, transaction policy, quorum, recovery, or end-to-end authorization.
CLM-SEC-002	
Confidence: High	
Accessed: 2026-07-13	NIST SP 800-57 provides key-management guidance covering lifecycle, protection, compromise, recovery, and cryptoperiod considerations. <i>Class:</i> Validated from documentation <i>Sources:</i> nist-sp-800-57 <i>Locations:</i> sec:security <i>Caveat:</i> System-specific custody policy and regulated operating roles still require design and approval.
CLM-SEC-003	
Confidence: High	
Accessed: 2026-07-13	NIST's Secure Software Development Framework defines practices for preparing, protecting, producing, and responding to software risks. <i>Class:</i> Validated from documentation <i>Sources:</i> nist-ssdf <i>Locations:</i> sec:security;sec:roadmap <i>Caveat:</i> The framework is a control baseline and does not replace system-specific threat modeling or evidence.
CLM-SEC-004	
Confidence: High	

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Table 62 – continued

Claim	Evidence record
Accessed: 2026-07-13	OpenZeppelin documents role-based access control and delayed administrative control patterns for smart contracts. <i>Class:</i> Validated from documentation <i>Sources:</i> openzeppelin-access-control <i>Locations:</i> sec:security;sec:technology-evaluation <i>Caveat:</i> Using a library does not prove that roles, upgrades, timelocks, monitoring, and emergency authority are correctly governed.
CLM-UNK-001	
Confidence: High	
Accessed: 2026-07-13	No reviewed public source establishes the current EWS service topology, systems of record, participant message contracts, data stores, queues, service levels, or recovery objectives. <i>Class:</i> Unknown / needs validation <i>Sources:</i> ews-zelle-p2p-2025;ews-fraud-rfi-2025 <i>Locations:</i> sec:problem-context;sec:roadmap <i>Caveat:</i> The reference architecture must not be read as a deployed-system description.
CLM-UNK-002	
Confidence: High	
Accessed: 2026-07-13	No reviewed public source establishes the ZelleUSD issuer entity, authorization, reserves, redemption, chain, contracts, custody, keys, controls, partners, circulation, or production schedule. <i>Class:</i> Unknown / needs validation <i>Sources:</i> ews-india-zelleusd-2026 <i>Locations:</i> sec:stablecoin-options;sec:roadmap <i>Caveat:</i> The announcement states that further details will follow and does not close these architecture questions.
CLM-UNK-003	
Confidence: High	
Accessed: 2026-07-13	Legal finality, customer completion, accounting close, loss allocation, data residency, disclosures, and regulatory roles are not publicly defined for the proposed corridors and instruments. <i>Class:</i> Unknown / needs validation <i>Sources:</i> genius-act-2025;ecfr-reg-e;ofac-virtual-currency-2021 <i>Locations:</i> sec:finality;sec:compliance;sec:roadmap <i>Caveat:</i> These items require role-, product-, corridor-, asset-, issuer-, chain-, and payout-specific decisions.
CLM-ZEL-001	
Confidence: High	
Accessed: 2026-07-13	Zelle is owned and operated by Early Warning Services. <i>Class:</i> Validated from public source <i>Sources:</i> ews-zelle-results-2026 <i>Locations:</i> sec:introduction;sec:problem-context <i>Caveat:</i> Ownership does not disclose platform topology or operating contracts.
CLM-ZEL-002	
Confidence: High	

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Table 62 – continued

Claim	Evidence record
Accessed: 2026-07-13	EWS reported that Zelle processed more than 1.2 trillion dollars and 4.2 billion transactions during 2025. <i>Class:</i> Validated from public source <i>Sources:</i> ews-zelle-results-2026 <i>Locations:</i> sec:introduction;sec:problem-context <i>Caveat:</i> EWS-reported annual totals do not establish peak throughput, workload distribution, or service levels.
CLM-ZEL-003	
Confidence: High	
Accessed: 2026-07-13	EWS reported more than 2300 participating financial institutions and reach to 80 percent of United States bank and credit-union accounts. <i>Class:</i> Validated from public source <i>Sources:</i> ews-community-2026;ews-zelle-results-2026 <i>Locations:</i> sec:problem-context <i>Caveat:</i> Exact active integration modes, versions, and participant capabilities are not public.
CLM-ZEL-004	
Confidence: High	
Accessed: 2026-07-13	Consumers primarily access Zelle through a participating financial institution’s mobile or online banking experience. <i>Class:</i> Validated from public source <i>Sources:</i> zelle-how-it-works;ews-brands <i>Locations:</i> sec:introduction;sec:problem-context <i>Caveat:</i> Authentication, limits, user interface, and support vary by institution.
CLM-ZEL-005	
Confidence: High	
Accessed: 2026-07-13	Zelle enrollment and payment addressing use an email address or United States mobile number associated with an eligible account. <i>Class:</i> Validated from public source <i>Sources:</i> zelle-how-it-works <i>Locations:</i> sec:problem-context <i>Caveat:</i> Public addressing behavior does not establish directory implementation or internal authority hierarchy.
CLM-ZEL-007	
Confidence: High	
Accessed: 2026-07-13	A 2025 EWS Zelle product brief depicts sending and receiving financial institutions settling through ACH or RTP. <i>Class:</i> Validated from public source <i>Sources:</i> ews-zelle-p2p-2025 <i>Locations:</i> sec:problem-context;sec:payment-lifecycle <i>Caveat:</i> The diagram does not prove every transaction and participant use the same route.
CLM-ZEL-008	
Confidence: High	
Accessed: 2026-07-13	EWS publicly describes network risk signals, enrollment controls, warnings, and participant responsibilities. <i>Class:</i> Validated from public source <i>Sources:</i> ews-fraud-rfi-2025 <i>Locations:</i> sec:problem-context;sec:compliance <i>Caveat:</i> Rules, models, features, ownership boundaries, and interfaces remain confidential.

Continued on next page

Table 62 – continued

Claim	Evidence record
CLM-ZEL-009	
Confidence: High	
Accessed: 2026-07-13	Zelle directs users of a participating institution's implementation to that bank or credit union for support. <i>Class:</i> Validated from public source <i>Sources:</i> zelle-support <i>Locations:</i> sec:problem-context <i>Caveat:</i> Loss allocation, dispute rules, and escalation interfaces are not established by the support page.
CLM-ZEL-010	
Confidence: High	
Accessed: 2026-07-13	Zelle warns users to pay people they know and trust and states that an authorized payment to an enrolled recipient generally cannot be canceled. <i>Class:</i> Validated from public source <i>Sources:</i> zelle-security <i>Locations:</i> sec:problem-context;sec:compliance <i>Caveat:</i> The page is not a complete statement of Regulation E, reimbursement, dispute, or legal-finality obligations.
CLM-ZEL-011	
Confidence: High	
Accessed: 2026-07-13	On October 24 2025 EWS announced an initiative to use stablecoins for cross-border Zelle money movement and intended availability to network financial institutions on equal terms. <i>Class:</i> Validated from public source <i>Sources:</i> ews-international-2025 <i>Locations:</i> sec:introduction;sec:problem-context <i>Caveat:</i> The announcement disclosed no chain, issuer, custody, reserve, ledger, contract, or API architecture.
CLM-ZEL-012	
Confidence: High	
Accessed: 2026-07-13	On June 11 2026 EWS announced India as the first country for United States-originated Zelle remittances with initial availability expected before the end of 2026. <i>Class:</i> Validated from public source <i>Sources:</i> ews-india-zelleusd-2026 <i>Locations:</i> sec:introduction;sec:problem-context <i>Caveat:</i> The announcement does not provide corridor implementation, partner, rail, legal, or service-level details.
CLM-ZEL-013	
Confidence: High	
Accessed: 2026-07-13	EWS announced ZelleUSD as its proprietary United States dollar-backed stablecoin for future international capabilities in other markets. <i>Class:</i> Validated from public source <i>Sources:</i> ews-india-zelleusd-2026 <i>Locations:</i> sec:introduction;sec:stablecoin-options <i>Caveat:</i> Announced or unveiled does not establish issuance, circulation, licensure, reserves, redemption, chain, contracts, custody, or production readiness.
CLM-ZEL-014	
Confidence: High	

Continued on next page

Table 62 – continued

Claim	Evidence record
Accessed: 2026-07-13	The June 2026 announcement does not state that the India route uses ZelleUSD. <i>Class:</i> Validated from public source <i>Sources:</i> ews-india-zelleusd-2026 <i>Locations:</i> sec:introduction;sec:stablecoin-options <i>Caveat:</i> Future announcements may add facts; the paper reflects the source wording as accessed on 2026-07-13.

Table 63: Source-to-claim index. Evidence keys match the annotated bibliography.

Evidence key	Related claim IDs	Paper locations
author-design	CLM-OPS-001, CLM-OPS-002, CLM-PROP-001, CLM-PROP-002, CLM-PROP-003, CLM-PROP-004, CLM-PROP-005, CLM-PROP-006, CLM-PROP-007	sec:architecture-overview, sec:component-design, sec:consistency, sec:finality, sec:internal-ledger, sec:java-architecture, sec:messaging, sec:non-goals, sec:payment-lifecycle, sec:payment-state-machine, sec:roadmap, sec:stablecoin-options, sec:technology-evaluation
aws-idempotent-apis	CLM-INT-001, CLM-PROP-007	sec:component-design, sec:consistency, sec:messaging
besu-docs	CLM-JAVA-001, CLM-PROP-006	sec:java-architecture, sec:technology-evaluation
besu-source	CLM-JAVA-002	sec:technology-evaluation
bis-annual-2026	CLM-DA-003	sec:problem-context, sec:stablecoin-options
bis-tokenised-ledger-2023	CLM-DA-003	sec:problem-context, sec:stablecoin-options
bitcoinj-docs	CLM-JAVA-009	sec:technology-evaluation
bitcoinj-source	CLM-JAVA-010	sec:technology-evaluation
circle-cctp	CLM-DA-004	sec:payment-lifecycle, sec:stablecoin-options
circle-usdc	CLM-DA-001	sec:stablecoin-options, sec:technology-evaluation
cloudevents-spec	CLM-INT-002	sec:messaging
corda-docs	CLM-JAVA-007	sec:technology-evaluation
corda-source	CLM-JAVA-008	sec:technology-evaluation
debezium-outbox	CLM-INT-005, CLM-PROP-007	sec:consistency, sec:messaging
ecfr-reg-e	CLM-REG-005, CLM-UNK-003	sec:compliance, sec:finality, sec:roadmap
eip-1967	CLM-DA-007	sec:security, sec:technology-evaluation
eip-20	CLM-DA-006	sec:stablecoin-options, sec:technology-evaluation
ethereum-pos	CLM-DA-005	sec:failure-model, sec:finality
ews-brands	CLM-ZEL-004	sec:introduction, sec:problem-context
ews-community-2026	CLM-ZEL-003	sec:problem-context
ews-fraud-rfi-2025	CLM-UNK-001, CLM-ZEL-008	sec:compliance, sec:problem-context, sec:roadmap
ews-india-zelleusd-2026	CLM-INF-002, CLM-UNK-002, CLM-ZEL-012, CLM-ZEL-013, CLM-ZEL-014	sec:architecture-overview, sec:introduction, sec:problem-context, sec:roadmap, sec:stablecoin-options
ews-international-2025	CLM-INF-002, CLM-ZEL-011	sec:architecture-overview, sec:introduction, sec:problem-context
ews-zelle-p2p-2025	CLM-INF-001, CLM-UNK-001, CLM-ZEL-007	sec:payment-lifecycle, sec:problem-context, sec:roadmap
ews-zelle-results-2026	CLM-ZEL-001, CLM-ZEL-002, CLM-ZEL-003	sec:introduction, sec:problem-context
fatf-virtual-assets-2025	CLM-REG-006	sec:compliance
fednow-about	CLM-RAIL-001	sec:finality, sec:problem-context
fednow-product-sheet	CLM-RAIL-001	sec:finality, sec:problem-context
fincen-cvc-2019	CLM-REG-003	sec:compliance
fireblocks-capabilities	CLM-CUST-001	sec:security
fireblocks-recovery	CLM-CUST-002	sec:security
genius-act-2025	CLM-REG-001, CLM-REG-002, CLM-UNK-003	sec:compliance, sec:finality, sec:roadmap, sec:stablecoin-options
hedera-docs	CLM-JAVA-005	sec:technology-evaluation
hierio-source	CLM-JAVA-006	sec:technology-evaluation
jep-444	CLM-JAVA-012	sec:java-architecture

Table 64 – continued

Evidence key	Related claim IDs	Paper locations
nacha-reversals	CLM-RAIL-003	sec:failure-model, sec:problem-context
nist-fips-140-3	CLM-SEC-001	sec:security
nist-sp-800-57	CLM-SEC-002	sec:security
nist-ssdf	CLM-SEC-003	sec:roadmap, sec:security
ofac-virtual-currency-2021	CLM-INF-002, CLM-REG-004, CLM-UNK-003	sec:architecture-overview, sec:compliance, sec:finality, sec:problem-context, sec:roadmap, sec:security
opentelemetry-spec	CLM-INT-003	sec:component-design, sec:roadmap
openzeppelin-access-control	CLM-SEC-004	sec:security, sec:technology-evaluation
paxos-pyusd	CLM-DA-002	sec:stablecoin-options
postgresql-transactions	CLM-INT-004	sec:consistency, sec:internal-ledger
spring-kafka-eos	CLM-JAVA-011, CLM-PROP-007	sec:consistency, sec:java-architecture, sec:messaging
tch-rtp	CLM-RAIL-002	sec:finality, sec:problem-context
web3j-docs	CLM-JAVA-003, CLM-PROP-006	sec:java-architecture, sec:technology-evaluation
web3j-source	CLM-JAVA-004	sec:technology-evaluation
zelle-how-it-works	CLM-INF-001, CLM-ZEL-004, CLM-ZEL-005	sec:introduction, sec:problem-context
zelle-security	CLM-ZEL-010	sec:compliance, sec:problem-context
zelle-support	CLM-ZEL-009	sec:problem-context

F. Figure descriptions and visual reading order

All canonical figures are original SVGs with title and description elements. Colors repeat semantics and are paired with labels, badges, border styles, lane labels, or shapes. The longer descriptions below carry relationships that may be difficult to infer from the visual alone.

1. **Context boundaries (figure 1).** The left PUBLIC boundary contains customers, participating institutions, documented network/alias roles and domestic rails. The right PROPOSED boundary retains those participants and adds a control plane. Custody, issuer, chain, FX and payout boxes remain external; UNKNOWN badges mark organization-specific selections.
2. **Layered architecture (figure 2).** Requests flow down from participants through intent/policy/orchestration to ledger/workflow authority, then through adapters and signing to external rails. Dashed observations and dotted reconciliation flow upward without granting providers mutation authority.
3. **Trust/deployment (figure 3).** Institution, operator, signer/custody, node, public network and provider boundaries have different administrators. Submission and observation use separable paths; operator action is a governed boundary.
4. **Components (figure 4).** Synchronous acceptance ends at durable intent. Commands coordinate policy, ledger, treasury and adapters. Observations enter workflow; reconciliation compares source records.
5. **Ledger (figure 5).** Internal accounts represent participant, treasury, in-transit, issuer, FX, payout, fee and suspense obligations. External statements reconcile to specific accounts rather than replacing them.
6. **Happy state (figure 6).** States progress only when entry evidence satisfies a guard. Economic chain finality, recipient credit and reconciliation are visibly separate.
7. **Exception state (figure 7).** Rejection, retryable/final failure, manual review and compensation branch from multiple lifecycle points. No generic failure or automatic loop exists.

8. **Cross-border sequence (figure 8).** Participant acceptance and policy precede reservation, signing and submission. Independent observation precedes payout authorization; notification and reconciliation follow their own evidence.
9. **Ambiguous submission (figure 9).** A lost response creates ambiguity. Independent inquiry and stable transaction identity decide whether to observe, safely retry after proved absence, or open a case. A duplicate participant request returns the original intent.
10. **Payout failure (figure 10).** Chain economic finality is already achieved, so recovery retains a payout payable and chooses inquiry, retry, alternative payout, refund/redemption or compensation; it does not reverse the chain.
11. **Transaction boundaries (figure 11).** Local stores have atomic boundaries. Broker, signer, chain and payout edges are non-atomic gaps handled by operation identity, inquiry and reconciliation.
12. **Outbox/inbox (figure 12).** Producer state/outbox and consumer inbox/effect each commit locally. At-least-once broker delivery is safe because event/effect identities are unique and replay is governed.
13. **Containment (figure 13).** Scoped switches prevent new exposure while leaving inquiry, observation, reconciliation and support available. Domestic routes remain isolated from digital-asset failure.
14. **Four finalities (figure 14).** Chain, legal, customer and accounting lanes have different evidence and owners. Their order can vary; no vertical line declares universal completion.
15. **Key authority (figure 15).** Application authorization and signer policy are independent. Exact bytes and transaction fields enter HSM/MPC/custody; keys never enter application memory. Independent observation checks the signed result.
16. **Compliance overlay (figure 16).** Compliance decisions recur from enrollment through reconciliation. Holds and case obligations can occur before or after submission without rewriting history.
17. **Java boundary and roadmap (figure 17).** The top panel keeps domain/control modules in Java and specialist components behind language-neutral APIs. The bottom panel shows seven evidence gates; arrows denote earned promotion, not elapsed time.

G. Build and verification notes

The canonical hierarchy, required tools, offline commands, reproducibility epoch, derivative checks, PDF inspection, and archive rebuild procedure are documented in `README.md` and `docs/BUILD.md`. The release verification report records actual versions, hashes, warnings, page count, font status, link checks, stale-output checks, and isolated archive build. Generated files identify their source and are never edited directly.

H. Glossary

Table 65. Cross-domain vocabulary used by this paper.

Term	Definition in this architecture
ABI (application binary interface)	Contract-call and event encoding boundary used by EVM tooling; conformance tests must prove exact bytes rather than trust a generated wrapper.
ACL (access-control list)	Explicit set of principals and permitted actions on a broker, service or resource; financial authorization also requires purpose, object, value and state constraints.
ACH (Automated Clearing House)	Batch-oriented United States bank-account payment network used here only as a documented domestic-rail option; its rules and reversal model differ from real-time rails and blockchains.
ACID (atomicity, consistency, isolation, durability)	Database transaction properties relied on for one local journal, operation record, versions and outbox; they do not extend to a broker, signer, chain, FX or payout provider.

Term	Definition in this architecture
ADR (architecture decision record)	Versioned decision artifact stating context, options, evidence, decision, owners, consequences, change triggers and supersession.
AML (anti-money laundering)	Risk-based policies, monitoring, investigation, reporting and controls applied according to the organization's role, jurisdiction, counterparties, asset and product.
API (application programming interface)	Versioned machine contract for participant or component commands, queries and evidence; it does not by itself define authority or atomicity.
AWS (Amazon Web Services)	Cloud provider whose official idempotent-API guidance is cited as technical documentation; the citation does not select AWS for deployment.
Accounting finality	The point at which required journals, valuations, external reconciliations, break dispositions and close controls satisfy the accountable accounting policy. Later corrections are new entries.
Available balance	Settled, policy-eligible balance minus active reservations, restrictions and other amounts unavailable for a new authorized operation.
BIS (Bank for International Settlements)	International institution whose policy research is cited for tokenized-money and settlement distinctions; its analysis is not a Zelle requirement.
Blockchain finality	A chain-risk policy determination that canonical transaction evidence meets the selected network, route, value and health threshold; not automatically legal or accounting finality.
Bridge	A mechanism that transfers or represents value across networks, often through lock/mint or burn/mint and additional validators/attesters. No generic bridge is in the initial pilot.
Business operation ID	Stable identifier for one intended financial effect, reused across delivery retries and linked to authorization, provider records, journals and reconciliation.
Chain reorganization (re-org)	Replacement of previously observed canonical blocks by another canonical history, potentially removing or changing a transaction's observed status.
CI/CD (continuous integration and continuous delivery/deployment)	Automated build, test, provenance, approval and release path; production promotion remains evidence-gated rather than automatic.
CCTP (Cross-Chain Transfer Protocol)	Circle's issuer-mediated USDC burn, attestation and mint mechanism, used only as a documented interoperability example rather than a selected generic bridge.
CFT (countering the financing of terrorism)	Controls paired with AML in risk-based frameworks; exact obligations depend on role, jurisdiction, counterparty, product and transaction.
CNCF (Cloud Native Computing Foundation)	Open-source foundation governing projects including OpenTelemetry; foundation governance does not by itself prove product fitness or support.
Confirmation	Evidence that a transaction appears in a block or provider record at a particular depth/status; an input to, not a synonym for, finality.
Custody	Safekeeping and controlled use of digital-asset keys/assets under defined legal, security, authorization, recovery and reporting obligations.
CVE (Common Vulnerabilities and Exposures)	Public identifier for a disclosed security vulnerability used in dependency, patch and incident processes; a CVE count alone is not a maturity score.
Customer-visible completion	Product-policy decision that permits the participant/customer experience to present a payment as completed based on required evidence and any explicitly absorbed interval risk.
Double entry	Accounting method in which every journal has equal debit and credit totals in the relevant unit, preserving a testable balance invariant.
DB (database)	Persistent data system; workflow DB and ledger DB mean separately owned transactional authorities, not necessarily distinct products in the pilot.
DLT (distributed ledger technology)	Shared ledger/protocol family that includes public and permissioned networks; the term says nothing by itself about legal claim, finality, privacy or operating fitness.

Term	Definition in this architecture
DX (developer experience)	Practical quality of APIs, tooling, local tests, diagnostics, documentation and failure visibility for engineers who build and operate a component.
Early Warning Services (EWS)	Owner and operator of Zelle according to the cited public record; the paper does not infer its private topology or vendors.
EIP / ERC (Ethereum Improvement Proposal / Ethereum Request for Comments)	Numbered Ethereum standards process and application-interface conventions; ERC-20 supplies token calls/events but not issuer economics or legal rights.
EVM (Ethereum Virtual Machine)	Execution environment and bytecode/contract model used by Ethereum-compatible networks; compatible execution does not make networks or assets equivalent.
Economic finality	This paper's chain-risk gate represented by <code>ECONOMICALLY_FINAL</code> : evidence and policy make further reorganization sufficiently remote for the approved next action, without making a legal conclusion.
FATF (Financial Action Task Force)	Intergovernmental standard-setting body whose virtual-asset guidance informs role- and jurisdiction-specific AML/CFT decisions; it is not a product implementation specification.
FIPS (Federal Information Processing Standard)	United States government standard family; FIPS 140-3 defines scoped cryptographic-module security requirements, not end-to-end custody governance.
Finality threshold	Versioned, route-specific evidence rule for a finality decision, including sources, network conditions, value bands and escalation.
Foreign exchange (FX)	Conversion and associated quote, fill, settlement, exposure and reconciliation between currencies/assets.
General ledger (GL)	Organization's authoritative accounting book to which the payment subledger reconciles and exports controlled totals and adjustments.
gRPC	Remote-procedure-call framework often used by node or provider APIs; transport choice does not decide financial authority or retry safety.
Hardware security module (HSM)	Tamper-resistant cryptographic module used to generate/protect/use keys under controlled interfaces and policy.
HTTP / JSON-RPC	Common request and remote-procedure-call transports used by participant and blockchain APIs; a successful response is evidence from one boundary, not proof of end-to-end effect.
Idempotency	Property that repeating the same authorized request/operation identity produces the same business result without duplicating its effect; conflicting payload reuse is rejected.
Issuer	Legal entity that creates and governs an asset and its terms, including applicable reserve, redemption and administrative controls.
JDK / JVM (Java Development Kit / Java Virtual Machine)	Java toolchain and managed runtime. The evaluated control plane uses them without requiring node, contract or custody implementations to be Java.
LTS (long-term support)	Vendor or community maintenance designation for a release line. Support terms, patch horizon and provider must be verified rather than inferred from the acronym.
Ledger	Internal authoritative double-entry record of the platform's business obligations, reservations, assets, liabilities, income, expenses and corrections.
Legal settlement finality	Counsel/product-policy determination of when an obligation is legally discharged or irrevocable under applicable instruments, agreements, roles and law. This paper provides no opinion.

Term	Definition in this architecture
Multiparty computation (MPC)	Cryptographic technique that can distribute signing authority across parties/devices without reconstructing a full key in one place; governance remains necessary.
NIST (National Institute of Standards and Technology)	United States standards organization cited for cryptographic key management, secure development and module requirements.
Node	Software/service that participates in or exposes a blockchain network's transactions, blocks, state and protocol. A node response is one evidence source.
On/off ramp	Service/process converting between fiat/bank value and digital assets, potentially involving issuer redemption, exchange, treasury and payout.
OFAC (Office of Foreign Assets Control)	United States Treasury office administering economic sanctions; applicability and screening decisions require current role, party, asset and jurisdiction facts.
Outbox	Record committed atomically with local business state and later published to messaging, closing the local database-to-broker commit gap without making publication synchronous.
Payout rail	Destination mechanism/provider that credits recipient fiat or account value and returns evidence under its own contract and failure model.
Posting	Immutable debit or credit line belonging to a balanced journal and identified by account, unit, operation, time and audit metadata.
PII (personally identifiable information)	Data that identifies or can be linked to a person; the design minimizes it, keeps it off public chains, and governs purpose, region, access and retention.
PITR (point-in-time recovery)	Database recovery capability to restore state to an approved time; restored state must still reconcile with immutable external effects that occurred afterward.
PDF (Portable Document Format)	Final publication format; LaTeX, CSV, SVG and BibTeX remain canonical and the PDF is a reproducible derivative.
Prefunding	Holding approved inventory/cash before payment execution so settlement does not depend on unsecured real-time credit.
PYUSD	Paxos-issued PayPal USD stablecoin, considered only as an external-asset diligence example under its current issuer terms and controls.
RACI (responsible, accountable, consulted, informed)	Ownership matrix used to assign decisions, incidents, evidence and approvals; it does not replace technical authorization.
RBAC / ABAC (role-based / attribute-based access control)	Authorization models using roles or contextual attributes. Financial operations also require purpose, value, quorum and exact-object constraints.
RDBMS (relational database management system)	Database class providing relational schema, constraints, transactions and SQL; the paper recommends the class conditionally rather than a procurement choice.
Reconciliation	Versioned comparison of internal records with independent external evidence, including tolerance, break, owner, aging and disposition.
Reservation	Ledger/control record that reduces available capacity for an intent without claiming final external movement.
RPO / RTO (recovery point / recovery time objective)	Approved maximum data-loss interval and restoration time targets. Exact values are organizational assumptions to test, not public Zelle facts.
RPC (remote procedure call)	Request/response interface that invokes an operation on another process; timeouts and success responses must be mapped to authority and inquiry semantics.
RTP (Real-Time Payments)	The Clearing House real-time payment network; cited public material describes its own interbank settlement properties, not every participant's customer policy.
SBOM (software bill of materials)	Machine-readable inventory of build dependencies used for vulnerability, license and provenance review.
SDK (software development kit)	Library/tooling that helps construct protocol calls or decode results; it is not automatically a node, signer, ledger or reliability boundary.

Term	Definition in this architecture
SLA (service-level agreement)	Contracted service commitment and remedies; it must be distinguished from measured objectives, end-to-end product behavior and legal finality.
SQL (Structured Query Language)	Relational data definition/query language used for ledger constraints, reconciliation and inspection; correctness still depends on schema and transaction design.
SRE (site reliability engineering)	Engineering discipline owning reliability objectives, telemetry, capacity, incident response and recovery together with domain owners.
Stablecoin	Token intended to maintain a reference value under issuer, reserve/redemption, contract, network and legal terms; interface similarity does not imply economic equivalence.
Suspense	Governed temporary account for an unresolved difference; it requires an owner, evidence, aging and approved resolution, not use as a balancing dump.
Tokenized deposit	Digital representation of a commercial-bank deposit claim under the issuing bank/network arrangement, distinct from a nonbank-issued stablecoin.
Transactional inbox	Consumer-owned record of processed event IDs/effects committed with the local effect to prevent duplicate application under at-least-once delivery.
Travel Rule	Risk-based regulatory information-sharing requirements that may apply to qualifying transfers depending on jurisdiction, role, counterparty and asset.
UI (user interface)	Customer or operator interaction surface; it presents derived status and governed actions but does not directly mutate ledger truth.
URI (Uniform Resource Identifier)	Identifier used for web and PDF links; the release verifies embedded URI schemes and spot-checks external destinations.
USDC	Circle-issued dollar stablecoin used as a documented external-asset example; selection still requires issuer, redemption, custody, network, liquidity and legal diligence.
UTXO (unspent transaction output)	Ledger model in which transactions consume and create uniquely referenced outputs; Corda and Bitcoin-related uses differ and do not imply EVM account semantics.
Wallet	Software/service and associated keys/address capable of controlling digital assets; the initial product does not expose one to consumers.